UNITED STATES
SECURITIES AND EXCHANGE COMMISSION
Washington, D.C. 20549

FORM 10-K

☒ ANNUAL REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934
For the Fiscal Year EndedDecember 31, 2022

☐ TRANSITION REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934
Commission File Number 1-39162

ARCONIC CORPORATION

(Exact name of Registrant as specified in its charter)

Delaware	84-2745636
(State of incorporation)	(I.R.S. Employer Identification No.)

201 Isabella Street, Pittsburgh, Pennsylvania 15212-5872
(Address of principal executive offices) (Zip code)

(412) 992-2500
(Registrant's telephone number, including area code)

Securities registered pursuant to Section 12(b) of the Act;

Title of each class	Trading Symbol	Name of each exchange on which registered
Common Stock, par value $0.01 per share	ARNC	New York Stock Exchange

Securities to be registered pursuant to Section 12(g) of the Act: None

Indicate by check mark if the registrant is a well-known seasoned issuer, as defined in Rule 405 of the Securities Act. Yes ☑ No ☐

Indicate by check mark if the registrant is not required to file reports pursuant to Section 13 or Section 15(d) of the Act. Yes ☐ No ☑

Indicate by check mark whether the registrant (1) has filed all reports required to be filed by Section 13 or 15(d) of the Securities Exchange Act of 1934 during the preceding 12 months (or for such shorter period that the registrant was to file such reports), and (2) has been subject to such filing requirements for the past 90 days. Yes ☑ No ☐

Indicate by check mark whether the registrant has submitted electronically every Interactive Data File required to be submitted pursuant to Rule 405 of Regulation S-T (§232.405 of this chapter) during the preceding 12 months (or for such shorter period that the registrant was required to submit such files). Yes ☑ No ☐

Indicate by check mark whether the registrant is a large accelerated filer, an accelerated filer, a non-accelerated filer, a smaller reporting company, or an emerging growth company. See the definitions of "large accelerated filer," "accelerated filer," "smaller reporting company," and "emerging growth company" in Rule 12b-2 of the Exchange Act.

Large accelerated filer ☑	Accelerated filer ☐
Non-accelerated filer ☐	Smaller reporting company ☐
	Emerging growth company ☐

If an emerging growth company, indicate by check mark if the registrant has elected not to use the extended transition period for complying with any new or revised financial accounting standards provided pursuant to Section 13(a) of the Exchange Act. ☐

Indicate by check mark whether the registrant has filed a report on and attestation to its management's assessment of the effectiveness of its internal control over financial reporting under Section 404(b) of the Sarbanes-Oxley Act (15 U.S.C. 7262(b)) by the registered public accounting firm that prepared or issued its audit report. ☑

If securities are registered pursuant to Section 12(b) of the Act, indicate by check mark whether the financial statements of the registrant included in the filing reflect the correction of an error to previously issued financial statements. ☐

Indicate by check mark whether any of those error corrections are restatements that required a recovery analysis of incentive-based compensation received by any of the registrant's executive officers during the relevant recovery period pursuant to §240.10D-1(b). ☐

Indicate by check mark whether the registrant is a shell company (as defined in Rule 12b-2 of the Act). Yes ☐ No ☑

The aggregate market value of the outstanding common stock, other than shares held by persons who may be deemed affiliates of the registrant, as of the last business day of the registrant's most recently completed second fiscal quarter was approximately $2.9 billion. As of February 17, 2023, there were 99,399,365 shares of common stock, par value $0.01 per share, of the registrant outstanding.

Documents incorporated by reference.

Part III of this Form 10-K incorporates by reference certain information from the registrant's definitive Proxy Statement for its 2023 Annual Meeting of Stockholders to be filed pursuant to Regulation 14A (2023 Proxy Statement).

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Forward-Looking Statements

This Annual Report on Form 10-K contains statements that relate to future events and expectations and, as such, constitute forward-looking statements within the meaning of the Private Securities Litigation Reform Act of 1995. Forward-looking statements include those containing such words as “anticipates,” “believes,” “could,” “estimates,” “expects,” “forecasts,” “goal,” “guidance,” “intends,” “may,” “outlook,” “plans,” “projects,” “seeks,” “sees,” “should,” “targets,” “will,” “would,” or other words of similar meaning. All statements that reflect Arconic Corporation’s expectations, assumptions, projections, beliefs or opinions about the future, other than statements of historical fact, are forward-looking statements, including, without limitation, statements relating to the condition of, or trends or developments in, the ground transportation, aerospace, building and construction, industrial, packaging and other end markets; Arconic Corporation's future financial results, operating performance, working capital, cash flows, liquidity and financial position; cost savings and restructuring programs; Arconic Corporation’s strategies, outlook, business and financial prospects; share repurchases; costs associated with pension and other postretirement benefit plans; projected sources of cash flow; potential legal liability; the impact of inflationary price pressures; and the potential impact of public health epidemics or pandemics, including the COVID-19 pandemic. These statements reflect beliefs and assumptions that are based on Arconic Corporation’s perception of historical trends, current conditions and expected future developments, as well as other factors Arconic Corporation believes are appropriate in the circumstances. Forward-looking statements are not guarantees of future performance, and actual results may differ materially from those indicated by these forward-looking statements due to a variety of risks, uncertainties and changes in circumstances, many of which are beyond Arconic Corporation’s control.

For a discussion of some of the specific factors that could cause actual results to differ materially from the information contained in this report, see the following sections of this report: "Summary of Risks Affecting our Business," Part I. Item 1A. "Risk Factors," Part II. Item 7. "Management’s Discussion and Analysis of Financial Condition and Results of Operations," including the disclosures under "Segment Information" and "Critical Accounting Policies and Estimates," and Note T to the Consolidated Financial Statements in Part II, Item 8. "Financial Statements and Supplementary Data." Market projections are subject to the risks discussed in this report and other risks in the market. Arconic Corporation disclaims any intention or obligation to update publicly any forward-looking statements, whether in response to new information, future events or otherwise, except as required by applicable law.

Unless otherwise specified or the context otherwise requires, when used in this Annual Report on Form 10-K, the terms “we,” “our,” “us,” “Arconic,” or the “Company” refer to Arconic Corporation and its subsidiaries. The term "ParentCo" refers to Arconic Inc. prior to our separation from Arconic Inc. on April 1, 2020, and “Howmet” refers to “Howmet Aerospace Inc.,” the name of ParentCo following the separation on April 1, 2020.

Summary of Risks Affecting our Business

Our business is subject to numerous risks. The following summary highlights some of the risks you should consider with respect to our business, financial condition and results of operations. This summary is not complete and the risks summarized below are not the only risks we face. You should review and consider carefully the risks and uncertainties described in more detail in Part I, Item 1A. "Risk Factors" of this Annual Report on Form 10-K, which includes a more complete discussion of the risks summarized below as well as a discussion of other risks related to our business and an investment in our common stock.

Business Risks – Global Conditions

- The markets for our products are cyclical and are influenced by a number of factors, including global economic conditions, that could have a material adverse effect on our business, financial condition or results of operations.
- Our business, results of operations, financial condition, liquidity and cash flows have been, and in the future could be, materially adversely affected by the effects of widespread public health epidemics/pandemics.
- Climate change, and evolving customer and stakeholder expectations, legal, regulatory and policy requirements, and market dynamics driven by climate change, could adversely affect our business, financial condition or results of operations.
- Governments of countries in which we operate could nationalize or expropriate private enterprises, or otherwise change their policies regarding private enterprise, which could adversely impact the value of our operations in those countries.
- We are exposed to economic factors, including inflation, fluctuations in aluminum prices, foreign currency exchange rates and interest rates, and currency controls in the countries in which we operate.
- Our global operations expose us to risks that could adversely affect our business, financial condition, results of operations, cash flows or the market price of our securities.
- An adverse decline in the liability discount rate, lower-than-expected investment return on pension assets and other factors could affect our results of operations or amount of pension funding contributions in future periods.

Business Risks – Competition and Customers

- We face significant competition, which may have an adverse effect on profitability.
- We could be adversely affected by the loss of key customers, the impact of supply chain disruptions or other economic conditions on our key customers, or significant changes in the business or financial condition of our customers.
- Our customers may reduce their demand for aluminum products in favor of alternative materials.
- We may face challenges to our intellectual property rights which could adversely affect our reputation, business and competitive position, financial condition and results of operations.

Business Risks – Operations and Product Development

- We could encounter manufacturing difficulties or other issues that impact product performance, quality or safety, which could affect our ability to supply customers or meet contractual obligations, reputation, business and financial condition and results of operations.
- Our business depends, in part, on our ability to meet increased customer demand successfully and to mitigate the impact of customer program cancellations, reductions and delays.
- A material disruption or limitation of our operations, particularly at one or more of our manufacturing facilities, could adversely affect our business.
- We may be unable to develop innovative new products or implement technology initiatives successfully.

Business Risks – Supply Chain

- Our business could be adversely affected by increases in the cost or volatility in the availability of aluminum or other raw materials.
- We are dependent on a limited number of suppliers for a substantial portion of our primary and scrap aluminum and certain other raw materials essential to our operations.

Business Risks – Strategy

- We may not be able to realize the expected benefits of our re-entry into the packaging market in the U.S. and other geographies.

- We may be unable to realize future targets or goals established for our business segments, or complete capital or other projects, at the levels, projected costs or by the dates targeted.
- Our business and growth prospects may be negatively impacted by limits in our capital expenditures.

Business Risks – Information Security and Internal Controls

- Information technology system failures, cyber-attacks and security breaches may threaten the integrity of our intellectual property and sensitive information, disrupt our business operations, and result in reputational harm and other negative consequences that could have a material adverse effect on our financial condition and results of operations.
- If we fail to maintain an effective system of internal controls, we may not be able to accurately report financial results or prevent fraud.

Risks Related to Employee and Labor Matters

- Labor disputes and difficulties retaining or hiring skilled employees could adversely affect our business, financial condition or results of operations.
- A failure to attract, retain or provide adequate succession plans for key personnel could adversely affect our operations and competitiveness.
- Failure to comply with domestic or international employment and related laws could result in penalties or costs that could have a material adverse effect on our business results.

Risks Related to Legal Proceedings and Government Regulations

- Product liability, product safety, personal injury, property damage, and recall claims and investigations may materially affect our financial condition and damage our reputation.
- We may be exposed to significant legal proceedings, investigations or changes in U.S. federal, state, local or foreign laws, regulations or policies.
- We are exposed to environmental and safety risks and are subject to a broad range of health, safety and environmental laws and regulations, which may result in substantial costs and liabilities.
- We are subject to privacy and data security/protection laws in the jurisdictions in which we operate and may be exposed to substantial costs and liabilities associated with such laws and regulations.
- Unanticipated changes in our tax provisions or exposure to additional tax liabilities could affect our future profitability.

Risks Related to Our Indebtedness

- We have significant debt obligations, and may in the future incur, additional debt obligations that could adversely affect our business and profitability and our ability to meet other obligations.
- Our failure to comply with the agreements relating to our outstanding indebtedness, including as a result of events beyond our control, could result in an event of default that could materially and adversely affect our business, financial condition, results of operations or cash flows.

Risks Related to the Separation

- We have a limited history of operating as an independent company and our historical financial information may not be a reliable indicator of our future results.
- In connection with the separation into two public companies, we and Howmet agreed to indemnify each other for certain liabilities. If we are required to pay under these indemnities to Howmet, our financial results could be negatively impacted. The Howmet indemnities may not be sufficient to hold us harmless from the full amount of liabilities for which Howmet has been allocated responsibility, and Howmet may not be able to satisfy its indemnification obligations in the future.
- Howmet may fail to perform under various transaction agreements that were executed as part of the separation.
- If the distribution, together with certain related transactions, does not qualify as a transaction that is generally tax-free for U.S. federal income tax purposes, we, as well as Howmet and Howmet's stockholders, could be subject to significant tax liabilities, and, in certain circumstances, we could be required to indemnify Howmet for material taxes and other related amounts pursuant to indemnification obligations under the tax matters agreement.

PART I

Item 1. Business.

Overview

Arconic Corporation ("Arconic" or the "Company") is a global leader in manufacturing aluminum sheet, plate, extrusions and architectural products and systems, serving primarily the ground transportation, aerospace, building and construction, industrial, and packaging end markets. We maintain a competitive position in our targeted markets through our global footprint of 20 primary manufacturing facilities, as well as various sales and service facilities, located across North America, Europe, the United Kingdom and China.

On April 1, 2020 (the “Separation Date”), ParentCo separated into two standalone, publicly-traded companies, Arconic and Howmet (the “Separation”). The spin-off company, Arconic, comprised the rolled aluminum products, aluminum extrusions, and architectural products operations of ParentCo, as well as the Latin America extrusions operations sold in April 2018 (collectively, the “Arconic Corporation Businesses”).

Sale of our Samara Facility

On November 15, 2022, we completed the sale of our Russian operations. For a discussion of the sale of our facility in Samara, Russia and the potential impact on our business, see “Part II, Item 7. "Management’s Discussion and Analysis of Financial Condition and Results of Operations—Results of Operations—Segment Information—Rolled Products,” and Note S to the Consolidated Financial Statements in Part II, Item 8. “Financial Statements and Supplementary Data.” For a discussion of legal proceedings involving our facility in Samara, Russia prior to the sale, see Note T to the Consolidated Financial Statements in Part II, Item 8. “Financial Statements and Supplementary Data” under the caption “Contingencies and Commitments—Contingencies—Litigation—Federal Antimonopoly Service Of The Russian Federation Litigation."

Description of the Business

We manage our business operations through three reportable segments: Rolled Products, Building and Construction Systems ("BCS") and Extrusions. We strive to make our portfolio of facilities operationally efficient and competitive in the industry. We are well positioned in attractive end markets, and our long-term contracts with customers enhance the strength and stability of our business and our earnings. We believe our extensive manufacturing experience and our commitment to quality and innovation are competitive advantages.

Rolled Products

Rolled products are used in the production of finished goods ranging from automotive body panels and airframes to industrial plate and brazing sheet. Sheet and plate are used extensively in the transportation industries as well as in building and construction and packaging. They are also used for industrial applications such as tooling plate for the production of plastic products.

Our Rolled Products segment produces a range of aluminum sheet and plate products for the following end markets:

Ground Transportation — provides specialty aluminum sheet and plate products, including auto body sheet, structural reinforcement, proprietary heat exchanger products like multilayer brazing sheet, trailer and cab structures, vehicle components, mold plate for wheels, and sheet for fuel tanks.

Aerospace — supplies a wide range of highly differentiated sheet and plate products that meet strict quality requirements for aerospace applications, including polished fuselage sheet, structural parts, aluminum-lithium stringers, and wing skins.

Industrial — supplies a diverse range of industrial solutions for applications that include mold and tooling plate for semiconductors; general engineering/machinery and injection molding applications; specialty finishes for appliances, cosmetic packaging and RVs; tread plate/sheet for toolbox and flooring applications; and circles for cookware.

Building and Construction — supplies a wide range of products serving both the commercial and residential end markets, including roofing, architectural composite panels, curtain walls, ventilated facades and ceiling panels, spacers, culvert pipes and gutters.

Packaging — serves the packaging end market in North America, primarily through our facility in Tennessee, and the markets in Asia and Mexico through our facility located in Bohai, China.

Competitive Conditions

Our Rolled Products segment is one of the leaders in many of the aluminum flat rolled end markets in which it participates, including the ground transportation, aerospace, industrial, building and construction, and packaging end markets. While Rolled Products participates in end markets where we believe we have a significant competitive advantage due to our extensive knowledge of our customers, advanced manufacturing capability, unique technology and/or differentiated products, in certain cases, our competitors are capable of making products similar to our products. We continuously work to maintain and enhance our competitive position through innovation: new alloys such as high-formability automotive alloys, aluminum lithium aerospace alloys, differentiated products such as our 5-layer brazing products and break-through processes such as A951™ bonding technology.

Some of our Rolled Products markets are global and some are more regionally focused. Participation in these markets by competitors varies. Additionally, there are a number of new competitors emerging, particularly in China and other developing economies. We expect that this competitive pressure will continue and increase in the future as customers seek to globalize their supply bases in order to reduce costs.

Principal Facilities

The table below sets forth our Rolled Products principal properties as of December 31, 2022.

Country	Location	Products
China	Kunshan	Sheet
	Qinhuangdao[(1)]	Sheet and Plate
Hungary	Székesfehérvár	Sheet, Plate, Slabs and Billets
United Kingdom	Birmingham	Plate
United States	Davenport, IA	Sheet and Plate
	Danville, IL	Sheet and Plate
	Hutchinson, KS	Sheet and Plate
	Lancaster, PA	Sheet and Plate
	Alcoa, TN	Sheet and Plate

(1) Leased property or partially leased property.

Building and Construction Systems

Our BCS segment manufactures differentiated products and building envelope solutions, including entrances, curtain walls, windows, composite panel and coil coated sheet. The business is focused on two product lines: architectural systems, which carry the Kawneer® brand, and architectural products, which carry the Reynobond® and Reynolux® brands. The BCS segment has competitive positions with respect to both product lines, attributable to its strong brand recognition, high quality products and strong relationships through the building and construction value chain.

As the inventor of the modern storefront more than 100 years ago, our Kawneer® branded architectural systems products include windows, doors and curtain walling. Kawneer is a premium brand, known for the breadth, depth and performance of its product portfolio and is a leading manufacturer of architectural systems in North America, with an established presence in Europe. Key customers of this market segment include fabricators and glazing subcontractors.

The Reynobond and Reynolux brands deliver innovative exterior and interior cladding and coil coated sheet solutions with end uses that include building façades, retail, sign and display, interior applications and various industrial applications. Reynobond is composite material that consists of an extruded core that is fused between two sheets of coil-coated aluminum and Reynolux is coil-coated aluminum sheet that can be sold in coil or flat-sheet form. Key customers include metal fabricators and installers.

BCS differentiates itself through its footprint in North America and Europe and by offering a broad portfolio of building envelope products that span the range of building end-use and building complexities. Architects, general contractors and fabricators consider BCS a go-to provider of products and systems with product lines that can address localized functional and building code requirements. We believe that our products and systems have a reputation for quality and reliability.

Competitive Conditions

In North America, our BCS segment primarily competes with manufacturers of products used in nonresidential construction. In Europe, it competes with manufacturers of products used in both residential and nonresidential construction. Our competitive advantage is based on strong brands, innovative products, customer intimacy and technical services.

In the architectural systems market we compete with regional competitors. The competitive landscape in the architectural systems market has been relatively stable since the mid-2000s, with the major competitors in North America and Europe remaining constant, despite some industry consolidation in North America during the late 2000s.

The primary product categories in architectural products are aluminum composite material and coil coated sheet. The market for our architectural products business is more global and primarily served by subsidiaries of larger multinational companies.

Principal Facilities

The table below sets forth our BCS principal properties as of December 31, 2022

Country	Location	Products
Canada	Lethbridge, Alberta	Architectural Systems
France	Merxheim[(1)]	Architectural Products
United Kingdom	Runcorn	Architectural Systems
United States	Springdale, AR	Architectural Systems
	Visalia, CA	Architectural Systems
	Eastman, GA	Architectural Products
	Bloomsburg, PA	Architectural Systems
	Cranberry, PA	Architectural Systems

(1) Leased property or partially leased property.

Principal facilities are listed, and do not include locations that serve as service centers or administrative offices. The service centers perform light manufacturing, such as assembly and fabrication of certain products.

Extrusions

Our Extrusions segment produces a range of extruded products, including automotive shapes (e.g., driveshafts, anti-lock brake housings, and turbo chargers), aerospace shapes (e.g., wing stringers, floor beams, fuselage, cargo), seamless tube, hollows, mortar fins, and high strength rod and bar. With process and product technologies that include large and small extrusion presses, integrated cast houses, horizontal heat treat furnaces, vertical heat treat furnaces, annealing furnaces, induction billet heating and ultrasonic inspection capabilities, our Extrusions segment serves a broad range of customers in several of our core end markets, including the following:

Ground Transportation — provides aluminum extrusions for applications that include driveshafts for the automotive market and aluminum frame rails for the commercial transportation market.

Aerospace — supplies a wide range of applications for commercial airframes.

Industrial — supplies a diverse range of industrial solutions for applications that include rods and bars for building supplies and other industrial applications.

Competitive Conditions

The Extrusions segment is a prominent supplier in many of the end markets in which it participates, including automotive (including driveshafts), aerospace, and industrial markets. Extrusions participates in end markets where we believe we have a significant competitive position due to our extensive knowledge of our customers, advanced manufacturing capability, unique technology and/or differentiated products. We continuously work to maintain and enhance our competitive position through innovation: new alloys such as aluminum lithium aerospace alloys and differentiated products.

Some of our Extrusions end markets are worldwide and some are more regionally focused. Participation in these segments by competitors varies. Additionally, there are a number of other competitors emerging, particularly in China and other developing economies. We expect that this competitive pressure will continue and increase in the future as customers seek to globalize their supply bases in order to reduce costs.

Principal Facilities

The table below sets forth our Extrusions principal properties as of December 31, 2022.

Country	Location	Products
Germany	Hannover[1]	Extrusions
United States	Lafayette, IN	Extrusions
	Massena, NY[2]	Extrusions

_____

(1) Leased property.

(2) Pursuant to certain agreements entered into in connection with the Separation, Arconic leased the Massena property from Howmet until such time as the property could be conveyed to Arconic. The property was conveyed to Arconic in the fourth quarter of 2022.

Principal facilities are listed and do not include locations that have been idled. In the second half of 2021, the Chandler, AZ facility was fully idled and its commercial operations were integrated into the Lafayette facility. The Company may temporarily restart one or more of the presses at Chandler over short periods of time to address customer demand requirements. See Item 7. “Management’s Discussion and Analysis of Financial Condition and Results of Operations—Results of Operations—Extrusions—Overview.”

End Market and Customer Revenues

We provide products to customers in a number of end markets, and the revenues from any end market may vary from year to year. For more information regarding revenues by major end markets served, see Note C to the Consolidated Financial Statements in Part II. Item 8. "Financial Statements and Supplementary Data." In addition, the demand for products in certain end markets is concentrated in a relatively small number of customers. We have multi-year contracts with many of our key customers, primarily in the aerospace and ground transportation end markets. These contracts indirectly expose us to changes in these customers' end markets, and the loss of sales under these contracts could have a material adverse effect on our business if such sales are not replaced by sales to other customers.

Customer and Distribution Channels

Rolled Products and Extrusions

Our Rolled Products and Extrusions segments have two primary sales channels: direct sales to our customers and sales to distributors.

Direct Sales

Our Rolled Products and Extrusions segments supply various customers all over the world through a direct sales force operating from individual facilities or sales offices. The direct sales channel typically serves very large, sophisticated customers and OEMs, but can service medium and small size customers as well. Long-standing relationships are maintained with leading companies in industries using aluminum rolled and extruded products. Supply contracts for large global customers generally range from one to five years in length and historically there has been a high degree of renewal business with these customers. As the manufacture of higher aluminum content vehicles continues to grow, we continue to develop long-term relationships with the automotive original equipment manufacturers (“OEMs”). In some cases, the products we supply are proprietary in nature. Further, certain industries, such as automotive and aerospace, and their related customers require suppliers to complete a rigorous qualification process. The ability to obtain and maintain these qualifications is an important part of doing business in these segments. A customer’s cost to switch and either find a new product or qualify a new supplier can be significant, so it is in both the customer’s and the supplier’s best interest to maintain these relationships.

Distributors

Our Rolled Products and Extrusions segments also sell their products through third-party distributors. Customers of distributors are typically widely dispersed, and sales through this channel are usually highly fragmented. Distributors sell mostly commodity or less specialized products into many end-use segments in smaller quantities.

Building and Construction Systems

Our BCS segment supplies architectural facade systems and products principally in North America and Europe but also globally through both direct sales and distributors. Its typical customers are installers or fabricators who purchase product on a project-by-project basis. Long-standing relationships are maintained with its leading customers. BCS also maintains an e-commerce platform for numerous standard architectural products for use by its North American customers and offers standard architectural products for purchase in its service centers.

Sources and Availability of Raw Materials

Important raw materials used by Arconic are: primary aluminum for remelting (sows, t-bars, and ingots, including high purity and off-grade), aluminum alloyed and unalloyed casthouse products (including rolling slab and billet), aluminum scrap, alloying materials (including, but not limited to, magnesium, copper, and zinc), aluminum coil, electricity, natural gas, coatings, lube oil, packaging materials, and resin. Generally, other materials are purchased from third-party suppliers under competitively priced supply contracts or bidding arrangements. We believe that the raw materials necessary to our businesses are and will continue to be available generally, although we experience and anticipate continued cost, availability and other impacts to our supply chain arising from, among other factors, the COVID-19 pandemic, geopolitical instability in areas where we or our suppliers operate, challenges with transportation and logistics, and inflationary price pressures.

Intellectual Property

We believe that our domestic and international intellectual property assets provide us with a significant competitive advantage, as we continue to strive to improve our products and processes. Our rights under our patents, as well as the products made and sold under them, are important to us as a whole, and to varying degrees, important to each business segment. The patents owned by us generally concern metal alloys, particular products, and manufacturing equipment or techniques. Our business as a whole is not, however, materially dependent on any single patent, trademark or other intellectual property. As a result of product development and technological advancement, we continue to pursue patent protection in select jurisdictions throughout the world. As of December 31, 2022, our worldwide patent portfolio consists of approximately 500 granted patents and 225 pending patent applications. Patent terms extend for varying periods based on the filing date or the grant date in the various countries where we have or have applied for patent protection. The actual protection afforded by a patent varies from country to country, with the most significant variations relating to the scope of patent protection and the legal remedies available.

With respect to domestic and foreign trademarks, we have many that have significant recognition within the markets that are served. Examples include the name “Arconic” and the Arconic symbol for aluminum products, Kawneer for building panels, and Reynobond and Reynolux for architectural products. As of December 31, 2022, our worldwide trademark portfolio consists of approximately 1,060 registered trademarks and 60 pending trademark applications. Our rights under our trademarks are important to us as a whole and, to varying degrees, important to each business segment. Trademark protection continues in some countries for as long as the mark is used and in others for as long as it remains registered. Registrations generally are periodically renewable for additional fixed terms.

Research and Development

We engage in research and development programs that include process and product development, and basic and applied research. The Arconic Technology Center, located in New Kensington, Pennsylvania, serves as the headquarters for our research and development efforts, with additional research and development facilities in Norcross, Georgia; Lancaster, Pennsylvania; Runcorn, United Kingdom; Vendargues, France; and Harderwijk, Netherlands. These facilities focus on innovation and have given us a leading position in the development of proprietary next-generation specialty alloys, products and manufacturing processes as evidenced by our robust intellectual property portfolio.

Government Regulations

As a global company, our business is subject to government regulations in various jurisdictions in which we operate. We believe we are materially compliant with all applicable government laws and regulations, and maintaining compliance, other than as noted below, is not expected to materially affect our capital expenditures, earnings or competitive position. Any new or amended laws or regulations that impose significant operational restrictions and compliance requirements may negatively

impact our business, capital expenditures, earnings, and competitive position. Additional information is included in Part I. Item 1A. "RiskFactors."

Our operations are subject to national, federal, state and local environmental, health and safety laws and regulations, including those regulating the discharge of materials into the environment, greenhouse gas emissions, hazardous materials in products, chemical usage and workplace safety requirements and protocols. We maintain a global environmental, health and safety program that includes policies and procedures; dedicated environmental, health and safety personnel; procedures for handling of waste materials and hazardous materials; monitoring, reporting and remediating environmental, health, and safety issues; compliance auditing; and training, among other measures.

Approved capital expenditures for new or existing facilities for environmental control are approximately $8 million for 2023. Additional information relating to environmental matters and other matters is included in Note T to the Consolidated Financial Statements included in Part II. Item 8. "Financial Statements and Supplementary Data" under the caption “Contingencies and Commitments — Contingencies - Environmental Matters.”

Other than the foregoing, we do not anticipate any material capital expenditures during fiscal 2023 related to compliance with environmental or other government regulations.

Sustainability and Environmental, Social and Governance (“ESG”)

Arconic is committed to being a good corporate citizen and living our core values by working with our stakeholders to create a better, more sustainable future. Arconic has a cross-functional ESG Council tasked with developing and implementing our sustainability strategy. The ESG Council regularly reports to executive leadership and to the Governance and Nominating Committee of the Board of Directors, which has general oversight responsibility for ESG matters, working in conjunction with other committees of the Board of Directors to which oversight of specific matters have been delegated. Functions responsible for development and implementation of our sustainability commitments, strategies and initiatives directly report to the Chief Executive Officer. In 2022, Arconic engaged a third party to advise in the development of long-term sustainability goals based on findings from the Materiality Assessment conducted in 2021. The Materiality Assessment included stakeholder engagement on a wide range of issues, including managing the impacts of our business on the climate and environment, ensuring supply chain sustainability, and fostering positive human capital dynamics. We expect continued engagement to enhance our understanding of stakeholder views, which help guide our ESG efforts. The use of the terms “material” or “materiality” in this section and in our 2022 ESG disclosures is not related to or intended to convey matters or facts that could be deemed “material” to a reasonable investor within the meaning of U.S. securities laws or similar requirements of other jurisdictions.

Human Capital Resources

At Arconic, we are committed to living our core values by protecting and improving the health and safety of our employees, our communities and the environment. We strive to cultivate an inclusive and diverse culture that advocates for equity, acts with integrity, and upholds high standards for human rights. We strengthen our global workforce by providing learning opportunities, employee engagement programs and talent development efforts that drive innovation, agility, people development and collaboration.

Our worldwide employment at the end of 2022 was approximately 11,550 employees located in 17 countries. The breakdown of our employees by region is as follows:

Region	Percentage of Total Population
Americas	74%
Europe, Middle East and Africa	18%
Asia	8%

Turnover

During 2022, the Company increased hiring initiatives to support the business’s organic growth strategy, including participation in various local and national recruiting events, encouraging and responding to employee feedback through its engagement survey, and leveraging social media campaigns as a vehicle to reach a broader base of existing and potential employees. As a result of these and other targeted efforts, in 2022 the Company’s employee base increased by approximately 590 employees. This net increase was offset by a reduction of approximately 2,900 employees resulting from the sale of our facility in Samara, Russia in November 2022.

Collective Bargaining Representation

We believe in freedom of association. We respect an individual's choice to be represented by – or not be represented by – a union or other legally authorized associations or organizations in accordance with the laws of the countries in which we operate. Many of our employees are represented by labor unions. We believe that relations with any applicable union representatives are generally constructive and professional. In the U.S., approximately 4,100 employees are represented by various labor unions. The largest collective bargaining agreement is the master collective bargaining agreement between us and the United Steelworkers ("USW"). The USW master agreement covers approximately 3,500 employees at four U.S. locations. The prior master collective bargaining agreement with the USW expired on May 15, 2022, and we entered into a new labor agreement that expires on May 15, 2026. There are seven other collective bargaining agreements in the U.S. with varying expiration dates. On a local basis, there are agreements between Arconic and unions with varying expiration dates that cover employees in Europe, North America, and Asia.

Governance

Oversight responsibility for the Company's ESG policies and strategies has been delegated by our Board of Directors to our Governance and Nominating Committee, which works in conjunction with other committees of the Board to which oversight of specific matters have been delegated. For example, the responsibilities of our Compensation and Benefits Committee of the Board of Directors include the oversight of talent management strategies, such as the Company's recruitment, development, employee engagement, promotion and retention programs; policies and practices promoting diversity, equity and inclusion ("DEI") within the Company; key metrics and objectives related to the Company's talent; and the succession plan for our executive officers. Our Board of Directors is also committed to our talent management and has oversight responsibility for our safety practices.

Diversity, Equity and Inclusion

We are committed to maintaining an inclusive environment where everyone feels valued, and we celebrate both the differences and similarities among our people. We also believe that diversity in all areas, including cultural background, experience and thought, is essential in making our Company stronger. Consistent with that belief, we conducted focused recruiting and retention initiatives that further demonstrate our ongoing commitment to DEI. The breakdown of our female and U.S. ethnic minority employees globally at the end of 2022, as members of management and as executives, is as follows:

	Total %	Management[3] %	Executive[4] %
Women[1]	17.7%	25.8%	30.0%
Minorities[2]	25.6%	13.1%	20.0%

(1) Percentages are on a global basis and include locations where Arconic has valid data. Some regions and countries, such as Germany, have privacy laws and regulations that may prevent Arconic from reporting on certain employee demographics and those regions or countries are not included in the global percentages of female employees.

(2) Percentages are on a U.S. basis only.
(3) Represents members of management, other than executives.
(4) Represents our leaders who serve in a Vice President or higher role.

Our Diversity, Equity, and Inclusion Council, chaired by our Chief Executive Officer, continues to advance our mission in partnership with our six employee resource groups ("ERGs") with senior executives volunteering as sponsors for each group. Our six ERGs – Arconic African Heritage Network, Arconic Hispanic Network, Arconic Next Generation Network, Arconic Veterans Network, Thrive Network (Women) and Spectrum (LGBTQ+) – reflect an inclusive, respectful and values-based company culture. All of our employees are encouraged to participate in these grassroots, employee-led organizations that:

- Drive employee engagement through community outreach;
- Provide learning and development opportunities for employees;
- Help position Arconic as a global employer of choice through strategic recruiting activities;
- Inform Company policies around diversity and inclusion; and
- Reinforce our brand through key external endorsements such as the Human Rights Campaign and the United Nations Women's Empowerment Principles.

With the momentum gained from our Grow Together campaigns in 2020 and 2021, we broadened our Grow Together scope in 2022 to integrate equity into our mission, demonstrating our commitment to create a work environment with equal opportunities for all. Grow Together continues to provide momentum to enable our DEI focus and encourages employees to take actions related to inclusion, diversity or social justice through learning activities and volunteering. Employees based in the U.S. also have the opportunity to donate to various non-profit organizations with a social equity mission aligned with and recommended by our ERGs and receive a matching contribution from the Arconic Foundation, an independently endowed

foundation. As a result, in 2022 the Arconic Foundation granted over $34,000 in matching contributions to organizations that are driving change with DEI initiatives. In addition, the Arconic Foundation granted over $960,000 to non-profits that support inclusivity and social equity in communities where Arconic operates.

Talent Development

We are committed to attracting, retaining, and developing talent that shares our values and demonstrates agility, results focus, and people development capability. All employees have access to a robust, ongoing career planning and performance management process that begins at recruitment and continues throughout their careers. Our twice-yearly succession-planning and calibration processes help us identify successors to business-critical roles while developing our high-potential employees. In addition, we offer management skills training to all supervisory employees, leadership development and coaching programs for mid- and senior-level leaders, and a variety of technical, business, and soft skills training programs to help our business units and resource units meet their strategic objectives.

Compensation and Benefits Programs

Our compensation and benefits programs are designed to provide comprehensive total rewards to attract, motivate, engage and retain a talented workforce. Our programs are competitive with our peers and emphasize performance-based compensation to align employee rewards with company performance. Benefits are a key component of our total rewards package. We offer a holistic benefits package designed to provide greater security for our employees and their families, which depending on country may include healthcare, life insurance, paid parental leave, disability, savings and retirement and various welfare benefit programs through plans we sponsor or through social programs in the countries where we operate.

Health and Safety

We value human life above all else and are committed to operating worldwide in a safe, responsible manner which respects the environment and the health of our employees, our customers and the communities where we operate. Our strong health and safety culture empowers our employees and contractors to take personal responsibility for their actions and the safety of their coworkers. Our focus on safety also includes an ongoing commitment to maintaining a secure work environment that respects the dignity and worth of every employee, which drives our continuous improvement approach in our robust safety programs. Our employees play an important role in actively supporting a workplace that strives to be free of violence, threats, intimidation, and harassment. This culture is supported by internal policies, standards, rules and procedures that clearly articulate our stringent expectations for working safely and maintaining a secure work environment in all of our facilities worldwide.

In response to the COVID-19 pandemic, we implemented significant changes that we determined were in the best interest of our employees, as well as the communities in which we operate. We established strict requirements for onsite work that is continually assessed against the Centers for Disease Control and other expert guidance to assure that we are providing a safe work environment. We continue to adapt our requirements in response to the continuing impacts of the COVID-19 pandemic, including new and developing variants, in the various regions in which we operate.

We are committed to achieving zero fatalities and keeping a major focus on fatality prevention. Each location maintained an active fatality prevention team and had zero employee and contractor fatalities in 2022. We have prioritized our risk management processes toward fatality and serious injury potential to focus on the most impactful hazards that have the potential for life-altering outcomes. In addition, during 2022, all of our key safety rates remained significantly below the most recent U.S. industry averages. At 0.48, our 2022 days away, restricted and transfer (“DART”) was less than 1 percent lower than our adjusted 2021 DART of 0.482. Our 2022 total recordable incident rate (“TRR”) of 1.39 (recorded in accordance with U.S. Occupational Safety and Health Administration (“OSHA”) record keeping requirements) was 35% higher than our adjusted 2021 TRR of 1.03. Our previously reported 2021 DART was 0.42 and 2021 TRR was 0.88. Both rates for 2021 were adjusted above to exclude safety rates for our facility in Samara, Russia that was sold in November 2022. At the time of the sale of our Samara facility in November 2022, our 2022 DART (including our Samara facility) was 0.41 versus our unadjusted 2021 DART of 0.42 and our 2022 TRR (including our Samara facility) was 1.11 versus our unadjusted 2021 TRR of 0.88.

Information about our Executive Officers

The following table sets forth information, as of February 17, 2023, regarding the individuals who are executive officers of Arconic.

Name	Age	Position
Timothy D. Myers	57	Chief Executive Officer
Erick R. Asmussen	56	Executive Vice President and Chief Financial Officer
Daniel G. Fayock	48	Executive Vice President, Chief Legal Officer and Secretary
Melissa M. Miller	51	Executive Vice President and Chief Human Resources Officer
Diana B. Perreiah	58	Executive Vice President, Rolled Products North America
Mark J. Vrablec	62	Executive Vice President and Chief Commercial Officer
Robert L. Woodall	56	Executive Vice President, Rolled Products International and Extrusions

Timothy D. Myers has served as President of Arconic Corporation since February 2020 and as Chief Executive Officer since April 2020. From October 2017 to April 2020, Mr. Myers served as Executive Vice President and Group President, Global Rolled Products, which was restructured in contemplation of the Separation to include the Extrusions and Building and Construction Systems businesses of ParentCo. From May 2016 to June 2019, he served as Executive Vice President and Group President of ParentCo's Transportation and Construction Solutions segment, which then comprised Arconic Wheel and Transportation Products and Building and Construction Systems and which segment was eliminated in the third quarter of 2019, with the Building and Construction Systems business then moved to the Global Rolled Products segment. Prior to that assignment, he was President of Alcoa Wheel and Transportation Products, from June 2009 to May 2016. Mr. Myers was Vice President and General Manager, Commercial Vehicle Wheels for the Alcoa Wheel Products business from January 2006 to June 2009. Mr. Myers joined Alcoa Inc. in 1991 as an automotive applications engineer in the Commercial Rolled Products Division, and held a series of engineering, marketing, sales and management positions with Alcoa Inc. and ParentCo since that time.

Erick R. Asmussen has served as Executive Vice President and Chief Financial Officer of Arconic Corporation since February 2020. Mr. Asmussen previously served as Senior Vice President and Chief Financial Officer of Momentive Performance Materials Inc. from May 2015 to July 2019. Prior to joining Momentive, Mr. Asmussen served as Vice President and Chief Financial Officer of GrafTech International, Ltd. from September 2013 to May 2015. Mr. Asmussen joined GrafTech in 1999 and served in multiple leadership roles, including Vice President of Strategy, Planning and Corporate Development, Worldwide Controller, Tax Director, and Treasurer. Prior to GrafTech, Mr. Asmussen worked in various financial positions with Corning Incorporated, AT&T Corporation, and Arthur Andersen LLP.

Daniel G. Fayock has served as Executive Vice President, Chief Legal Officer and Corporate Secretary of Arconic Corporation since April 2022. Mr. Fayock previously served as Assistant General Counsel and Secretary of PPG Industries, Inc. from September 2018 to March 2022, where he was responsible for advising the board of directors and executive team and serving as counsel to various businesses on legal and strategic issues. Prior to this position, Mr. Fayock served as Corporate Counsel from January 2015 to September 2018 and previously in a number of senior legal positions over his 15 years with PPG Industries with a broad spectrum of progressive legal responsibilities. Prior to joining PPG Industries, Mr. Fayock was a senior attorney with Allegheny Energy after starting his legal career as a corporate associate with K&L Gates.

Melissa M. Miller has served as Executive Vice President and Chief Human Resources Officer of Arconic Corporation since April 2020. From October 2017 until April 2020, Ms. Miller served as Vice President of Human Resources for ParentCo's Global Rolled Products business, which was restructured in contemplation of the Separation to include the Extrusions and Building and Construction Systems (BCS) businesses of ParentCo. From May 2016 until October 2017, Ms. Miller served as Vice President of Human Resources for the business segment that comprised the BCS business and Arconic Wheel & Transportation Products. From June 2011 until February 2016, Ms. Miller served as Global Human Resources Director of the BCS business. Ms. Miller joined Alcoa Inc. in 2005 and held multiple leadership roles at Alcoa Inc. and ParentCo with a broad spectrum of progressive HR responsibilities, including HR strategy and delivery, talent management, workforce planning, succession planning, employee engagement, campus partnerships, HR technology, growth in emerging markets, merger integrations, turnarounds and employee/labor relations. Prior to joining Alcoa Inc., Ms. Miller served in several HR-related roles at Marconi Communications (formerly known as FORE Systems) for more than seven years.

Diana B. Perreiah has served as Executive Vice President, Rolled Products North America of Arconic Corporation since August 2022. Ms. Perreiah served as President, Building and Construction Systems at ParentCo from May 2015 to March 2020

and continued in that role at Arconic Corporation from the Separation Date to August 2022. Ms. Perreiah joined Alcoa Inc. in 1986 as a systems analyst and subsequently held a variety of positions with Alcoa Inc. and ParentCo, including Operations Manager, North American Extrusions, and several positions in corporate, including Global Information Services, and in the internal manufacturing excellence group. She joined the Building and Construction Systems group in 2009 as Vice President, Business Operations, Kawneer North America. Since then, she has held a series of leadership positions within Building and Construction Systems, including Vice President & General Manager, Kawneer North America.

Mark J. Vrablec has served as Executive Vice President and Chief Commercial Officer of Arconic Corporation since April 2020. From February 2019 to April 2020, Mr. Vrablec served as Vice President for ParentCo's Global Rolled Products business, which was restructured in contemplation of the Separation to include the Extrusions and Building and Construction Systems businesses. From July 2017 to February 2019, Mr. Vrablec served as Vice President, Global Rolled Products Commercial and Business Development. From November 2016 to July 2017, Mr. Vrablec served as President of the Aerospace and Automotive Products business, holding the same role for Alcoa Inc. from October 2015 until November 2016. From September 2011 until October 2015, Mr. Vrablec served as President of Alcoa Inc.'s Aerospace, Transportation and Industrial Rolled Products business. Mr. Vrablec joined Alcoa Inc. in 1982 as a metallurgist, and has held a series of quality assurance, operations, and management positions with Alcoa Inc. and ParentCo since that time.

Robert L. Woodall has served as Executive Vice President, Rolled Products International and Extrusions of Arconic since August 2022. Mr. Woodall served as President, Global Rolled Products Europe and Asia from April 2020 to August 2022, and assumed responsibility for Extrusions in June 2022. From July 2018 to April 2020, he served as Director of Global Plant Operations for Global Rolled Products at ParentCo, and prior to July 2018, he served as Vice President of Operations at ParentCo. Mr. Woodall joined Alcoa in 1989 as a mechanical engineer and subsequently held a variety of engineering, maintenance, production, and management positions with Alcoa Inc. and ParentCo, including Plant Manager, Pt. Henry, Australia; Director of Australian Rolled Products; and Manufacturing Director for Arconic Davenport Works and Satellite locations.

Available Information

The Company's Internet address is www.arconic.com. Arconic makes available free of charge on or through its website its Annual Report on Form 10-K, Quarterly Reports on Form 10-Q, Current Reports on Form 8-K, and amendments to those reports filed or furnished pursuant to Section 13(a) or 15(d) of the Securities Exchange Act of 1934, as amended ("Exchange Act"), as soon as reasonably practicable after the Company electronically files such material with, or furnishes it to, the Securities and Exchange Commission ("SEC"). The information on the Company's website is not a part of, or incorporated by reference in, this Annual Report on Form 10-K.

Item 1A. Risk Factors.

Our business, financial condition and results of operations may be impacted by a number of factors. In addition to the factors discussed elsewhere in this report, the following risks and uncertainties could materially harm our business, financial condition, or results of operations, including causing our actual results to differ materially from those projected in any forward-looking statements. The following list of significant risk factors is not all-inclusive or necessarily in order of importance. Additional risks and uncertainties not presently known to us or that we currently deem immaterial also may materially adversely affect us in future periods

Business Risks – Global Conditions

The markets for our products are cyclical and are influenced by a number of factors, including global economic conditions, that could have a material adverse effect on our business, financial condition or results of operations.

We are subject to cyclical fluctuations in global economic conditions and lightweight metals end-use markets. Many of our products are sold to industries that are cyclical, such as the aerospace, automotive, commercial transportation and building and construction industries, and the demand for our products are sensitive to, and quickly impacted by, demand for the finished goods manufactured by our customers in these industries, which may change as a result of changes in regional or worldwide economies, currency exchange rates, energy prices or other factors beyond our control.

In particular, we derive a significant portion of our revenue from products sold to the aerospace industry, which can be cyclical and reflective of changes in the general economy. The commercial aerospace industry is historically driven by the demand from commercial airlines for new aircraft. The U.S. and international commercial aviation industries may face challenges arising from competitive pressures and fuel costs. Demand for commercial aircraft is influenced by airline industry profitability, trends in airline passenger traffic, the state of U.S., regional and world economies, the ability of aircraft purchasers to obtain required financing and numerous other factors, including the effects of terrorism, health and safety concerns,

environmental constraints imposed upon aircraft operators, the retirement of older aircraft, the performance and cost of alternative materials, and technological improvements to aircraft.

Further, the demand for our ground transportation products is driven by the number of vehicles produced by automotive and commercial transportation manufacturers and volume of aluminum content per vehicle. The automotive industry is sensitive to general economic conditions, including credit markets and interest rates, and consumer spending and preferences regarding vehicle ownership and usage, vehicle size, configuration and features. Automotive and commercial transportation sales and production can also be affected by other factors, including supply chain disruptions, the age of the vehicle fleet and related scrap rates, labor relations issues, fuel prices, regulatory requirements, government initiatives, trade agreements, interest rates, health and safety concerns and levels of competition both within and outside of the aluminum industry.

Our products are used in a variety of industrial applications, including mold and tooling plate for semiconductors; general engineering/machinery and injection molding applications; specialty finishes for appliances, cosmetic packaging, and vehicle components; tread plate and sheet; and building and construction products. Common alloy sheet, which is a significant portion of the total industrial products market, is particularly sensitive to the volume of imports of common alloys into the U.S. The implementation of anti-dumping and countervailing duties imposed on Chinese common alloy sheet during 2018 has led to a significant decrease in the volume of imports from China. That decrease was followed by a significant increase in imports of common alloy into the U.S. from other countries, which led to softening prices. In 2021, the U.S. government imposed new anti-dumping and countervailing duties on imports of common alloy aluminum sheet from 16 additional countries. The anti-dumping and countervailing duties have led to improved pricing in the U.S. for common alloy sheet, though the long-term impact of the tariffs on import levels and pricing, as well as the likelihood of extension of such duties, remains difficult to predict.

We are unable to predict the future course of industry variables, the strength of the U.S., regional or global economies, or the effects of government actions. Negative economic conditions, such as a major economic downturn, a prolonged recovery period, or disruptions in the financial markets, could have a material adverse effect on our business, financial condition or results of operations.

Our business, results of operations, financial condition, liquidity and cash flows have been, and in the future could be, materially adversely affected by the effects of widespread public health epidemics/pandemics.

Outbreaks of contagious diseases, public health epidemics or pandemics (including the COVID-19 pandemic, which resulted in travel restrictions, governmental restrictions on certain activities, and shutdown of certain businesses globally, Sudden Acute Respiratory Syndrome, Avian Influenza, H1N1 virus, or the Ebola virus) or other adverse public health developments in countries where we, our employees, customers and suppliers operate could have a material and adverse effect on our business, results of operations, financial condition, liquidity and/or cash flows. Any such epidemics or pandemics could experience resurgences in cases, communicability or severity as a result of the development of different variants, as with the COVID-19 pandemic, which could extend the magnitude or duration of the adverse impact on our operations. The extent to which any such outbreak affects our operations over time is highly uncertain and beyond our control, and is dependent on a variety of factors, including the duration and severity of the initial outbreak or subsequent variants, the imposition of governmental quarantine or other public health measures, the availability of vaccines or other medical remedies and preventive measures, and determinations regarding, among other things, health and safety, demand for specific products, and broader economic conditions. Many of the actions that may be taken to mitigate the impact of an epidemic or pandemic, including declarations of states of emergency, governmental quarantines, shelter-in-place and stay-at-home orders, social distancing requirements, business closures and staged procedures for reopening, manufacturing restrictions and a prolonged period of travel, commercial and/or other similar restrictions and limitations, are highly likely to impact our business and the business of many of our customers, and therefore are likely to magnify the risks of a material adverse impact on our business, results of operations, financial condition, liquidity and/or cash flows, as well as on our business strategies and initiatives. In addition, the impact of any epidemic or pandemic, and the related restrictions, may differ in the areas in which our products are manufactured, distributed or sold, or may change on short notice in response to new variants or other circumstances and, accordingly, any such impact on our operations or the operations of our customers and suppliers is difficult to predict. Because we rely on supply chain continuity, restrictions in one location may materially impact operations in multiple locations, and the impact of an epidemic or pandemic in one location may have a disproportionate effect on our operations in the future.

An epidemic or pandemic subjects our operations, financial performance and financial condition to a number of risks, including, but not limited to modification of business practices, including idling or production decreases at our facilities and workforce reductions; actions we may take associated with the safety and welfare of our employees, including increased costs associated with recruiting, hiring, training and supervising new employees or employees required to perform new roles, maintaining high levels of employee awareness of and compliance with our internal procedures and regulatory requirements, and implementing employee health and safety initiatives; lower demand for our products should our customers experience labor shortages, supply chain issues or other operational impacts; concessions or contract modifications that we may grant to our

customers; supply chain disruption, including labor shortages, unavailability of or price volatility for raw materials or energy, and transportation and logistics challenges; volatility in financial and commodities markets adversely impacting asset valuations, including pension assets; and the impact on our customers' businesses related to the financial condition of or other restrictions on the end users of their products or services, including air travel, construction and the trends in the purchases of automobiles, industrial products and other manufactured goods. In addition, we may experience decreased availability of liquidity under our asset-based lending credit facility (the "ABL Credit Facility" and the credit agreement forming a part of the ABL Credit Facility, the "ABL Credit Agreement"), which is based on a borrowing base calculation based on our financial results and cash from operations, or credit rating downgrades that could adversely affect our cost of funding and related margins, liquidity, competitive position, access to capital markets, and ability to refinance indebtedness on favorable terms. The magnitude of the adverse effects of these and other risks on our business, results of operations, financial condition, liquidity and cash flows will vary depending on the duration and severity of an epidemic or pandemic and the responses of governmental authorities, suppliers, customers and Arconic. A sustained impact to our operations and financial results may require material impairments of our assets including, but not limited to, inventory, goodwill, intangible assets, long-lived assets, right-of-use assets, and deferred income tax assets.

An epidemic or pandemic may also exacerbate other risks disclosed in this Annual Report on Form 10-K, including, but not limited to, risks related to global economic conditions and inflation, competition, loss of customers, costs of supplies, manufacturing difficulties and disruptions, our credit profile, our credit ratings and interest rates. In addition, a future epidemic or pandemic may also affect our operating and financial results in a manner that is not presently known to us, or present significant risks to our business, results of operations, financial condition, liquidity and/or cash flows that are different from the risks presented by prior epidemics or pandemics.

Climate change, and evolving customer and stakeholder expectations, legal, regulatory and policy requirements, and market dynamics driven by climate change, could adversely affect our business, financial condition or results of operations.

There are inherent climate-related risks in various regions where we conduct business. Global climate change is resulting, and is expected to continue to result, in certain natural disasters and adverse weather conditions, such as drought, wildfires, storms, tornados, hurricanes, blizzards, changes in sea-levels, flooding, and extreme temperatures, occurring more frequently or with greater intensity and unpredictability. Such conditions could result in disruptions to any facility or surrounding community directly impacted by a climate-related event, including physical damage resulting in shutdowns and requiring repair or our employees' unavailability to work, and could also adversely impact our suppliers, customers, and shipping and transportation networks. These disruptions could make it more difficult and costly for us to produce and deliver our products, obtain raw materials or other supplies, maintain our critical corporate functions, and could reduce customer demand for our products.

In addition, customers, communities, investors and other stakeholders are increasingly focusing on environmental issues, including climate change and the carbon footprint of businesses throughout our supply chain. Changing customer preferences may result in increased demands regarding, among other matters, the source of aluminum, alloying metals and other materials used in our products, demand for increased use of recycled materials in our products, the manner in which power we consume is generated, our use and treatment of water and other natural resources, and the packing materials and shipping methods we use to deliver our products. In order to respond to these demands, we may need to make changes to our facilities, operations or production methods, or increase research and development efforts, any of which are likely to result in significant additional costs. Additional costs, or diminished customer demand for our products, loss of market share, or reputational damage resulting from our failure to satisfy customer preferences or to meet evolving investor, stakeholder or industry expectations, could have a material adverse effect on our business, operating results and financial condition.

Additionally, concerns over climate change have resulted in ongoing public pressure to address, and to adopt legal and regulatory requirements designed to address, climate change, including regulating greenhouse gas emissions (and the establishment of enhanced internal processes or systems to track emissions), policies mandating or promoting the use of renewable or zero-carbon energy and sustainability initiatives, and additional taxes on or other costs related to fuel and energy. For example, in January 2021, the U.S. recommitted to the Paris Agreement and in April 2021, President Biden announced targets to reduce U.S. greenhouse gas emissions. If newly enacted laws or regulations, or newly adopted policies or initiatives, are more stringent than current requirements, we, our suppliers and our customers may be subject to increased compliance burdens, incur significant additional costs, or experience disruption in the sourcing of materials and the manufacturing and distribution of products, any of which could have a material adverse effect on our business, financial condition or results of operations.

Governments of countries in which we operate could nationalize or expropriate private enterprises, or otherwise change their policies regarding private enterprise, which could adversely impact the value of our operations in those countries.

Certain countries in which we operate are subject to political, social, diplomatic and economic uncertainty. The governments of these countries may develop or implement political, social or economic policies contrary to, or reversing current policies encouraging, private enterprise, economic decentralization and/or outside investment in such countries. Changes in policies, the enactment of new laws or regulations, or changes in the interpretation of current policies, laws or

regulations could result in, among other impacts, the imposition of confiscatory taxation, trade sanctions or embargoes, our inability to protect our intellectual property rights, renegotiation or nullification of existing agreements or property rights, restrictions on currency conversion, restrictions or prohibitions on the payment of dividends or distributions, or the nationalization or expropriation of private enterprises, including our operations. Any of these occurrences could have a material adverse effect on our business, financial condition or results of operations, and nationalization or expropriation could result in the loss of all revenues generated in, and the entire value or our investment in, such countries.

We are exposed to economic factors, including inflation, fluctuations in aluminum prices, foreign currency exchange rates and interest rates, and currency controls in the countries in which we operate.

We have experienced, and continue to experience, inflationary pressures on the prices of aluminum, materials, transportation, energy and labor. In an inflationary environment, such as the current economic environment, our ability to implement customer pricing adjustments or surcharges to pass-through or offset the impacts of inflation may be limited. Continued inflationary pressures could reduce our profit margins and profitability.

Other economic factors, including fluctuations in foreign currency exchange rates and interest rates, competitive factors in the countries in which we operate, and continued volatility or deterioration in the global economic and financial environment could affect our revenues, expenses and results of operations. Changes in the valuation of the U.S. dollar against other currencies, including the Euro, British pound, Canadian dollar, and Chinese yuan (renminbi), may affect our profitability as some important inputs are purchased in other currencies, while our products are generally sold in U.S. dollars.

Our ABL Credit Facility bears interest at rates equal to the Secured Overnight Financing Rate (“SOFR”) plus a credit spread adjustment, plus a margin. Accordingly, we will be subject to risk from changes in interest rates on the variable component of the rate.

We also face risks arising from the imposition of cash repatriation restrictions and exchange controls. Cash repatriation restrictions and exchange controls may limit our ability to convert foreign currencies into U.S. dollars or to remit dividends and other payments by our foreign subsidiaries or businesses located in or conducted within a country imposing restrictions or controls. While we currently have no need, and do not intend, to repatriate or convert cash held in countries that have significant restrictions or controls in place, should we need to do so to fund our operations, we may be unable to repatriate or convert such cash, or be unable to do so without incurring substantial costs. We currently have substantial operations in countries that have, or that may in the future impose, cash repatriation restrictions or exchange controls in place, including China, and, if we were to need to repatriate or convert such cash, these controls and restrictions may have an adverse effect on our operating results and financial condition.

Our global operations expose us to risks that could adversely affect our business, financial condition, results of operations, cash flows or the market price of our securities.

We have operations or activities in numerous countries and regions outside the U.S., including Europe, the United Kingdom, Canada, and China. As a result, our global operations are affected by economic, political and other conditions in the foreign countries in which we do business as well as U.S. laws regulating international trade, including:

- economic and commercial instability risks, including those caused by sovereign and private debt default, corruption, and changes in local government laws, regulations and policies, such as those related to tariffs, sanctions and trade barriers (including tariffs imposed by the U.S. as well as retaliatory tariffs imposed by the European Union or other foreign entities), import or export restrictions, taxation, environmental regulations, production curtailments, exchange controls, employment regulations and repatriation of assets or earnings;
- the ongoing uncertainty regarding the United Kingdom’s withdrawal from the European Union (known as “Brexit”) and its impact and long-term effects on trade, imports and exports, tariffs and currencies, and our relationships with customers and suppliers;
- geopolitical risks such as political instability, civil unrest, expropriation, nationalization of properties by a government, imposition of sanctions, and renegotiation or nullification of existing agreements;
- the potential for increased tensions between the U.S. and countries in which we operate, including China and European Union nations;
- the global impact of the ongoing conflict between Russia and Ukraine, including tariffs, economic sanctions and import-export restrictions imposed by either nation, and retaliatory actions by the other nation;
- war or terrorist activities;
- kidnapping of personnel;

- major public health issues such as an outbreak of a pandemic or epidemic, such as the COVID-19 pandemic, which could cause disruptions in our operations, workforce, supply chain and/or customer demand;
- shipping, freight and supply chain disruptions;
- difficulties enforcing contractual rights and intellectual property, including a lack of remedies for misappropriation, in certain jurisdictions;
- changes in trade and tax laws that may impact our operations and financial condition and/or result in our customers being subjected to increased taxes, duties and tariffs and reduce their willingness to use our services in countries in which we are currently manufacturing products we supply to them;
- rising labor costs;
- labor unrest, including strikes;
- compliance with antitrust and competition regulations;
- compliance with foreign labor laws, which generally provide for increased notice, severance and consultation requirements compared to U.S. laws;
- aggressive, selective or lax enforcement of laws and regulations by national governmental authorities;
- compliance with the Foreign Corrupt Practices Act and other anti-bribery and corruption laws;
- compliance with U.S. laws concerning trade, including the International Traffic in Arms Regulations, the Export Administration Regulations, and the sanctions, regulations and embargoes administered by the U.S. Department of Treasury's Office of Foreign Assets Control;
- imposition of currency controls;
- compliance with data privacy regulations; and
- adverse tax laws and audit rulings.

Although the effect of any of the foregoing factors is difficult to predict, any one or more of them could adversely affect our business, financial condition, or results of operations. Our international operations subject us to complex and dynamic laws and regulations that, in some cases, could result in conflict or inconsistency between applicable laws and/or legal obligations. While we believe we have adopted appropriate risk management, compliance programs and insurance arrangements to mitigate the associated risks, such measures may provide inadequate protection against costs, penalties, liabilities or other potential risks such as loss of export privileges or repatriation of assets that may arise from such events.

An adverse decline in the liability discount rate, lower-than-expected investment return on pension assets and other factors could affect our results of operations or amount of pension funding contributions in future periods.

We provide defined benefit pension and retiree healthcare benefits to eligible employees and retirees. Our results of operations may be negatively affected by the amount of expense we record for our pension and other postretirement benefit plans, reductions in the fair value of plan assets, significant changes in market interest rates, investment losses or lower than expected returns on plan assets, and other factors. We calculate income or expense for our plans using actuarial valuations in accordance with accounting principles generally accepted in the United States of America ("GAAP").

These valuations reflect assumptions about financial market and other economic conditions, which may change based on changes in key economic indicators. The most significant year-end assumptions used to estimate pension or other postretirement benefit income or expense for the following year are the discount rate applied to plan liabilities and the expected long-term rate of return on plan assets. In addition, we are required to make an annual measurement of plan assets and liabilities, which may result in a significant charge to stockholders' equity. For a discussion regarding how our financial statements can be affected by pension and other post-retirement benefits accounting policies, see Note B to the Consolidated Financial Statements in Part II. Item 8, and Part II. Item 7. "Management's Discussion and Analysis of Financial Conditions and Results of Operations--Critical Accounting Policies and Estimates." Although GAAP expense and pension funding contributions are impacted by different regulations and requirements, the key economic factors that affect GAAP expense would also likely affect the amount of cash or securities we would contribute to the pension plans. The defined benefit pension plans were underfunded as of December 31, 2022 by $578 million based on actuarial methods and assumptions in accordance with GAAP. In the event that actual results differ from the actuarial assumptions, the funded status of our defined benefit plans and future cash contributions may increase or decrease. See Part II. Item 7. "Management's Discussion and Analysis of Financial Condition and Results of Operations--Contractual Obligations and Off-Balance Sheet Arrangements--Contractual Obligations" and "--Obligations for Operating Activities" for additional information regarding expected contributions and benefit payments in 2023.

Business Risks – Competition and Customers

We face significant competition, which may have an adverse effect on profitability.

The markets for our products are highly competitive. Our competitors include a variety of both U.S. and non-U.S. companies in all major markets. We may also face competition from emerging competitors, particularly in China and other developing economies, as customers seek to globalize their supply bases in order to reduce costs. New product offerings, new technologies in the marketplace or new facilities may compete with or replace our products. The willingness of customers to accept substitutes for our products, the ability of large customers to exert leverage in the marketplace to affect the pricing for our products, and technological advancements or other developments by or affecting our competitors or customers could adversely affect our business, financial condition or results of operations. See Part I. Item 1. Business "Description of the Business—Rolled Products—Competitive Conditions," "—Extrusions—Competitive Conditions," and "—Building and Construction Systems—Competitive Conditions" for additional information about competition in the markets for our products.

In addition, we may face increased competition due to industry consolidation. As companies attempt to strengthen or maintain their market positions in an evolving industry, they could be acquired or merged. Companies that are strategic alliance partners in some areas of our business may acquire or form alliances with our competitors, thereby reducing their business with us. Industry consolidation may result in stronger competitors who are better able to obtain favorable terms from suppliers or who are better able to compete as sole-source vendors for customers. Consolidation within our customer base may result in customers who are better able to command increased leverage in negotiating prices and other terms of sale, which could adversely affect our profitability. Moreover, if, as a result of increased leverage, customers require us to reduce our pricing such that our gross margins are diminished, we could decide not to sell certain products to a particular customer, or not to sell certain products at all, which would decrease our revenue. Consolidation within our customer base may also lead to reduced demand for our products, a combined entity replacing our products with those of our competitors, and cancellations of orders. The result of these developments could have a material adverse effect on our business, operating results and financial condition.

We could be adversely affected by the loss of key customers, the impact of supply chain disruptions or other economic conditions on our key customers, or significant changes in the business or financial condition of our customers.

We have long-term contracts with a significant number of our customers, some of which are subject to renewal, renegotiation or re-pricing at periodic intervals or upon changes in competitive supply conditions. Our failure to successfully renew, renegotiate or favorably re-price such agreements, or a material deterioration in or termination of these customer relationships, could result in a reduction or loss in customer purchase volume or revenue.

Additionally, a significant downturn or deterioration in the business or financial condition or loss of a key customer could affect our financial results. Our customers may experience unavailability or price volatility of raw materials, key components or other supply chain disruptions, weak demand for their products,delays in the certification or launch of new products, inability to achieve expected future orders in China or other markets, labor strikes, diminished liquidity or credit unavailability that negatively impact the customer's ability to make full or timely payment or that require us to restructure payment terms, or other difficulties in their businesses, any of which could significantly reduce demand for our products. These impacts may be unique to one customer or a small number of customers. For example, Boeing has experienced negative impacts in recent periods, including the temporary reduction in the production rate and subsequent temporary suspension of production of the 737 MAX aircraft, and delays in certification of the 737 MAX aircraft in China; and reduced passenger travel during the COVID-19 pandemic and the resulting elevated inventory of airframes and fuselages. These events have negatively impacted sales of aluminum sheet and plate products that we produce for Boeing airplanes and may continue to negatively impact sales for future periods. The impacts may also be experienced by a broader group of customers within an end market that we serveFor example, in 2020, the impact of the COVID-19 pandemic and other global factors contributed to a severe shortage in semiconductors used in automotive manufacturing. As a result, automotive customers were forced to curtail production at various intervals and across many different product lines resulting in reductions in the volume of vehicles manufactured. For 2020, 2021 and 2022, global vehicle production was reduced by more than 10 million vehicles, resulting in a significant reduction in sales of aluminum sheet and extrusions products that we produce for a number of key automotive customers. Customer initiatives to recover lost volume as business and economic conditions improve may not be successfully implemented, or may be implemented over extended time periods. Depending on the nature, extent and duration of negative business or economic conditions impacting our customers, our financial condition and results of operations may be adversely affected acutely or over a longer period of time.

Our customers may also change their business strategies or modify their business relationships with us, including to reduce the amount of our products they purchase or to switch to alternative suppliers. If our customers reduce, terminate or delay

purchases from us due to the foregoing factors or otherwise and we are unsuccessful in replacing such business in whole or in part or replace it with less profitable business, our financial condition and results of operations may be adversely affected.

Our customers may reduce their demand for aluminum products in favor of alternative materials.

Certain applications of our aluminum-based products compete with products made from other materials, such as steel, titanium, plastics, glass and composites. The willingness of customers to pursue materials other than aluminum often depends upon the desire to achieve specific performance attributes. For example, the commercial aerospace industry has used and continues to evaluate the further use of alternative materials to aluminum, such as titanium and composites, in order to further reduce the weight and increase the fuel efficiency of aircraft. The automotive industry, while motivated to reduce vehicle weight through the use of aluminum, may substitute with steel or other materials for certain applications. The packaging industry continues to experience advances in alternative materials, such as plastics, glass and organic or compostable materials, which may compare favorably to aluminum with respect to preservation of food and beverage quality and recyclability. Further, the decision to use aluminum may be impacted by aluminum prices or compatibility of aluminum with other materials used by a customer in a given application. The willingness of customers to accept other materials in lieu of aluminum could adversely affect the demand for certain of our products, and thus adversely affect our business, financial condition or results of operations.

We may face challenges to our intellectual property rights which could adversely affect our reputation, business and competitive position, financial condition and results of operations.

We own important intellectual property, including patents, trademarks and copyrights. Our intellectual property plays an important role in maintaining our competitive position in a number of the markets that we serve. Our competitors may develop technologies that are similar or superior to our proprietary technologies or design around the patents we own or license. Despite our controls and safeguards, our technology may be misappropriated by our employees, our competitors or other third parties. The pursuit of remedies for any misappropriation of our intellectual property is expensive and the ultimate remedies may be deemed insufficient. Further, in jurisdictions where the enforcement of intellectual property rights is less robust, the risk of misappropriation of our intellectual property increases despite efforts we undertake to protect it. Developments or assertions by or against us relating to intellectual property rights, and any inability to protect or enforce our rights sufficiently, could adversely affect our business and competitive position, financial condition and results of operations.

Business Risks – Operations and Product Development

We could encounter manufacturing difficulties or other issues that impact product performance, quality or safety, which could affect our ability to supply customers or meet contractual obligations, reputation, business and financial condition and results of operations.

The manufacture of many of our products is a highly exacting and complex process. Problems may arise during manufacturing for a variety of reasons, including equipment malfunction or unforeseen maintenance outages, failure to follow specific protocols, specifications and procedures, including those related to quality or safety, unavailability of raw materials, supply chain interruptions, natural disasters, health pandemics (including the COVID-19 pandemic), labor shortages or unrest, and environmental factors. Such problems could have an adverse impact on our ability to fulfill customer orders or meet contractual obligations, including with respect to quantity, delivery times, or product quality, specifications or performance, which could result in recalls, customer penalties, contract cancellation and product liability exposure. Because of approval, license and qualification requirements applicable to manufacturers and/or their suppliers, alternatives to mitigate manufacturing disruptions may not be readily available to us or our customers. Accordingly, manufacturing problems, product defects or other risks associated with our products could result in significant costs related to remediation and other liabilities that could have a material adverse effect on our business, financial condition or results of operations, including the payment of potentially substantial monetary damages, fines or penalties, as well as negative publicity and damage to our reputation, which could adversely impact product demand and customer relationships.

Our business depends, in part, on our ability to meet increased customer demand successfully and to mitigate the impact of customer program cancellations, reductions and delays.

We are under contract to supply aluminum sheet, plate and extrusions for a number of new and existing commercial and general aviation aircraft programs, as well as aluminum sheet and extrusions for a number of aluminum-intensive automotive vehicle programs. Many of these programs are scheduled for production increases over the next several years. In addition, we expect customer demand for packaging materials to continue to increase. If we fail to meet production levels or encounter difficulty or unexpected costs in meeting such levels, it could have a material adverse effect on our business, financial condition

or results of operations. Similarly, program cancellations, reductions or delays could also have a material adverse effect on our business.

A material disruption or limitation of our operations, particularly at one or more of our manufacturing facilities, could adversely affect our business.

If our operations, particularly one of our manufacturing facilities, were to be disrupted as a result of significant equipment failures, natural disasters, adverse weather conditions, power outages, fires, floods, explosions, terrorism, theft, sabotage, public health crises, labor shortages or disputes or other reasons, we may be unable to effectively meet our obligations to or demand from our customers, which could adversely affect our financial performance.

Our operations depend on the continued and efficient functioning of our facilities, including critical equipment. Despite our routine maintenance programs for our facilities and equipment, we may experience periods of reduced production or production delays due to planned and unplanned equipment outages at our facilities. Our facilities also require significant capital improvements, including upgrades to or replacements of equipment, from time to time. Repairs, equipment replacement or other facility improvement projects may also result in periods of reduced or delayed production. Supply chain issues and labor shortages may impact our ability to obtain parts, materials or replacement equipment necessary to make repairs or improvements, and may increase the time required to complete such repairs or improvements. Unforeseen delays or unavailability of parts or materials could significantly increase the costs associated with repairs or improvements.

Interruptions in production could result in significant increases in our costs and reductions in our sales. Any interruption in production capability could require us to incur costs for premium freight, make substantial capital expenditures or purchase alternative material at higher costs to fill customer orders, which could negatively affect our profitability and financial condition. Furthermore, because customers may be dependent on planned deliveries from us, customers that have to reschedule their own production due to our delivery delays may be able to pursue financial claims against us, and we may incur costs to correct such problems in addition to any liability resulting from such claims. While we maintain property damage insurance that we believe to be adequate to provide for reconstruction of facilities and equipment, as well as business interruption insurance to mitigate certain of our losses resulting from significant production interruption or shutdown caused by an insured loss, not all events leading to a disruption of operations are covered events. Additionally, any recovery under our insurance policies may not fully offset the lost profits or increased costs that may be experienced during the disruption of operations, which could adversely affect our business, results of operations, financial condition and cash flow.

We may be unable to develop innovative new products or implement technology initiatives successfully.

Our competitive position and future performance depend, in part, on our ability to:

- identify and evolve with emerging technological and broader industry trends in our target end-markets;
- identify and successfully execute on a strategy to remain an essential and sustainable element of our customers' supply chains;
- fund, develop, manufacture and bring innovative new products and services to market quickly and cost-effectively;
- monitor disruptive technologies and understand customers' and competitors' abilities to deploy those disruptive technologies; and
- achieve sufficient return on investment for new products based on capital expenditures and research and development spending.

We continuously work on new developments for strategic projects, including alloy development, engineered finishes and product design, high speed continuous casting and rolling technology and other advanced manufacturing technologies. For more information on our research and development programs, see Part I, Item 1. Business "Research and Development."

In spite of our expenditure of financial resources and dedicated effort to develop innovative new products and services, we may not be able to successfully differentiate our products or services from those of our competitors or match the level of research and development spending of our competitors, including those developing technology to displace our current products. In addition, we may not be able to adapt to evolving markets and technologies or achieve and maintain technological advantages. There can be no assurance that any of our new products or services, development programs or technologies will be commercially adopted or beneficial to us.

Business Risks – Supply Chain

Our business could be adversely affected by increases in the cost or volatility in the availability of aluminum or other raw materials.

We derive a significant portion of our revenue from aluminum-based products. The price of primary aluminum has historically been subject to significant cyclical price fluctuations, and the timing of changes in the market price of aluminum is largely unpredictable. Although the pricing of most of our products is generally intended to pass substantially all the risk of metal price fluctuations on to our customers or is otherwise hedged, there are situations where we are unable to pass on the entire cost of increases to our customers and there is a potential time lag on certain products between increases in costs for aluminum and the point when we can implement a corresponding increase in price to our customers and/or there are other timing factors that may result in our exposure to certain price fluctuations which could have a material adverse effect on our business, financial condition or results of operations. Further, since metal prices fluctuate among the various exchanges, our competitors may enjoy a metal price advantage from time to time.

We may be adversely affected by changes in the availability or cost of other raw materials (including, but not limited to, copper, magnesium, silicon and zinc), as well as labor costs, energy costs and freight costs associated with transportation of materials. Prices for and the availability of materials necessary for production may fluctuate due to a number of factors, including inflationary pressure, supply shortages and disruptions caused by geopolitical or global health events, such as the COVID-19 pandemic. The availability and costs of certain raw materials necessary for the production of our products may also be influenced by private or government entities, including mergers and acquisitions, changes in world politics or regulatory requirements (such as human rights regulations, environmental regulations or production curtailments), labor relations between the producers and their work forces, labor shortages, political instability in exporting nations, export quotas, sanctions, new or increased import duties, countervailing or anti-dumping duties, infrastructure and transportation issues, market forces of supply and demand, and inflation. In addition, from time to time, commodity prices may fall rapidly. When this happens, suppliers may withdraw capacity from the market until prices improve, which may cause periodic supply interruptions. We may be unable to offset fully the effects of material shortages or higher costs through customer price increases, productivity improvements or cost reduction programs. Shortages or price fluctuations in raw materials could have a material adverse effect on our operating results.

We are dependent on a limited number of suppliers for a substantial portion of our primary and scrap aluminum and certain other raw materials essential to our operations.

We have supply arrangements with a limited number of suppliers for aluminum and other raw materials. We maintain annual or long-term contracts for a majority of our supply requirements, and for the remainder we depend on spot purchases. From time to time, increasing aluminum demand levels have caused regional supply constraints in the industry and further increases in demand levels could exacerbate these issues. Such constraints could impact our production or force us to purchase primary metal and other supplies from alternative sources, which may not be available in sufficient quantities or may only be available on terms that are less favorable to us. Further, there can be no assurance that we will be able to renew, or obtain replacements for, any of our long-term contracts when they expire on terms that are as favorable as our existing agreements or at all. Additionally, we could have exposure if a key supplier in a particular region is unable to deliver sufficient quantities of a necessary material on a timely basis. A significant interruption in the operations of a key supplier could jeopardize the ability of plants in that region to operate at capacity, which could in turn have a material adverse effect on our financial condition, results of operations and cash flow. In addition, a significant downturn in the business or financial condition of our significant suppliers exposes us to the risk of default by the supplier on our contractual agreements, and this risk is increased by weak and deteriorating economic conditions on a global, regional or industry sector level.

We also depend on scrap aluminum for our operations and acquire our scrap inventory from numerous sources. These suppliers generally are not bound by long-term contracts and have no obligation to sell scrap metal to us. In periods of low inventory prices, suppliers may elect to hold scrap until they are able to charge higher prices. Additionally, the purity and attributes of scrap material can vary significantly, which could result in a shortage of useable scrap metal. If an adequate supply of scrap metal is not available to us, we would be unable to recycle metals at desired volumes and our results of operations, financial condition and cash flows could be materially adversely affected.

Business Risks – Strategy

We may not be able to realize the expected benefits of our re-entry into the packaging market in the U.S. and other geographies.

We have made a significant investment of management time and financial resources in our re-entry into the U.S. packaging market, including capital investments in machinery, application of research and development resources to developing innovations in packaging materials, re-tooling portions of our rolled products capacity to produce materials designed to suit the needs of customers in the packaging market, supplementing our workforce to fulfill capacity, and engaging with a new customer base that has different needs than our aerospace, automotive, and industrial customers. In addition, the competitive landscape in the packaging market involves not only some of our current key competitors, with whom we compete for customers, labor and materials including scrap, but also new competitors offering alternative packaging materials, particularly plastics and glass products, many of whom are larger and more established in the packaging market than we are. Our ability to realize the benefits of our strategic decision to re-enter the packaging market could be impacted by availability of labor, raw materials or scrap necessary to produce sufficient volume to satisfy customer demands, unforeseen outages at our facilities that serve the packaging market, or unexpected costs. If we are unable to realize the projected benefits of our re-entry into the packaging market in the U.S. or other geographies, our financial condition and results of operations may be materially adversely affected.

We may be unable to realize future targets or goals established for our business segments, or complete capital or other projects at the levels, projected costs or by the dates targeted.

From time to time, we may announce future targets or goals for our business, which are based on our then current expectations, estimates, forecasts and projections about the operating environment, economies and markets in which we operate. Future targets and goals reflect our beliefs and assumptions and our perception of historical trends, then current conditions and expected future developments, as well as other factors appropriate in the circumstances. As such, targets and goals are inherently subject to significant business, economic, competitive and other uncertainties and contingencies regarding future events, including the risks discussed therein. The actual outcome may be materially different from projected or target outcomes, and there can be no assurance that any targets or goals established by us will be accomplished at the levels or by the dates targeted, if at all. Failure to achieve our targets or goals may have a material adverse effect on our business, financial condition, results of operations or the market price of our securities.

In addition, the implementation of our business strategy may involve the entry into and the execution of complex capital or other projects, which place significant demands on our management and personnel, and the completion and ultimate impact of such projects on our operations may depend on numerous factors beyond our control. There can be no assurance that such projects will be completed within budgeted costs, on a timely basis, or at all, whether due to the risks described herein, or other factors. The failure to complete a material project as planned, a significant delay in a material project, increased or unforeseen costs associated with a project, or the failure of a project to have the projected impact on our business or operations, whatever the cause, could have an adverse effect on our business, financial condition, or results of operations.

Our business and growth prospects may be negatively impacted by limits in our capital expenditures.

We require substantial capital to invest in growth opportunities and to maintain and prolong the life and capacity of our existing facilities. Insufficient cash generation or capital project overruns or delays may negatively impact our ability to fund as planned our sustaining and return-seeking capital projects. Over the long term, our ability to take advantage of improved market conditions or growth opportunities in our businesses may be constrained by earlier capital expenditure restrictions, which could adversely affect the long-term value of our business and our position in relation to our competitors.

We may be unable to realize the expected benefits from acquisitions, divestitures, joint ventures and strategic alliances.

We have made, and may continue to plan and execute, acquisitions and divestitures and take other actions to grow our business or streamline our portfolio. There is no assurance that anticipated benefits will be realized. Acquisitions present significant challenges and risks, including our effective integration of the acquired business, unanticipated costs and liabilities, and the ability to realize anticipated benefits, such as growth in market share, revenue or margins, at the levels or in the timeframe expected. We may be unable to manage acquisitions successfully. Additionally, adverse factors may prevent us from realizing the benefits of our growth projects, including unfavorable global economic conditions, currency fluctuations, or unexpected delays in target timelines.

With respect to portfolio optimization actions such as divestitures, curtailments and closures, we may face barriers to exit from unprofitable businesses or operations, including high exit costs or objections from customers, suppliers, unions, local or

national governments, or other stakeholders. In addition, we may retain unforeseen liabilities for divested entities or businesses, including, but not limited to, if a buyer fails to honor all commitments. Our business operations are capital intensive, and curtailment or closure of operations or facilities may include significant charges, including employee separation costs, asset impairment charges and other measures.

In addition, we have participated in, and may continue to participate in, joint ventures, strategic alliances and other similar arrangements from time to time. Although we have, in connection with past and existing joint ventures, sought to protect our interests, joint ventures and strategic alliances inherently involve special risks. Whether or not we hold majority interests or maintains operational control in such arrangements, our partners may:

- have economic or business interests or goals that are inconsistent with or opposed to ours;
- exercise veto rights to block actions that we believe to be in our or the joint venture's or strategic alliance's best interests;
- take action contrary to our policies or objectives with respect to investments; or
- as a result of financial or other difficulties, be unable or unwilling to fulfill their obligations under the joint venture, strategic alliance or other agreements, such as contributing capital to expansion or maintenance projects.

There can be no assurance that acquisitions, growth investments, divestitures, closures, joint ventures, strategic alliances or similar arrangements will be undertaken or completed in their entirety as planned or that they will be beneficial to us, whether due to the above-described risks, unfavorable global economic conditions, increases in construction costs, currency fluctuations, political risks, or other factors.

A decline in our financial performance or outlook or a deterioration in our credit profile could negatively impact our access to the capital markets and commercial credit, reduce our liquidity, and increase our borrowing costs.

We have significant capital requirements and may require, in the future, the issuance of debt to fund our operations and contractual commitments or to pursue strategic acquisitions. A decline in our financial performance or outlook due to internal or external factors could affect our access to, and the availability or cost of, financing on acceptable terms and conditions. There can be no assurance that we will have access to the capital markets on terms we find acceptable.

Major credit rating agencies evaluate our creditworthiness and give us specified credit ratings. These credit ratings are based on a number of factors, including our financial strength and financial policies as well as our strategies, operations and execution. These credit ratings are limited in scope, and do not address all material risks related to investment in us, but rather reflect only the view of each rating agency at the time the rating is issued. Nonetheless, the credit ratings we receive will impact our borrowing costs as well as the terms upon which we will have access to capital. Failure to obtain sufficiently high credit ratings could adversely affect the interest rate in future financings, our liquidity or our competitive position, and could also restrict our access to capital markets.

There can be no assurance that one or more of the credit rating agencies will not take negative actions with respect to our ratings in the future. Increased debt levels, macroeconomic conditions, a deterioration in our debt protection metrics, a contraction in our liquidity, or other factors could potentially trigger such actions. A credit rating agency may lower, suspend or withdraw entirely a rating or place it on negative outlook or watch if, in that rating agency's judgment, circumstances so warrant. A downgrade of our credit ratings by one or more credit rating agencies could result in adverse consequences, including: adversely impact the market price of our securities; adversely affect existing financing; limit access to the capital (including commercial paper) or credit markets or otherwise adversely affect the availability of other new financing on favorable terms, if at all; result in more restrictive covenants in agreements governing the terms of any future indebtedness that we incur; increase the cost of borrowing or fees on undrawn credit facilities; or result in vendors or counterparties seeking collateral or letters of credit from us.

Limitations on our ability to access the global capital markets, a reduction in our liquidity or an increase in borrowing costs could materially and adversely affect our ability to maintain or grow our business, which in turn may adversely affect our financial condition, liquidity and results of operations.

Business Risks – Information Security and Internal Controls

Information technology system failures, cyber-attacks and security breaches may threaten the integrity of our intellectual property and sensitive information, disrupt our business operations, and result in reputational harm and other negative consequences that could have a material adverse effect on our financial condition and results of operations.

We rely on our information technology systems to manage and operate our business, process transactions, and summarize our operating results. Our information technology systems are subject to damage or interruption from power outages, computer, network and telecommunications failures, computer viruses, and catastrophic events, such as fires, floods, earthquakes, tornadoes, hurricanes, acts of war or terrorism, and usage errors by employees. If our information technology systems are damaged or cease to function properly, we may have to make a significant investment to fix or replace them, and we may suffer loss of critical data and interruptions or delays in our operations. Any material disruption in our information technology systems, or delays or difficulties in implementing or integrating new systems or enhancing current systems, could have an adverse effect on our business, financial condition or results of operations.

We believe that we face the threat of cyber-attacks due to the industries we serve, the locations of our operations and our technological innovations. These cyber-attacks may range from uncoordinated individual attempts to sophisticated and targeted measures, known as advanced persistent threats, directed at us and our customers, suppliers and vendors. Cyber-attacks and security breaches may include, but are not limited to, attempts to access information, computer viruses, denial of service and other electronic security breaches, any of which could manipulate or improperly use our systems or networks, compromise confidential information, destroy or corrupt data, or otherwise disrupt our operations. We have experienced cybersecurity attacks in the past, including breaches of our information technology systems in which information was taken, and may experience them in the future, potentially with more frequency or sophistication. Based on information known to date, past attacks have not had a material impact on our financial condition or results of operations. However, due to the evolving nature of cybersecurity threats, the scope and impact of any future incident cannot be predicted.

We continually work to safeguard our systems and mitigate potential risks, and our enterprise risk management program and disclosure controls and procedures include elements intended to ensure that we analyze potential disclosure obligations arising from cyber-attacks and security breaches. However, there is no assurance that these safeguards and controls will be sufficient to detect, prevent, engage in timely response to, or report cyber-attacks or security breaches. The occurrence of cyber-attacks or security breaches could negatively impact our reputation and competitive position and could result in litigation with third parties, regulatory action, loss of business, diminution of the value of investments in research and development, potential liability and increased remediation costs, any of which could have a material adverse effect on our financial condition and results of operations.

If we fail to maintain an effective system of internal controls, we may not be able to accurately report financial results or prevent fraud.

We are subject to reporting and other obligations under the Exchange Act, the Sarbanes-Oxley Act, the Dodd-Frank Wall Street Reform and Consumer Protection Act, and the regulations of the New York Stock Exchange, and to the requirements of Section 404 of Sarbanes-Oxley which requires management to establish and maintain internal control over financial reporting and disclosure controls and procedures. Internal control over financial reporting is a process designed to provide reasonable assurance regarding the reliability of financial reporting and the preparation of financial statements in accordance with generally accepted accounting principles. Internal controls are also important in the prevention and detection of fraudulent activity. Disclosure controls and procedures are processes designed to ensure that information required to be disclosed is communicated to management and reported in a timely fashion. A control system, no matter how well designed and operated, can provide only reasonable, not absolute, assurance with respect to the reliability of reporting, including financial reporting and financial statement preparation. If we are not able to maintain effective internal control over financial reporting or disclosure controls and procedures, or other accounting, financial management or reporting systems or procedures, or experience difficulties or delays in the implementation of systems or controls, we may not be able to accurately report our financial results or prevent fraud, and in some cases may be required to restate financial results. As a result, stockholders could lose confidence in our financial and other public reporting, could result in adverse regulatory consequences and/or loss of investor confidence, which could limit our ability to access the global capital markets and could have a material adverse effect on our business, financial condition, results of operations, cash flows or the market price of our securities. In addition, the remediation of any ineffective internal controls could result in unforeseen expenses.

Risks Related to Employee and Labor Matters

Labor disputes and difficulties retaining or hiring skilled employees could adversely affect our business, financial condition or results of operations.

The continuity of our operations depends on maintaining a skilled workforce. We have previously experienced shortages of qualified labor due in part to the COVID-19 pandemic and other economic factors, and may see similar labor shortages in the future. Any labor shortage could decrease our ability to effectively produce and deliver product and to achieve our strategic objectives. In addition, any potential labor shortage may result in increased expenses related to hiring and retention of qualified employees.

Furthermore, a significant portion of our employees are represented by labor unions in a number of countries under various collective bargaining agreements with varying durations and expiration dates. While we previously have been successful in renegotiating our collective bargaining agreements with various unions, we may not be able to satisfactorily renegotiate all collective bargaining agreements in the U.S. and other countries when they expire. In addition, existing collective bargaining agreements may not prevent a strike or work stoppage at our facilities in the future. We may also be subject to general country strikes or work stoppages unrelated to our business or collective bargaining agreements. Any such work stoppages could have a material adverse effect on our business, financial condition or results of operations.

A failure to attract, retain or provide adequate succession plans for key personnel could adversely affect our operations and competitiveness.

Our existing operations and development projects require highly skilled executives and staff with relevant industry and technical experience. Our inability to attract and retain such people may adversely impact our ability to meet project demands adequately and fill roles in existing operations. Skills shortages in engineering, manufacturing, technology, construction and maintenance contractors and other labor market inadequacies may also impact activities. These shortages may adversely impact the cost and schedule of development projects and the cost and efficiency of existing operations.

In addition, the continuity of key personnel and the preservation of institutional knowledge are vital to the success of our growth and business strategy. The loss of key members of management and other personnel could significantly harm our business, and any unplanned turnover, or failure to develop adequate succession plans for key positions, could deplete our institutional knowledge base, result in loss of technical expertise, delay or impede the execution of our business plans and erode our competitiveness.

Failure to comply with domestic or international employment and related laws could result in penalties or costs that could have a material adverse effect on our business results.

We are subject to a variety of domestic and foreign employment laws, such as the Fair Labor Standards Act (which governs such matters as minimum wages, overtime and other working conditions), state and local wage laws, the Employee Retirement Income Security Act, and regulations related to health and safety, discrimination, organizing, whistleblowing, classification of employees, privacy, severance payments, citizenship requirements, and healthcare insurance mandates. Allegations that we have violated such laws or regulations could damage our reputation and lead to fines from or settlements with federal, state or foreign regulatory authorities or damages payable to employees, which could have a material adverse impact on our operations and financial condition.

Risks Related to Legal Proceedings and Government Regulations

Product liability, product safety, personal injury, property damage, and recall claims and investigations may materially affect our financial condition and damage our reputation.

The manufacture and sale of our products exposes us to potential product liability, personal injury, property damage and related claims. These claims may arise from our alleged failure to meet product specifications, design flaws in our products, malfunction of our products, misuse of our products, use of our products in an unintended, unapproved or unrecommended manner, or use of our products with systems not manufactured or sold by us. New data and information, including information about the ways in which our products are used, may lead regulatory authorities, government agencies or other entities or organizations to publish guidelines or recommendations, or impose restrictions, related to the manufacturing or use of our products.

In the event that a product of ours fails to perform as expected, regardless of fault, or is used in an unexpected manner, and such failure or use results in, or is alleged to result in, bodily injury and/or property damage or other losses, we may be subject to product liability lawsuits and other claims, or may be required or requested by our customers to participate in a recall or other corrective action involving such product. In addition, if a product of ours is perceived to be defective or unsafe, sales of our products could be diminished, our reputation could be adversely impacted, and we could be subject to further liability claims. Moreover, events that give rise to actual, potential or perceived product safety concerns could expose us to government investigations or regulatory enforcement action.

There can be no assurance that we will be successful in defending any such proceedings or that insurance available to us will be sufficient to cover any losses associated with such proceedings. An adverse outcome in one or more of these proceedings or investigations could have a material adverse effect on our business, financial condition or profitability; impose substantial monetary damages and/or non-monetary penalties; result in additional litigation, regulatory investigations or other proceedings involving us; result in loss of customers; require changes to our products or business operations; damage our reputation and/or negatively impact the market price of our common stock. Even if we successfully defend against these types of claims, we could still be required to spend a substantial amount of money in connection with legal proceedings or investigations with respect to such claims; our management could be required to devote significant time, attention and operational resources responding to and defending against these claims and responding to these investigations; and our reputation could suffer. Product liability claims and related lawsuits and investigations, product recalls, and allegations of product safety or quality issues, regardless of their validity or ultimate outcome, may have a material adverse effect on our business, financial condition and reputation and on our ability to attract and retain customers.

For further discussion of potential liability associated with some of our products, including proceedings and investigations relating to the June 13, 2017 fire at the Grenfell Tower in London, U.K., see Note T to the Consolidated Financial Statements in Part II. Item 8. "Financial Statements and Supplementary Data" under the caption "Contingencies and Commitments - Contingencies."

We may be exposed to significant legal proceedings, investigations or changes in U.S. federal, state, local or foreign laws, regulations or policies.

Our results of operations or liquidity in a particular period could be affected by new or increasingly stringent laws, regulatory requirements or interpretations, or outcomes of significant legal proceedings or investigations adverse to us. We may experience an unfavorable change in effective tax rates or become subject to unexpected or rising costs associated with business operations or provision of health or welfare benefits to employees due to changes in laws, regulations or policies.

We are subject to a variety of legal and regulatory compliance risks in the U.S. and abroad in connection with our business and products. These risks include, among other things, potential claims relating to product liability, product testing, health and safety, environmental matters, employment matters, required record keeping and record retention, compliance with securities laws, intellectual property rights, government contracts and taxes, insurance or commercial matters, as well as compliance with U.S. and foreign laws and regulations governing import and export, anti-bribery, antitrust and competition, sales and trading practices, human rights and modern slavery, sourcing of raw materials, third-party relationships, supply chain operations and the manufacture and sale of products. We may be a party to litigation in a foreign jurisdiction where geopolitical risks might influence the ultimate outcome of such litigation. We could be subject to fines, penalties, damages (in certain cases, treble damages), or suspension or debarment from government contracts.

The global and diverse nature of our operations means that these risks will continue to exist, and additional legal proceedings and contingencies may arise from time to time. While we believe we have adopted appropriate risk management and compliance programs to address and reduce these risks, including insurance arrangements with respect to these risks, such measures may provide inadequate protection against liabilities that may arise. In addition, various factors or developments can lead us to change current estimates of liabilities or make such estimates for matters previously insusceptible to reasonable estimates, such as a significant judicial ruling or judgment, a significant settlement, significant regulatory developments or changes in applicable law. A future adverse ruling or settlement or unfavorable changes in laws, regulations or policies, or other contingencies that we cannot predict with certainty could have a material adverse effect on our financial condition, results of operations or cash flows in a particular period. Litigation and compliance efforts may require substantial attention from management and could result in significant legal expenses, settlement costs or damage awards that could have a material impact on our financial position, results of operations and cash flows. For additional information regarding our legal proceedings, including proceedings and investigations relating to the June 13, 2017 fire at the Grenfell Tower in London, U.K., see Note T to the Consolidated Financial Statements in Part II, Item 8. "Financial Statements and Supplementary Data" under the Caption "Contingencies and Commitments - Contingencies."

We are exposed to environmental and safety risks and are subject to a broad range of health, safety and environmental laws and regulations, which may result in substantial costs and liabilities.

Our operations worldwide are subject to numerous complex and increasingly stringent health, safety and environmental laws and regulations. The costs of complying with such laws and regulations, including participation in assessments and cleanups of sites, as well as internal voluntary programs, are significant and will continue to be so for the foreseeable future. Failure to comply with such laws and regulations could result in significant penalties or criminal liability. Environmental laws may impose cleanup liability on owners and occupiers of contaminated property, including present, past or divested properties, regardless of whether the owners and occupiers caused the contamination or whether the activity that caused the contamination was lawful at the time it was conducted. Environmental matters for which we may be liable may arise in the future at our present sites, at sites owned or operated by our predecessors or affiliates, at sites that we may acquire in the future, or at third-party sites used by our predecessors or affiliates for material and waste handling and disposal. Compliance with health, safety and environmental laws and regulations, including remediation obligations, may prove to be more challenging and costly than we anticipate. Our results of operations or liquidity in a particular period could be affected by certain health, safety or environmental matters, including remediation costs and damages related to certain sites as well as other health and safety risks relating to our operations and products. Additionally, evolving regulatory standards and expectations can result in increased litigation and/or increased costs, including increased remediation costs, all of which can have a material and adverse effect on our financial condition, results of operations and cash flows.

In addition, the heavy industrial activities conducted at our facilities present a significant risk of injury or death to our employees, customers or third parties that may be on site. We have experienced serious injuries in the past, notwithstanding the safety protocols, practices and precautions we take. Our operations are subject to regulation by various federal, state and local agencies in the U.S. and regulation by foreign government entities abroad responsible for employee health and safety, including OSHA. From time to time, we have incurred fines for violations of various health and safety standards. While we maintain insurance and have in place policies to minimize such risks, we may nevertheless be unable to avoid material liabilities for any injury or death that may occur in the future. These types of incidents may not be covered by or may exceed our insurance coverage and could have a material adverse effect on our results of operations and financial condition or result in negative publicity and/or significant reputational harm.

We are subject to privacy and data security/protection laws in the jurisdictions in which we operate and may be exposed to substantial costs and liabilities associated with such laws and regulations.

The regulatory environment surrounding information security and privacy is increasingly demanding, with frequent imposition of new and changing requirements as well as significant fines for non-compliance. While Arconic has appropriate processes and procedures in place regarding compliance with existing data protection regulations in other countries, such as the European Union's General Data Protection Regulation, the California Privacy Rights Act, and China's Personal Information Protection Law, further changes may be necessary to ensure those processes and procedures will be adequate to implement additional state and country specific requirements. Compliance with changes in privacy and information security laws and standards may result in significant expense due to increased investment in technology and the development of new operational processes, which could have a material adverse effect on our financial condition and results of operations. In addition, the payment of potentially significant fines or penalties in the event of a breach of privacy and information security laws, as well as the negative publicity associated with such a breach, could damage our reputation and adversely impact product demand and customer relationships.

Unanticipated changes in our tax provisions or exposure to additional tax liabilities could affect our future profitability.

We are subject to income taxes in both the U.S. and various non-U.S. jurisdictions. Our domestic and international tax liabilities are dependent upon the distribution of income among these different jurisdictions. Changes in applicable domestic or foreign tax laws and regulations, or their interpretation and application, including the possibility of retroactive effect, could affect our tax expense and profitability. Our tax expense includes estimates of additional tax that may be incurred for tax exposures and reflects various estimates and assumptions. The assumptions include assessments of our future earnings that could impact the valuation of our deferred tax assets. Our future results of operations could be adversely affected by changes in the effective tax rate as a result of a change in the mix of earnings in countries with differing statutory tax rates, changes in our overall profitability, changes in tax legislation and rates, changes in generally accepted accounting principles, changes in the valuation of deferred tax assets and liabilities, the results of tax audits and examinations of previously filed tax returns or related litigation and continuing assessments of our tax exposures. Any failure to comply with all such tax laws or regulations could subject us to liability.

Corporate tax law changes continue to be analyzed in the U.S. and in many other jurisdictions. The Organisation for Economic Co-operation and Development and current U.S. presidential administration have proposed changes that could

adversely impact the taxation of corporations in the U.S. and abroad. Any change to the U.S. corporate tax system could have a substantial impact, positive or negative, on our future effective tax rate, cash tax expenditures, and deferred tax assets and liabilities.

Risks Related to Our Indebtedness

We have significant debt obligations, and may in the future incur, additional debt obligations that could adversely affect our business and profitability and our ability to meet other obligations.

On February 7, 2020, we completed an offering for $600 million of 6.125% (fixed rate) Senior Secured Second-Lien Notes due 2028 (the “2028 Notes”).On May 13, 2020, we completed an offering of $700 million principal amount of 6.0% Senior Secured First-Lien Notes due 2025 (the "2025 Notes"). Also on May 13, 2020, we entered into the ABL Credit Agreement, which provides for a senior secured asset-based revolving credit facility in an aggregate principal amount of $800 million, including a letter of credit sub-facility, a swingline loan sub-facility and an accordion feature allowing the Company to request one or more increases to the revolving commitments in an aggregate principal amount up to $350 million. On March 3, 2021, we completed an offering for an additional $300 million principal amount of the 2028 Notes, which were issued under the indenture governing the existing 2028 Notes. We increased the aggregate principal amount of the ABL Credit Agreement in February 2022 to $1.2 billion. We may also incur additional indebtedness in the future, including by drawing under the ABL Credit Facility.

This significant amount of debt could potentially have important consequences to us and our debt and equity investors, including:

- requiring a substantial portion of our cash flow from operations to make interest payments;
- making it more difficult to satisfy debt service and other obligations;
- increasing the risk of a future credit ratings downgrade of our debt, which could increase future debt costs and limit the future availability of debt financing;
- increasing our vulnerability to general adverse economic and industry conditions;
- reducing the cash flow available to fund capital expenditures and other corporate purposes and to grow our business;
- limiting our flexibility in planning for, or reacting to, changes in our business and the industry;
- placing us at a competitive disadvantage relative to our competitors that may not be as highly leveraged with debt; and
- limiting our ability to borrow additional funds as needed or take advantage of business opportunities as they arise, pay cash dividends or repurchase securities.

Subject to the restrictions in the indenture governing the 2025 Notes, the indenture governing the 2028 Notes and the ABL Credit Agreement, we, including our subsidiaries, have the ability to incur significant additional indebtedness. Although the terms of the 2025 Notes indenture, the 2028 Notes indenture and the ABL Credit Facility include restrictions on the incurrence of additional indebtedness, these restrictions are subject to a number of important exceptions, and indebtedness incurred in compliance with these restrictions could be substantial. Adding new debt to our current debt levels could intensify the related risks that we and our subsidiaries face now or may face in the future. In addition, our actual cash requirements in the future may be greater than expected. Our cash flow from operations may not be sufficient to repay all of the outstanding debt as it becomes due, and we may not be able to borrow money, sell assets or otherwise raise funds on acceptable terms, or at all, to refinance our debt.

Our indebtedness restricts our current and future operations, which could adversely affect our ability to respond to changes in our business and manage our operations.

The terms of the 2025 Notes indenture, the 2028 Notes indenture and the ABL Credit Agreement includea number of restrictive covenants that impose significant operating and financial restrictions on us, including restrictions on our ability to, among other things:

- make investments, loans, advances, guarantees and acquisitions;
- dispose of assets;
- incur or guarantee additional debt and issue certain disqualified equity interests and preferred stock;
- make certain restricted payments, including a limit on dividends on equity securities or payments to redeem, repurchase or retire equity securities or other indebtedness;
- engage in transactions with affiliates;
- enter into certain restrictive agreements;
- create liens on assets to secure debt; and
- consolidate, merge, sell or otherwise dispose of all or substantially all of our or a subsidiary guarantor’s assets.

These covenants limit our operational flexibility and could prevent us from taking advantage of business opportunities as they arise, growing our business or competing effectively. In addition, the ABL Credit Facility contains a financial maintenance covenant applicable when the excess availability is less than the greater of (a) 10% of the lesser of (x) the aggregate amount of the commitments under the ABL Credit Facility and (y) the borrowing base and (b) $50.0 million. In such circumstances, we would be required to maintain a fixed charge coverage ratio of not less than 1.00 to 1.00. Our ability to draw under the ABL Credit Facility could be impacted by a number of factors, including but not limited to any impact by disruptions to our operations and financial performance.

The ABL Credit Facility also provides for "springing control" over the cash in our deposit accounts constituting ABL priority collateral for the ABL Credit Facility, and such cash management arrangement includes a cash sweep at any time that excess availability under the ABL Credit Facility is less than the greater of (x) 12.5% of the lesser of the borrowing base and the aggregate amount of the commitments under the ABL Credit Facility at such time and (y) $62.5 million for five consecutive business days. Such cash sweep, if in effect, will cause all our available cash in deposit accounts subject to such "springing control" to be applied to outstanding borrowings under our ABL Credit Facility. If we satisfy the conditions to borrowings under the ABL Credit Facility while any such cash sweep is in effect, we may be able to make additional borrowings under the ABL Credit Facility to satisfy our working capital and other operational needs. If we do not satisfy the conditions to borrowing, we will not be permitted to make additional borrowings under our ABL Credit Facility, and we may not have sufficient cash to satisfy our working capital and other operational needs.

Our ability to comply with these agreements may be affected by events beyond our control, including prevailing economic, financial and industry conditions. These covenants could have an adverse effect on our business by limiting our ability to take advantage of financing, merger and acquisition or other opportunities. The breach of any of these covenants or restrictions could result in a default under the 2025 Notes indenture, the 2028 Notes indenture or the ABL Credit Agreement.

Our failure to comply with the agreements relating to our outstanding indebtedness, including as a result of events beyond our control, could result in an event of default that could materially and adversely affect our business, financial condition, results of operations or cash flows.

If there were an event of default under any of the agreements relating to our outstanding indebtedness, including the 2025 Notes indenture, the 2028 Notes indenture and the ABL Credit Agreement, we may not be able to incur additional indebtedness and the holders of the defaulted indebtedness could cause all amounts outstanding with respect to that indebtedness to be immediately due and payable. We cannot assure you that our assets or cash flow would be sufficient to fully repay our outstanding indebtedness if accelerated upon an event of default, which could have a material adverse effect on our ability to continue to operate as a going concern. Further, if we are unable to repay, refinance or restructure our secured indebtedness, the holders of such indebtedness could proceed against the collateral securing that indebtedness. In addition, any event of default under or declaration of acceleration under one debt instrument also could result in an event of default under one or more of the agreements governing our other indebtedness.

Risks Related to the Separation

We have a limited history of operating as an independent company and our historical financial information may not be a reliable indicator of our future results.

The historical information included in this Annual Report on Form 10-K for periods prior to the Separation refers to Arconic as operated by and integrated with ParentCo for those periods. Our historical financial information is derived from ParentCo's accounting records and is presented on a standalone basis as if Arconic was independent of ParentCo. Accordingly, the historical information does not necessarily reflect the financial condition, results of operations or cash flows that we would have achieved as a separate, publicly traded company during the periods presented or those that we will achieve in the future primarily as a result of significant changes in our cost structure, management, financing and business operations as a result of operating as a company separate from ParentCo. For additional information about the past financial performance of our business, see Part II, Item 7. "Management's Discussion and Analysis of Financial Condition and Results of Operations," and Part II, Item 8. "Financial Statements and Supplementary Data."

In connection with the Separation, we and Howmet have agreed to indemnify each other for certain liabilities. If we are required to pay under these indemnities to Howmet, our financial results could be negatively impacted. The Howmet indemnities may not be sufficient to hold us harmless from the full amount of liabilities for which Howmet has been allocated responsibility, and Howmet may not be able to satisfy its indemnification obligations in the future.

Pursuant to the separation agreement and certain other agreements between ParentCo and us, each party has agreed to indemnify the other for certain liabilities, in each case for uncapped amounts. Indemnities that we may be required to provide Howmet are not subject to any cap, may be significant and could negatively impact our business. Third parties could also seek to hold us responsible for any of the liabilities that Howmet has agreed to retain. Any amounts we are required to pay pursuant to these indemnification obligations and other liabilities could require us to divert cash that would otherwise have been used in furtherance of our operating business. Further, the indemnities from Howmet for our benefit may not be sufficient to protect us against the full amount of such liabilities, and Howmet may not be able to fully satisfy its indemnification obligations.

Moreover, even if we ultimately succeed in recovering from Howmet any amounts for which we are held liable, we may be temporarily required to bear these losses ourselves. Each of these risks could negatively affect our business, results of operations and financial condition.

Howmet may fail to perform under various transaction agreements that were executed as part of the Separation.

In connection with the Separation, we and ParentCo have entered into the separation agreement and various other agreements, including a tax matters agreement, an employee matters agreement, intellectual property license agreements, an agreement relating to the Davenport, Iowa plant, metal supply agreements and real estate and office leases. We will rely on Howmet to satisfy its performance and payment obligations under these agreements. If Howmet is unable or unwilling to satisfy its obligations under these agreements, including its indemnification obligations, we could incur operational difficulties and/or losses.

If the distribution, together with certain related transactions, does not qualify as a transaction that is generally tax-free for U.S. federal income tax purposes, we, as well as Howmet and Howmet's stockholders, could be subject to significant tax liabilities, and, in certain circumstances, we could be required to indemnify Howmet for material taxes and other related amounts pursuant to indemnification obligations under the tax matters agreement.

It was a condition to the distribution of our common stock to ParentCo stockholders in connect with the Separation that ParentCo receive an opinion of its outside counsel, satisfactory to the ParentCo Board of Directors, regarding the qualification of the distribution, together with certain related transactions, as a "reorganization" within the meaning of Sections 355 and 368(a)(1)(D) of the U.S. Internal Revenue Code. The opinion of counsel was based upon and relied on, among other things, various facts and assumptions, as well as certain representations, statements and undertakings of ParentCo and us, including those relating to the past and future conduct of ParentCo and us. If any of these facts, assumptions, representations, statements or undertakings was, or becomes, inaccurate or incomplete, or if ParentCo breaches its or we breach any of our respective representations or covenants contained in the separation agreement and certain other agreements and documents or in any documents relating to the opinion of counsel, the opinion of counsel may be invalid and the conclusions reached therein could be jeopardized.

Notwithstanding receipt of the opinion of counsel, the U.S. Internal Revenue Service ("IRS") could determine that the distribution and/or certain related transactions should be treated as taxable transactions for U.S. federal income tax purposes if it determines that any of the representations, assumptions or undertakings upon which the opinion of counsel was based are false or have been violated. In addition, the opinion of counsel represented the judgment of such counsel and is not binding on the IRS or any court, and the IRS or a court may disagree with the conclusions in the opinion of counsel. Accordingly, notwithstanding receipt of the opinion of counsel, there is no assurance that the IRS will not assert that the distribution and/or certain related transactions do not qualify for tax-free treatment for U.S. federal income tax purposes or that a court would not sustain such a challenge. In the event the IRS were to prevail with such challenge, we, as well as ParentCo and ParentCo's stockholders, could be subject to significant U.S. federal income tax liability.

If the distribution were to fail to qualify as a transaction that is generally tax-free for U.S. federal income tax purposes under Sections 355 and 368(a)(1)(D) of the U.S. Internal Revenue Code, in general, for U.S. federal income tax purposes, ParentCo would recognize taxable gain as if it had sold the our common stock in a taxable sale for its fair market value, and ParentCo stockholders who received our shares in the distribution would be subject to tax as if they had received a taxable distribution equal to the fair market value of such shares.

Under the tax matters agreement entered into between ParentCo and us in connection with the separation, we generally are required to indemnify Howmet for any taxes resulting from the separation (and any related costs and other damages) to the extent such amounts resulted from (1) an acquisition of all or a portion of our equity securities or assets, whether by merger or otherwise (and regardless of whether we participated in or otherwise facilitated the acquisition), (2) certain of our other actions or failures to act, or (3) any of our representations, covenants or undertakings contained in the separation agreement and certain other agreements and documents or in any documents relating to the opinion of counsel being incorrect or violated. Any such indemnity obligations could be material. In addition, we, Howmet, and the respective subsidiaries may continue to incur certain tax costs in connection with the separation, including non-U.S. tax costs resulting from transactions (including the internal reorganization) in non-U.S. jurisdictions, which may be material.

Risks Related to Our Common Stock

We cannot be certain that an active trading market for our common stock will be sustained and our stock price may fluctuate significantly.

For many reasons, including the risks identified in this Annual Report on Form 10-K, the market price of our common stock may be volatile. These factors may result in short-term or long-term negative pressure on the value of our common stock. The market price of our common stock may fluctuate significantly due to a number of factors, some of which may be beyond our control, including:

- sales of a significant number of our shares or other shifts in our investor base;
- actual or anticipated fluctuations in our operating results;
- changes in earnings estimated by securities analysts or our ability to meet those estimates;
- the operating and stock price performance of comparable companies;
- changes to the regulatory and legal environment under which we operate;
- actual or anticipated fluctuations in commodities prices; and
- domestic and worldwide economic conditions.

Actions of activist shareholders could have an adverse effect on our business.

Companies across a variety of industries are experiencing an increase in shareholder activism, particularly shareholder proposals regarding ESG and DEI matters. If we are required to respond to shareholder proposals (including the implementation of any proposals), proxy contests or other actions by activist shareholders, we could incur significant expense, disruptions to our operations and diversion of the attention of management and our employees. In addition, perceived uncertainties as to our future direction, strategy or leadership created as a consequence of activist shareholder initiatives may result in reputational damage, which could negatively impact relationships with customers, suppliers and strategic partners, impair our ability to attract and retain employees, and cause volatility in our stock price.

Individual stockholders' percentage of ownership of our common stock may be diluted in the future.

In the future, individual stockholders' percentage of ownership in our common stock may be diluted because of equity issuances for acquisitions, capital market transactions or otherwise. In addition, from time to time, we grant stock-based awards to our directors, officers and employees. Such awards will have a dilutive effect on the number of our shares outstanding, and therefore on our earnings per share, which could adversely affect the market price of our common stock.

We cannot guarantee the timing, amount or payment of dividends on our common stock.

The initiation, timing, declaration, amount and payment of future dividends to our stockholders falls within the discretion of our Board of Directors. The Board of Directors' decisions regarding the payment of dividends depends on many factors, such as our financial condition, earnings, capital requirements, debt service obligations, covenants associated with certain of our debt service obligations, industry practice, legal requirements, regulatory constraints and other factors that our Board of Directors deems relevant. For more information, see Part II, Item 5. "Market for Registrant's Common Equity, Related Stockholder Matters and Issuer Purchases of Equity Securities."

Anti-takeover provisions could enable us to resist a takeover attempt by a third party and limit the power of our stockholders.

Our amended and restated certificate of incorporation and amended and restated bylaws contain, and Delaware law contains, provisions that are intended to deter coercive takeover practices and inadequate takeover bids by making such practices or bids unacceptably expensive to the bidder and to encourage prospective acquirers to negotiate with our Board of Directors rather than to attempt a hostile takeover. These provisions include, among others:

- the ability of our remaining directors to fill vacancies on our Board of Directors that do not arise as a result of removal by stockholders;
- limitations on stockholders' ability to call a special stockholder meeting;
- rules regarding how stockholders may present proposals or nominate directors for election at stockholder meetings; and
- the right of our Board of Directors to issue preferred stock without stockholder approval.

In addition, we are subject to Section 203 of the Delaware General Corporate Law (the "DGCL"), which could have the effect of delaying or preventing a change of control that stockholders may favor. Section 203 provides that, subject to limited exceptions, persons that acquire, or are affiliated with persons that acquire, more than 15% of the outstanding voting stock of a Delaware corporation may not engage in a business combination with that corporation, including by merger, consolidation or acquisitions of additional shares, for a three-year period following the date on which that person or any of its affiliates becomes the holder of more than 15% of the corporation's outstanding voting stock.

We believe these provisions will protect our stockholders from coercive or otherwise unfair takeover tactics by requiring potential acquirers to negotiate with our Board of Directors and by providing our Board of Directors with more time to assess any acquisition proposal. These provisions are not intended to make us immune from takeovers; however, these provisions will apply even if the offer may be considered beneficial by some stockholders and could delay or prevent an acquisition that our Board of Directors determines is not in our best interests and our stockholders' best interests. These provisions may also prevent or discourage attempts to remove and replace incumbent directors.

Our amended and restated certificate of incorporation designates the state courts within the State of Delaware as the sole and exclusive forum for certain types of actions and proceedings that may be initiated by our stockholders, which could discourage lawsuits against us and our directors and officers.

Our amended and restated certificate of incorporation provides that unless the Board of Directors otherwise determines, the state courts within the State of Delaware (or, if no state court located within the State of Delaware has jurisdiction, the federal district court for the District of Delaware) will be the sole and exclusive forum for any derivative action or proceeding brought on behalf of us, any action asserting a claim for or based on a breach of a fiduciary duty owed by any of our current or former directors or officers to us or our stockholders, including a claim alleging the aiding and abetting of such a breach of fiduciary duty, any action asserting a claim against us or any of our current or former directors or officers arising under any provision of the DGCL or our amended and restated certificate of incorporation or amended and restated bylaws, any action asserting a claim relating to or involving us governed by the internal affairs doctrine, or any action asserting an "internal corporate claim" as that term is defined in Section 115 of the DGCL.

To the fullest extent permitted by law, this exclusive forum provision applies to state and federal law claims, including claims under the federal securities laws, including the Securities Act of 1933, as amended ("Securities Act"), and the Exchange Act, although our stockholders will not be deemed to have waived our compliance with the federal securities laws and the rules and regulations thereunder. The enforceability of similar choice of forum provisions in other companies' certificates of incorporation has been challenged in legal proceedings, and it is possible that, in connection with claims arising under federal securities laws or otherwise, a court could find the exclusive forum provision contained in the amended and restated certificate of incorporation to be inapplicable or unenforceable.

This exclusive forum provision may limit the ability of our stockholders to bring a claim in a judicial forum that our stockholders find favorable for disputes with us or our directors or officers, which may discourage such lawsuits against us and our directors and officers. Alternatively, if a court were to find this exclusive forum provision inapplicable to, or unenforceable in respect of, one or more of the specified types of actions or proceedings described above, we may incur additional costs associated with resolving such matters in other jurisdictions, which could negatively affect our business, results of operations and financial condition.

Item 1B. Unresolved Staff Comments.

None.

Item 2. Properties.

As of December 31, 2022, we maintain 20 primary manufacturing facilities, as well as various sales and service facilities, located across North America, Europe, the United Kingdom, and China. Our principal office and corporate center is located at 201 Isabella Street, Suite 400, Pittsburgh, Pennsylvania 15212-5858. The Arconic Technology Center which serves as the

headquarters for our research and development efforts is located at 100 Technical Drive, New Kensington, Pennsylvania 15069-0001.

Arconic believes that its facilities are suitable and adequate for its operations. Although no title examination of properties owned by Arconic has been made for the purpose of this Annual Report on Form 10-K, the Company knows of no material defects in title to any such properties. Arconic leases some of its facilities as provided in Part 1. Item 1. "Business—Description of the Business—Rolled Products—Principal Facilities," "—Building and Construction Systems—Principal Facilities" and "—Extrusions—Principal Facilities." See Note B and Note P to the Consolidated Financial Statements in Part II, Item 8. "Financial Statements and Supplementary Data."

Arconic has active plants and holdings in each of its segments. See Part I. Item 1. "Business—Description of the Business—Rolled Products—Principal Facilities," "—Building and Construction Systems—Principal Facilities" and "—Extrusions—Principal Facilities."

Item 3. Legal Proceedings.

The information set forth inNote T to the Consolidated Financial Statements in Part II, Item 8. "Financial Statements and Supplementary Data" under the caption “Contingencies and Commitments - Contingencies” is incorporated herein by reference.

Item 4. Mine Safety Disclosures.

Not applicable.

PART II

Item 5. Market for Registrant’s Common Equity, Related Stockholder Matters and Issuer Purchases of Equity Securities.

The Company’s common stock is listed on the New York Stock Exchange under the trading symbol “ARNC.”

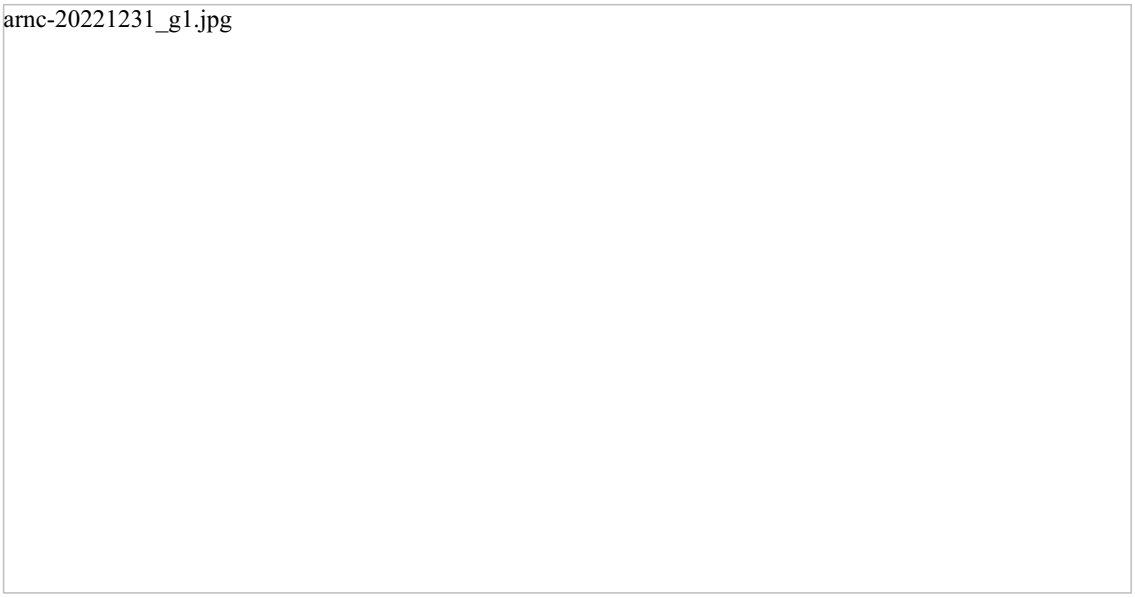
As of February 17, 2023, there were approximately 8,700 holders of record of shares of the Company's common stock. Because many of Arconic’s shares are held by brokers and other institutions on behalf of stockholders, the Company is unable to estimate the total number of stockholders represented by these recordholders.

Arconic did not declare or pay any dividends from April 1, 2020 through December 31, 2022. The timing, declaration, amount of, and payment of any dividends is within the discretion of the Company’s Board of Directors and depends upon many factors, including Arconic’s financial condition, earnings, capital requirements of Arconic’s operating subsidiaries, covenants associated with certain of Arconic’s debt service obligations, legal requirements, regulatory constraints, industry practice, ability to access capital markets, and other factors deemed relevant by the Company’s Board of Directors. Moreover, if Arconic determines to pay any dividends in the future, there can be no assurance that Arconic will continue to pay such dividends or the amount of such dividends.

Stock Performance Graph

The following graph compares the performance of the Company's common stock from market close on April 1, 2020 (beginning of "regular way" trading) to December 31, 2022 with (1) the Standard & Poor's (S&P) 500® Index, and (2) the S&P 1000® Materials Index, a group of 59 companies categorized by S&P as active in the "materials" market sector. The graph assumes, in each case, an initial investment of $100 on April 1, 2020, and the reinvestment of dividends, as applicable. The graph, table and related information shall not be deemed to be "filed" with the SEC, nor shall such information be incorporated by reference into future filings under the Securities Act of 1933 or Securities Exchange Act of 1934, each as amended, except to the extent that the Company specifically incorporates it by reference into such filing.

CUMULATIVE TOTAL RETURN
Based on an initial investment of $100 on
April 1, 2020 with dividends reinvested, as applicable



Source: S&P Capital IQ

	April 1, 2020	December 31, 2020	December 31, 2021	December 31, 2022
Arconic Corporation	$ 100.00	$ 430.64	$ 477.02	$ 305.78
S&P 500® Index	100.00	152.04	192.92	155.41
S&P 1000® Materials Index	100.00	175.00	222.10	210.92

Issuer Purchases of Equity Securities

Period	Total number of shares purchased	Average price paid per share	Total number of shares purchased as part of publicly announced plans or programs*	Approximate dollar value of shares that may yet be purchased under the plans or programs*
October 1 - October 31, 2022	—	$ —	—	$ —
November 1 - November 30, 2022	468,979	$ 21.59	468,979	$ 190,000,000
December 1 - December 31, 2022	1,602,856	$ 22.56	1,602,856	$ 154,000,000
Total for quarter ended December 31, 2022	2,071,835		2,071,835	

* On November 16, 2022, Arconic announced that its Board of Directors approved a share repurchase program authorizing the Company to repurchase up to $200 million of common stock for a two-year period expiring November 17, 2024. Repurchases under the program may be made from time to time, as the Company deems appropriate, based on a variety of factors such as price, capital position, liquidity, financial performance, alternative uses of capital, and overall market conditions. There can be no assurance as to the number of shares the Company will purchase, if any. The share repurchase program may be increased or otherwise modified, renewed, suspended or terminated by the Company at any time, without prior notice. In connection with the establishment of this share repurchase program, the prior $300 million repurchase program, which the Company completed in August 2022, was terminated. This program is intended to comply with Rule 10b5-1 and all purchases shall be made in compliance with Rule 10b-18, including without limitation the timing, price, and volume restrictions thereof.

Item 6. Reserved.

Item 7. Management's Discussion and Analysis of Financial Condition and Results of Operations.

(dollars in millions, except per-share amounts and per-metric ton amounts; shipments in thousands of metric tons (kmt))

Overview

Our Business

Arconic Corporation ("Arconic" or the "Company") is a manufacturer of fabricated aluminum products, including sheet and plate, extrusions, and architectural products and systems, with a primary focus on the ground transportation, aerospace, building and construction, industrial products, and packaging end markets. The Company has 20 primary operating locations in 7 countries around the world, situated in the United States, Canada, China, France, Germany, Hungary, and the United Kingdom. Arconic's previous operations in Russia were divested in November 2022 (see Rolled Products within Segment Information under Results of Operations below).

Management Review of 2022 and Outlook for the Future

Arconic had strong sales growth in 2022 resulting primarily from completion of our re-entry into the North American packaging end market and continued recovery in the aerospace end market, while we navigated challenges associated with our facility in Russia that was ultimately divested in November 2022 and the operational challenges and production outages experienced at the North American rolling mills in the second half of the year. We implemented actions to help offset inflation in alloying materials, energy prices and freight costs through increased pricing. We returned capital to shareholders as we repurchased $185 of shares of the Company's common stock and continued to progress high-return, low-risk organic capital projects in the Rolled Products segment that are expected to drive future EBITDA growth starting in 2023.

In 2022, Sales of $8,961 rose 19% from 2021, reflecting higher aluminum prices and favorable product pricing and mix that drove double digit increases in all of our end markets. The Company recorded a net loss of $182, or $1.75 per share, compared to a net loss of $397, or $3.65 per share, in 2021. The 2022 results included a pre-tax charge of $306 ($304 after-tax) for the loss on sale of the Russian operations and a pre-tax charge of $92 ($70 after-tax) for the impairment of long-lived assets in the Extrusions segment.

We ended the year with a cash balance of $261, and total liquidity of $1,450 (comprised of Cash and cash equivalents of $261 and undrawn availability of $1,189 under the Company's ABL Credit Agreement).

As we look forward to 2023, we anticipate continued growth in packaging volumes and ongoing recovery in the aerospace end-market. Growth in North America is also expected to continue in the ground transportation and building and construction end markets and is anticipated to occur in the industrial products end market, while these end markets in Europe are anticipated to improve later in the year. Based on current internal and external projections of build rates and leading indicators in the markets we serve, our expectations for sales by major end market, excluding the impact of changes in aluminum prices, in 2023 follow. For the ground transportation end market, we expect sales to be relatively flat to an increase of approximately 5% in 2023 compared with 2022, driven by continued strength in North American automotive, offset partly by weakness in European commercial transportation and automotive. For the industrial products end market, we anticipate sales to be relatively flat to an increase of approximately 10% in 2023 compared with 2022, driven by improved operations in North America, as production outages that reduced output in 2022 have been addressed and the Company's rolling mills work through backlog, while European recovery is anticipated in the second half of the year. For the building and construction end market, we anticipate sales to be relatively flat to an increase of approximately 5% in 2023 compared with 2022, as North American non-residential spend is expected to continue driving growth while the current European weakness is expected to improve later in the year. For the packaging end market, we expect sales to increase by approximately 5% to 10% in 2023 compared with 2022, as incremental North American can sheet production is expected to drive growth. For the aerospace end market, we expect sales to grow by approximately 25% to 30% in 2023 compared with 2022, as the market continues to recover, driven by increased aircraft build rates and consumer air travel.

Results of Operations

The discussion and analysis that follows includes a comparison of Arconic's results of operations between the 2022 and 2021 annual periods for both the total company and each reportable segment. Please refer to the Results of Operations section included in Management's Discussion and Analysis of Financial Condition and Results of Operations in Part II Item 7 of the Company's Annual Report on Form 10-K for the year ended December 31, 2021 (filed on February 22, 2022) for a discussion and analysis of Arconic's results of operations between the 2021 and 2020 annual periods for both the total company and each reportable segment.

Earnings Summary

Sales. Sales were $8,961 in 2022 compared to $7,504 in 2021, an increase of $1,457, or 19%. The increase was principally due to favorable impacts from higher aluminum prices, aluminum hedging activities, product pricing, and product mix, all of which were partially offset by the sale of the Company's Russian operations and unfavorable foreign currency movements driven by a weaker euro.

In March 2021, the Company entered into a settlement agreement with a customer related to the terms of an existing long-term contract. As a result, the customer agreed to pay Arconic $18, which was recognized over the applicable three-year period. Accordingly, the Company's sales in 2022 and 2021 included $6 and $12, respectively, associated with this settlement.

Cost of Goods Sold. COGS was $8,032 in 2022 compared to $6,573 in 2021, an increase of $1,459, or 22%. COGS as a percentage of Sales was 89.6% in 2022 compared to 87.6% in 2021. This percentage was negatively impacted by higher costs for direct materials, including alloying materials, energy, and transportation; operational issues and production outages affecting the Rolled Products segment; higher aluminum prices (underlying metal price is contractually passed-through to most customers at cost); an unfavorable impact related to environmental remediation matters (change of $39); and the sale of the Company's Russian operations. These negative impacts were partially offset by favorable product pricing driven by pricing pressures as a result of inflation.

Additionally, on May 14, 2022, the Company and the United Steelworkers reached a tentative four-year labor agreement covering approximately 3,300 employees at four U.S. locations; the previous labor agreement expired on May 15, 2022. The tentative agreement was ratified by the union employees on June 1, 2022. In 2022, Arconic recognized $19 in Cost of goods sold primarily for a one-time signing bonus for the covered employees.

In the 2022 fourth quarter, Arconic recorded both a $61 charge and a $53 benefit in Cost of goods sold to establish a liability for a potential settlement and a receivable for anticipated insurance reimbursement, respectively, related to a litigation matter. See the Reynobond PE matter in the Litigation section of Note T to the Consolidated Financial Statements in Part II Item 8 of this Form 10-K.

Selling, General Administrative, and Other Expenses. SG&A expenses were $246, or 2.7% of Sales, in 2022 compared to $247, or 3.3% of Sales, in 2021. The decrease of $1 was largely attributable to a benefit for an expected insurance reimbursement for legal fees related to a litigation matter and a reduction in stock-based compensation expense including the reversal of expense previously recognized in 2021 related to the performance condition portion of the 2021 performance stock units ($4), partially offset by higher labor costs, selling commission expense, and consulting fees.

Research and Development Expenses. R&D expenses were $37 in 2022 compared to $34 in 2021. The increase was primarily driven by increased costs for contracted services and utilities as a result of inflation.

Provision for Depreciation and Amortization. The provision for depreciation and amortization was $237 in 2022 compared to $253 in 2021. The decrease of $16 was mainly caused by the absence of depreciation resulting from an impairment charge on Properties, plants and equipment in the Extrusions segment taken in the 2022 third quarter (See Extrusions in Segment Information below), the sale of the Company's operations in Russia on November 15, 2022 (See Rolled Products in Segment Information below), and asset retirements.

Impairment of Goodwill. In 2022, after completing its annual review of goodwill for impairment, management determined there was no goodwill impairment for any of the Company's reporting units. In 2021, the Company recognized a charge of $65 in connection with its annual review of goodwill for impairment related to the Extrusions reporting unit. In accordance with Arconic's accounting policy, this review is performed in the fourth quarter each calendar year. Accordingly, this charge was recognized in the fourth quarter of 2021. See Goodwill in Critical Accounting Policies and Estimates below for additional information.

Restructuring and Other Charges. In 2022 and 2021, Restructuring and other charges were comprised of a net charge of $456 and $624, respectively See Note E to the Consolidated Financial Statements in Part II Item 8 of this Form 10-K for additional information.

Interest Expense. Interest expense was $104 in 2022 compared to $100 in 2021. The increase of $4, or 4%, was primarily due to $87 in weighted-average borrowings under the ABL Credit Facility in 2022 (no borrowings occurred in 2021) and a $300 debt offering completed in March 2021, somewhat offset by higher capitalized interest. See Financing Activities under Liquidity and Capital Resources below for additional information related to these financing transactions.

Other Expenses, Net. Other expenses, net was $41 in 2022 compared to $67 in 2021. SeeNote G to the Consolidated Financial Statements in Part II Item 8 of this Form 10-K for additional information.

(Benefit) Provision for Income Taxes. The Company's effective tax rate, including discrete items, was 5.7% (benefit on a loss) in 2022 and 13.5% (benefit on a loss) in 2021. See Note I to the Consolidated Financial Statements in Part II Item 8 of this Form 10-K for a reconciliation of the effective tax rate for each of these years to the U.S. federal statutory rate of 21%.

Segment Information

Arconic's operations consist of three reportable segments: Rolled Products, Building and Construction Systems, and Extrusions. Segment performance under the Company's management reporting system is evaluated based on several factors; however, the primary measure of performance is Segment Adjusted EBITDA (Earnings before interest, taxes, depreciation, and amortization).

The Company calculates Segment Adjusted EBITDA as Total sales (third-party and intersegment) minus each of (i) Cost of goods sold, (ii) Selling, general administrative, and other expenses, and (iii) and Research and development expenses, plus each of (i) Stock-based compensation expense, (ii) Metal price lag, and (iii) Unrealized (gains) losses on mark-to-market hedging instruments and derivatives (see below). Arconic's Segment Adjusted EBITDA may not be comparable to similarly titled measures of other companies' reportable segments.

Effective in the first quarter of 2022, management modified the Company's definition of Segment Adjusted EBITDA to exclude the impact of unrealized gains and losses on mark-to-market hedging instruments and derivatives. This modification was deemed appropriate as Arconic is considering entering into additional hedging instruments in future reporting periods if favorable conditions exist to mitigate cost inflation. Certain of these instruments may not qualify for hedge accounting resulting in unrealized gains and losses being recorded directly to Sales or Cost of goods sold, as appropriate (i.e., mark-to-market). Additionally, this change was also applied to derivatives that do not qualify for hedge accounting for consistency purposes. The Company does not have a regular practice of entering into contracts that are treated as derivatives for accounting purposes. Ultimately, this change was made to maintain the transparency and visibility of the underlying operating performance of Arconic's reportable segments. Prior to this change, the Company had a limited number of hedging instruments and derivatives that did not qualify for hedge accounting, the unrealized impact of which was not material to Arconic's Segment Adjusted EBITDA performance measure. Accordingly, prior period information presented was not recast to reflect this change.

Segment Adjusted EBITDA for all reportable segments totaled $729 in 2022, $757 in 2021, and $648 in 2020. The following information provides Sales and Segment Adjusted EBITDA for each reportable segment for each of the three years in the period ended December 31, 2022. See Note D to the Consolidated Financial Statements in Part II Item 8 of this Form 10-K for additional information.

Rolled Products

	2022	2021	2020
Third-party sales	$ 7,313	$ 6,187	$ 4,335
Intersegment sales	44	33	19
Total sales	$ 7,357	$ 6,220	$ 4,354
Segment Adjusted EBITDA	$ 581	$ 655	$ 527
Third-party aluminum shipments (kmt)	1,372	1,404	1,179

Overview. The Rolled Products segment produces aluminum sheet and plate for a variety of end markets. Sheet and plate are sold directly to customers and through distributors related to the aerospace, ground transportation, packaging, building and construction, and industrial products (mainly used in the production of machinery and equipment and consumer durables) end markets. While the customer base for flat-rolled products is large, a significant amount of sales of sheet and plate is to a relatively small number of customers. Prices for these products are generally based on the price of metal plus a premium for adding value to the aluminum to produce a semi-finished product, resulting in a business model in which the underlying price of metal is contractually passed-through to customers. Generally, the sales and costs and expenses of this segment are transacted in the local currency of the respective operations, which are the U.S. dollar and, to a lesser extent, each of the following: the Russian ruble (see below), the Chinese yuan, the euro, and the British pound.

On November 15, 2022, the Company completed the sale of all of its operations in Russia to Promishlennie Investitsii LLC, the majority owner of VSMPO-AVISMA Corporation. Arconic's former operations in Russia were comprised of one principal location in Samara, which manufactured sheet, plate, extrusions, and forgings across all of the Company's end markets. The Samara facility generated third-party sales of $903, $968, and $705, in 2022, 2021, and 2020, respectively, and, at the time of divestiture, had approximately 2,900 employees. See Note S to the Consolidated Financial Statements in Part II Item 8 of this Form 10-K for additional information.

Sales. Third-party sales for the Rolled Products segment increased $1,126, or 18%, in 2022 compared to 2021. The improvement was largely attributable to higher aluminum prices (see below), price increases for the pass-through of certain inflation impacts, favorable product mix, and favorable impacts from aluminum hedging activities partially offset by the divestiture of the Company's Russian operations, unfavorable foreign currency movements due to a weaker euro, and a net decrease in volumes (see below).

The higher aluminum prices were mostly driven by a 9% rise in the average LME aluminum price and increases in regional premiums, including a 13% improvement in the average Midwest premium (United States).

The net decrease in volumes was negatively impacted by the absence of shipments from November 15, 2022 through December 31, 2022 due to the sale of this segment's former operations in Russia (see Overview above). Otherwise, the overall net volume for this segment slightly increased in 2022 compared to 2021, largely driven by improvements in the packaging, aerospace, and automotive component of ground transportation end markets. Volume related to the packaging end market increased significantly as the can sheet operation at the Tennessee rolling mill essentially reached full capacity by the end of the 2022 second quarter. Higher volume associated with the aerospace end market was driven by continued recovery from the COVID-19 pandemic. An improvement in volume for the automotive component of ground transportation was due to slow recovery from the global semiconductor chip shortage. These higher volumes were mostly offset by declines in the industrial products, building and construction, and the commercial transportation component of ground transportation end markets, which were impacted by supply chain disruptions. Additionally, the Rolled Products segment experienced operational challenges and production outages associated with electrical and mechanical issues at the Tennessee and Lancaster rolling mills and disruptions in both the Tennessee and Davenport casting operations.

In March 2021, the Company entered into a settlement agreement with a customer related to the terms of an existing long-term contract. As a result, the customer agreed to pay Arconic $18, which was recognized over the applicable three-year period. Accordingly, in 2022 and 2021, Rolled Products' sales included $6 and $12, respectively, associated with this settlement.

Segment Adjusted EBITDA. Segment Adjusted EBITDA for the Rolled Products segment decreased $74, or 11%, in 2022 compared to 2021. The decline was primarily due to higher costs for alloying materials, energy, transportation, and maintenance all due to inflation, increased expenses for labor as this segment increases its workforce to address current and future volume growth, the impact of the previously mentioned operational challenges and production outages, and the divestiture of the Company's Russian operations. These higher costs were partially offset by customer price increases due to adjustments for inflation impacts and a benefit derived from the absence of below normal absorption of fixed costs that occurred in the 2021 first quarter.

Building and Construction Systems

	2022	2021	2020
Third-party sales	$ 1,245	$ 1,011	$ 963
Segment Adjusted EBITDA	$ 195	$ 130	$ 137

Overview. The Building and Construction Systems segment manufactures products that are used primarily in the non-residential building and construction end market. These products include integrated aluminum architectural systems and architectural extrusions, which are sold directly to customers and through distributors. A limited amount of this segment's product sales is directly impacted by metal pricing, which is a pass-through to the related customers. Generally, the sales and costs and expenses of this segment are transacted in the local currency of the respective operations, which are the U.S. dollar and, to a lesser extent, each of the following: the euro, the British pound, and Canadian dollar.

The Company is evaluating strategic options for the businesses that comprise the Building and Construction Systems segment. SeeNote S to the Consolidated Financial Statements in Part II Item 8 of this Form 10-K for additional information.

Sales. Third-party sales for the Building and Construction Systems segment increased $234, or 23%, in 2022 compared to 2021. The improvement was mostly due to multiple product price increases implemented since March 2021 across the entire portfolio to address inflationary cost pressures. Additionally, a positive impact from higher aluminum prices was virtually offset by a negative impact from unfavorable foreign currency movements due to a weaker euro.

Segment Adjusted EBITDA. Segment Adjusted EBITDA for the Building and Construction Systems segment increased $65, or 50%, in 2022 compared to 2021. The improvement was principally driven by favorable product pricing, partially offset by higher costs for aluminum, alloying materials, maintenance, and transportation, all due to inflation.

Extrusions

	2022	2021	2020
Third-party sales	$ 409	$ 306	$ 381
Segment Adjusted EBITDA	$ (47)	$ (28)	$ (16)
Third-party aluminum shipments (kmt)	38	35	40

Overview. The Extrusions segment produces a range of extruded and machined parts for the aerospace, ground transportation, and industrial products end markets. These products are sold directly to customers and through distributors. Prices for these products are generally based on the price of metal plus a premium for adding value to the aluminum to produce a semi-finished product, resulting in a business model in which the underlying price of metal is contractually passed-through to customers. Generally, the sales and costs and expenses of this segment are transacted in the local currency of the respective operations, which are the U.S. dollar and, to a lesser extent, the euro.

In the 2022 third quarter, management initiated a business review of the Extrusions segment aimed at identifying alternatives to improve the financial performance of this segment in future periods. Management continues to assess alternatives and no decisions or commitments were made as of December 31, 2022. In connection with this review, the Company updated its five-year strategic plan, the results of which indicated that there is an expected decline in the forecasted financial performance for the Extrusions segment (and asset group), including continued forecasted losses. As such, management evaluated the recoverability of the long-lived assets of the Extrusions asset group by comparing the aggregate carrying value to the undiscounted future cash flows. The result of this evaluation was that the long-lived assets were deemed to be impaired as the aggregate carrying value exceeded the undiscounted future cash flows. The impairment charge was measured as the difference between the aggregate carrying value and aggregate fair value of the long-lived assets. Fair value was determined using an orderly liquidation methodology for the machinery and equipment and a sales comparison approach for the land and structures. Significant judgments and assumptions were applied in estimating the fair value of the long-lived assets, including the use of market-based information specific to the machinery and equipment and information from recent sales or current listings of comparable properties for the land and structures. As a result, the Company recorded an impairment charge of $92, composed of $90 for Properties, plants, and equipment and $2 for intangible assets (included within Other noncurrent assets), in Restructuring and other charges on the Company's Statement of Consolidated Operations. See Note E to the Consolidated Financial Statements in Part II Item 8 of this Form 10-K for additional information.

In the 2021 first quarter, management approved the idling of the casthouse at the Lafayette (Indiana) plant. Additionally, in the 2021 second quarter, management approved the idling of the remaining operations (primarily small presses) at the Chandler

(Arizona) plant. These actions were deemed necessary to address the then-depressed demand related to the aerospace end market and identified operational inefficiencies in the Extrusions portfolio. The Lafayette casthouse action is near completion. Ultimately, the casthouse function will be fully outsourced to third-party vendors. The Chandler action was completed in the 2021 third quarter. The commercial operations related to Chandler were integrated into Lafayette for the foreseeable future. In 2021, the Company incurred certain charges related to these decisions such as inventory write-downs, severance costs, and customer qualification costs. These items were not material, individually or in the aggregate, and were reported in Corporate (i.e., not included in Extrusion's Segment Adjusted EBITDA). The Company may temporarily restart one or more of the presses at Chandler over short periods of time to address customer demand requirements.

Sales. Third-party sales for the Extrusions segment increased $103, or 34%, in 2022 compared to 2021. The improvement was largely attributable to favorable product mix, primarily due to higher volumes for aerospace; price increases due to adjustments for inflation impacts and higher aluminum prices; and a net increase in volumes, mostly driven by aerospace due to the lessened impact of the COVID-19 pandemic.

Segment Adjusted EBITDA. Segment Adjusted EBITDA for the Extrusions segment decreased $19, or 68%, in 2022 compared to 2021. The decline was mostly related to higher costs for aluminum, maintenance, energy, transportation, and outside services. The higher costs were the result of both increased pricing due to inflation and usage due to operational issues. These negative impacts were mostly offset by customer price increases due to adjustments for inflation impacts. Overall, operational issues are driving underperformance in this segment.

Reconciliation of Total Segment Adjusted EBITDA to Consolidated Net Loss Attributable to Arconic Corporation

For the year ended December 31,	2022	2021	2020
Total Segment Adjusted EBITDA	$ 729	$ 757	$ 648
Unallocated amounts:			
Corporate expenses[1]	(29)	(33)	(24)
Stock-based compensation expense	(15)	(22)	(23)
Metal price lag[2]	17	(16)	(27)
Unrealized gains on mark-to-market hedging instruments and derivatives	6	—	—
Provision for depreciation and amortization	(237)	(253)	(251)
Impairment of goodwill	—	(65)	—
Restructuring and other charges[3],[4]	(456)	(624)	(188)
Other[5]	(62)	(36)	(55)
Operating (loss) income	(47)	(292)	80
Interest expense	(104)	(100)	(118)
Other expenses, net	(41)	(67)	(70)
Benefit (Provision) for income taxes	11	62	(1)
Net income attributable to noncontrolling interest	(1)	—	—
Consolidated net loss attributable to Arconic[1]	$ (182)	$ (397)	$ (109)

(1) Corporate expenses are composed of general administrative and other expenses of operating the corporate headquarters and other global administrative facilities. The amounts presented for all periods prior to second quarter 2020 include an allocation of ParentCo's corporate expenses, including research and development expenses, for the portion of the period prior to the Separation Date.

(2) Metal price lag represents the financial impact of the timing difference between when aluminum prices included in Sales are recognized and when aluminum purchase prices included in Cost of goods sold are realized. This adjustment aims to remove the effect of the volatility in metal prices and the calculation of this impact considers applicable metal hedging transactions.

(3) In 2022, Restructuring and other charges includes a loss of$306 related to the sale of the Company's operations in Russia and a $92 asset impairment charge related to the Extrusions segment. See Note E to the Consolidated Financial Statements in Part II Item 8 of this Form 10-K for additional information.

(4) In 2022, 2021, and 2020, Restructuring and other charges includes a $47, $584, and $199, respectively, charge for the settlement of certain employee retirement benefits virtually all of which were within the United States and the United

Kingdom. See Note H to the Consolidated Financial Statements in Part II Item 8 of this Form 10-K for additional information.

(5) Other includes certain items that impact Cost of goods sold and Selling, general administrative, and other expenses on the Company's Statement of Consolidated Operations that are not included in Segment Adjusted EBITDA. In 2022, Other includes costs related to environmental remediation charges of $27 (see Environmental Matters in Note T to the Consolidated Financial Statements in Part II Item 8 of this Form 10-K for additional information) and costs related to the new labor agreement of $19 (see Note H to the Consolidated Financial Statements in Part II Item 8 of this Form 10-K for additional information). These charges were recorded in Costs of goods sold on Arconic's Statement of Consolidated Operations.

Environmental Matters

See the Environmental Matters section of Note T to the Consolidated Financial Statements in Part II Item 8 of this Form 10-K.

Liquidity and Capital Resources

Arconic's primary future cash needs are centered on operating activities, including working capital, as well as recurring and strategic capital expenditures. The Company's ability to fund its cash needs depends on its ongoing ability to generate and raise cash in the future. Although management believes that Arconic's future cash from operations, together with the Company's access to capital markets, will provide adequate resources to fund Arconic's operating and investing needs, the Company's access to, and the availability of, financing on acceptable terms in the future will be affected by many factors, including: (i) Arconic's credit rating; (ii) the liquidity of the overall capital markets; and (iii) the current state of the economy. There can be no assurances that the Company will continue to have access to capital markets on terms acceptable to Arconic.

Cash provided from operations and financing activities is expected to be adequate to cover the Company's operational and business needs over the next 12 months. For an analysis of long-term liquidity, see Contractual Obligations and Off-Balance Sheet Arrangements below.

At December 31, 2022, the Company's Cash and cash equivalents were $261, of which $110 was held outside the United States. Arconic has a number of commitments and obligations related to the Company's operations in various foreign jurisdictions, resulting in the need for cash outside the United States. Management continuously evaluates the Company's local and global cash needs for future business operations, which may influence future repatriation decisions.

The discussion and analysis that follows includes a comparison of Arconic's cash flows between the 2022 and 2021 annual periods. Please refer to the Liquidity and Capital Resources section included in Management's Discussion and Analysis of Financial Condition and Results of Operations in Part II Item 7 of the Company's Annual Report on Form 10-K for the year ended December 31, 2021 (filed on February 22, 2022) for a discussion and analysis of Arconic's cash flows between the 2021 and 2020 annual periods.

Operating Activities

Cash provided from operations was $338 in 2022 compared with cash used for operations of $407 in 2021.

In 2022, cash provided from operations was mostly comprised of a positive add-back for non-cash transactions in earnings of $753, partially offset by an unfavorable change in working capital of $222 and a net loss of $181. The change in working capital was largely driven by inventory build due to operational disruptions at the Company's North American rolling facilities.

In 2021, cash used for operations was comprised of an unfavorable change in working capital of $505, pension contributions of $458 (see below), and a net loss of $397, partially offset by a positive add-back for non-cash transactions in earnings of $953. The impact in working capital was largely driven by higher volumes and higher aluminum prices due to an increase in both the average LME price and regional premiums.

In January 2021, the Company contributed a total of $200 to its two funded U.S. defined benefit pension plans, comprised of the estimated minimum required funding for 2021 of $183 and an additional $17. Additionally, in April 2021, the Company contributed a total of $250 to its two funded U.S. defined benefit pension plans to maintain the funding level of the remaining plan obligations not transferred under a group annuity contract (see U.S. Pension Plan Annuitizations in Note H to the Consolidated Financial Statements in Part II Item 8 of this Form 10-K for additional information). This contribution was funded with the net proceeds from a March 2021 debt offering (see 2021 Debt Activity in Financing Activities below).

In addition, in 2022 and 2021, the Company received $6 and $12, respectively, related to a settlement agreement reached with a customer in March 2021. See Sales under Results of Operations above for additional information.

Financing Activities

Cash used for financing activities was $196 in 2022 compared with cash provided from financing activities of $135 in 2021. The use of cash in 2022 was mostly due to $185 for the repurchase of 6,935,507 shares of the Company's common stock (see Share Repurchase Programs below). In 2021, the source of cash was due to $314 in net proceeds from the issuance of new indebtedness (see 2021 Debt Activity below), somewhat offset by $161 for the repurchase of 4,912,505 shares of the Company's common stock (see Share Repurchase Programs below).

2022 Activity—On February 16, 2022, the Company amended its five-year credit agreement, dated May 13, 2020, with a syndicate of lenders named therein and Deutsche Bank AG New York Branch as administrative agent (the "ABL Credit Agreement") which provides for a senior secured asset-based revolving credit facility (the "ABL Credit Facility") to be used, generally, for working capital or other general corporate purposes. The ABL Credit Agreement was amended to increase the revolving commitments under the ABL Credit Facility to $1,200 from $800. Additionally, the accordion feature of the ABL Credit Facility was revised to provide for the Company to request a further increase to the revolving commitments in an aggregate principal amount equal to the greater of $350 and the excess of the borrowing base over the ABL Credit Facility commitments. Furthermore, the LIBOR-based floating interest rate was replaced with a term SOFR-based interest rate, plus a credit spread adjustment equal to 0.10%, 0.15%, or 0.25% per annum for SOFR-based borrowings with interest periods of one month, three months, or six months, respectively, under the ABL Credit Facility. Arconic paid $1 in upfront costs associated with these amendments.

In 2022, the Company borrowed $275 and repaid $275 under the ABL Credit Facility. These borrowings were designated as SOFR loans with either an initial one-month or three-month interest period. In 2022, the weighted-average interest rate and weighted-average days outstanding of the borrowings was 4.3% and 90 days, respectively.

2021 Debt Activity—On March 3, 2021, the Company completed a Rule 144A (U.S. Securities Act of 1933, as amended) debt offering for an additional $300 aggregate principal amount of 6.125% Senior Secured Second-Lien Notes due 2028 (the "Additional 2028 Notes"). The Additional 2028 Notes were issued under the indenture governing Arconic's existing 6.125% Senior Secured Second-Lien Notes due 2028. Other than with respect to the date of issuance and issue price, the Additional 2028 Notes are treated as a single series with and have the same terms as the referenced existing notes. The Additional 2028 Notes were sold at 106.25% of par (i.e., a premium) and, after reflecting a discount to the initial purchasers of the Additional 2028 Notes, the Company received $315 in net proceeds from the debt offering. Arconic used the net proceeds of this issuance to fund an annuitization of certain U.S. defined benefit pension plan obligations (see Note H to the Consolidated Financial Statements in Part II Item 8 of this Form 10-K for additional information). The premium ($19) and costs to complete the financing ($5) were deferred and are being amortized to interest expense over the term of the Additional 2028 Notes. The amortization of the premium is reflected as a reduction to interest expense and the amortization of the costs to complete the financing is reflected as an addition to interest expense. Interest on the Additional 2028 Notes is paid semi-annually in February and August, and commenced August 15, 2021.

Descriptions of the 2028 Notes, 2025 Notes, and ABL Credit Agreement are set forth inNote Q to the Consolidated Financial Statements in Part II Item 8 of this Form 10-K.

Share Repurchase Programs—On May 4, 2021, Arconic announced that its Board of Directors approved a share repurchase program authorizing the Company to repurchase shares of its outstanding common stock up to an aggregate transactional value of $300 over a two-year period expiring April 28, 2023. In 2022 and 2021, Arconic repurchased 4,863,672 and 4,912,505 shares, respectively, of the Company's common stock for $139 and $161, respectively, resulting in the completion of the total authorization under this program in August 2022. In connection with the establishment of a new repurchase program (see below), this repurchase program was terminated.

On November 16, 2022, Arconic announced that its Board of Directors approved a share repurchase program authorizing the Company to repurchase shares of its outstanding common stock up to an aggregate transactional value of $200 over a two-year period expiring November 17, 2024. In 2022, Arconic repurchased 2,071,835 shares of the Company's common stock for $46 under this program. See Note K to the Consolidated Financial Statements in Part II Item 8 of this Form 10-K for additional information.

Ratings—Arconic's cost of borrowing and ability to access the capital markets are affected not only by market conditions but also by the ratings assigned to Arconic and its debt by the major credit rating agencies. As of December 31, 2022, the following are the most recent ratings for Arconic and its outstanding debt.

Moody's Investor Service (Moody's) has assigned a Ba3 rating to both the Company and the 2028 Notes and a Ba1 rating to the 2025 Notes. Additionally, Moody's has given these ratings a stable outlook.

Standard and Poor's Global Ratings (S&P) has assigned a BB rating to the Company, a B+ rating to the 2028 Notes, and a BB+ rating to the 2025 Notes. Additionally, S&P has given these ratings a stable outlook.

Fitch Ratings (Fitch) has assigned a BB+ rating to both the Company and the 2028 Notes and a BBB- rating to both the 2025 Notes and the ABL Credit Facility. On September 9, 2022, Fitch revised the rating outlook from stable to positive.

Investing Activities

Cash used for investing activities was $214 in 2022 compared with $181 in 2021.

The use of cash in 2022 reflects capital expenditures of $245, more than half of which was attributable to sustaining spend at the U.S. rolling mills and the remaining amount due to growth spend at the Davenport and Lancaster mills, slightly offset by $27 in net proceeds from the sale of the Company's operations in Russia (see Note S to the Consolidated Financial Statements in Part II Item 8 of this Form 10-K).

The use of cash in 2021 reflects capital expenditures of $184, more than half of which was attributable to sustaining spend at the U.S. rolling mills.

Contractual Obligations and Off-Balance Sheet Arrangements

Contractual Obligations. Arconic is required to make future payments under various contracts, including long-term purchase obligations, lease agreements, and financing arrangements. The Company also has commitments to make contributions to its funded pension plans, provide payments for pension (unfunded) and other postretirement benefit plans, and fund capital projects. As of December 31, 2022, a summary of Arconic's outstanding contractual obligations is as follows (these contractual obligations are grouped in the same manner as they are classified in the Statement of Consolidated Cash Flows in order to provide a better understanding of the nature of the obligations and to provide a basis for comparison to historical information):

	Total	2023	2024-2025	2026-2027	Thereafter
Operating activities:					
Raw material purchase obligations*	$ 966	$ 880	$ 86	$ —	$ —
Energy-related purchase obligations	83	15	29	22	17
Other purchase obligations	37	14	10	6	7
Operating leases	137	40	55	23	19
Interest related to debt	408	97	173	110	28
Pension contributions - funded plans	367	38	148	181	—
Pension benefit payments - unfunded plans	71	7	14	14	36
Other postretirement benefit payments	261	27	54	53	127
Layoff and other restructuring payments	1	1	—	—	—
Uncertain tax positions	—	—	—	—	—
Financing activities:					
Debt	1,600	—	700	—	900
Dividends to stockholders	—	—	—	—	—
Investing activities:					
Capital projects	208	154	51	3	—
Totals	$ 4,139	$ 1,273	$ 1,320	$ 412	$ 1,134

* Subsequent to December 31, 2022, the Company entered into additional aluminum supply contracts totaling approximately $765, of which approximately $375, $195, and $195 is expected to be purchased in 2023, 2024, and 2025, respectively. Additionally, through February 20, 2023, Arconic is currently negotiating terms for further aluminum supply contracts totaling approximately $2,580.

Obligations for Operating Activities

Raw material purchase obligations consist mostly of aluminum with expiration dates ranging from less than one year to two years. Energy-related purchase obligations consist primarily of electricity and natural gas contracts with expiration dates ranging from one year to five years. Many of these purchase obligations contain variable pricing components, and, as a result, actual cash payments may differ from the estimates provided in the preceding table.

Operating leases represent multi-year obligations for certain land and buildings, plant equipment, vehicles, and computer equipment.

Interest related to debt is based on stated interest rates on debt with maturities that extend to 2028 (see the 2022 Debt Activity and 2021 Debt Activity sections of Financing Activities under Liquidity and Capital Resources above).

Pension contributions (funded plans) and pension (unfunded plans) and other postretirement benefit payments are based on actuarial estimates using current assumptions for, among others, discount rates, long-term rate of return on plan assets, rate of compensation increases, and/or health care cost trend rates. It is Arconic's policy to contribute amounts to funded pension plans sufficient to meet the minimum requirements set forth in applicable country employee benefit and tax regulations, including ERISA for U.S. plans. Management has determined that it is not practicable to present pension contributions (funded plans) and both pension (unfunded plans) and other postretirement benefit payments beyond 2027 and 2032, respectively.

Layoff and other restructuring payments primarily relate to severance costs.

Uncertain tax positions taken or expected to be taken on an income tax return may result in additional payments to tax authorities. As of December 31, 2022, there was no balance of uncertain tax positions and as such, no related interest and penalties were accrued. If a tax authority agrees with the tax position taken or expected to be taken or the applicable statute of limitations expires, then additional payments will not be necessary.

Obligations for Financing Activities

The debt amount in the preceding table represents the principal amounts of all outstanding long-term debt, which have maturities that extend to 2028 (see the 2022 Debt Activity and 2021 Debt Activity sections of Financing Activities under Liquidity and Capital Resources above).

As of December 31, 2022, Arconic had 99,432,194 outstanding shares of common stock. Dividends on common stock are subject to authorization by the Company's Board of Directors. Arconic did not declare or pay any dividends from the Separation Date through December 31, 2022.

On November 16, 2022, Arconic announced that its Board of Directors approved a share repurchase program authorizing the Company to repurchase shares of its outstanding common stock up to an aggregate transactional value of $200, depending on cash availability, market conditions, and other factors. This program has a predetermined expiration date of November 17, 2024; however, the Company is under no contractual obligation to make such repurchases. Accordingly, amounts have not been included in the preceding table. In 2022, the Company repurchased 2,071,835 shares of its common stock for $46 under this program. See the Common Stock section of Note K to the Consolidated Financial Statements in Part II Item 8 of this Form 10-K.

Obligations for Investing Activities

Capital projects in the preceding table only include amounts approved by management as of December 31, 2022. Funding levels may vary in future years based on anticipated construction schedules of the projects. It is expected that significant expansion projects will be funded through various sources, including cash provided from operations. Total capital expenditures are anticipated to be approximately $275 in 2023.

Off-Balance Sheet Arrangements. Arconic Corporation has outstanding bank guarantees, letters of credit, and surety bonds related to outstanding responsibilities and obligations. See the Commitments section of Note T to the Consolidated Financial Statements in Part II Item 8 of this Form 10-K.

Critical Accounting Policies and Estimates

The preparation of Arconic's Consolidated Financial Statements in accordance with accounting principles generally accepted in the United States of America requires management to make certain estimates based on judgments and assumptions regarding uncertainties that may affect the amounts reported in the Consolidated Financial Statements and disclosed in the Notes to the Consolidated Financial Statements. Areas that require such estimates include the review of properties, plants, and

equipment and goodwill for impairment, and accounting for each of the following: environmental and litigation matters; pension and other postretirement employee benefit obligations; stock-based compensation; and income taxes.

Management uses historical experience and all available information to make these estimates, and actual results may differ from those used to prepare the Company's Consolidated Financial Statements at any given time. Despite these inherent limitations, management believes that Management's Discussion and Analysis of Financial Condition and Results of Operations and the Consolidated Financial Statements, including the Notes to the Consolidated Financial Statements, provide a meaningful and fair representation of the Company.

A summary of Arconic's significant accounting policies is included inNote B to the Consolidated Financial Statements in Part II Item 8 of this Form 10-K. Management believes that the application of these policies on a consistent basis enables the Company to provide the users of the Consolidated Financial Statements with useful and reliable information about Arconic's operating results and financial condition.

Prior to the Separation Date, the Company did not operate as a separate, standalone entity. Arconic's operations were included in ParentCo's financial results. Accordingly, in all periods prior to the Separation Date, the Company's Consolidated Financial Statements were prepared from ParentCo's historical accounting records and were presented on a standalone basis as if the Arconic Corporation Businesses had been conducted independently from ParentCo. Such Consolidated Financial Statements include the historical operations that were considered to comprise the Arconic Corporation Businesses, as well as certain assets and liabilities that were historically held at ParentCo's corporate level but were specifically identifiable or otherwise attributable to the Arconic Corporation Businesses. The Critical Accounting Policies described below reflect any incremental judgments and assumptions made by management in the preparation of the Company's Consolidated Financial Statements prior to the Separation Date.

Properties, Plants, and Equipment. Properties, plants, and equipment are reviewed for impairment whenever events or changes in circumstances indicate that the carrying value of such assets may not be recoverable. Recoverability of assets is determined by comparing the estimated undiscounted net cash flows of the related operations (asset group) to the carrying value of the associated assets. An impairment loss would be recognized when the carrying value of the assets exceeds the estimated undiscounted net cash flows of the asset group. The amount of the impairment loss to be recorded is calculated as the excess of the carrying value of the assets over their fair value, with fair value determined using the best information available, which generally is a discounted cash flow (DCF) model. In certain circumstances, Arconic may use the services of a third-party to assist with other valuation techniques. The determination of what constitutes an asset group, the associated estimated undiscounted net cash flows, and the estimated useful lives of the assets also require significant judgments.

Goodwill. Goodwill is not amortized; it is reviewed for impairment annually (in the fourth quarter) or more frequently if indicators of impairment exist or if a decision is made to sell, exit, or realign a business. A significant amount of judgment is involved in determining if an indicator of impairment has occurred. Such indicators may include, among others, deterioration in general economic conditions, negative developments in equity and credit markets, adverse changes in the markets in which an entity operates, increases in input costs that have a negative effect on earnings and cash flows, or a trend of negative or declining cash flows over multiple periods. The fair value that could be realized in an actual transaction may differ from that used to evaluate goodwill for impairment.

Goodwill is allocated among and evaluated for impairment at the reporting unit level, which is defined as an operating segment or one level below an operating segment. Arconic has three reporting units, the Rolled Products segment, the Building and Construction Systems segment, and the Extrusions segment (full impairment of goodwill in 2021 – see below). At the time of the Company's annual 2022 review of goodwill for impairment, the carrying value of goodwill for Rolled Products and Building and Construction Systems was $232 and $67, respectively.

In reviewing goodwill for impairment, an entity has the option to first assess qualitative factors to determine whether the existence of events or circumstances leads to a determination that it is more likely than not (greater than 50%) that the estimated fair value of a reporting unit is less than its carrying amount. If an entity elects to perform a qualitative assessment and determines that an impairment is more likely than not, the entity is then required to perform a quantitative impairment test (described below), otherwise no further analysis is required. An entity also may elect not to perform the qualitative assessment and, instead, proceed directly to the quantitative impairment test. The ultimate outcome of the goodwill impairment review for a reporting unit should be the same whether an entity chooses to perform the qualitative assessment or proceeds directly to the quantitative impairment test.

Arconic determines annually, based on facts and circumstances, which of its reporting units will be subject to the qualitative assessment. For those reporting units where a qualitative assessment is either not performed or for which the conclusion is that an impairment is more likely than not, a quantitative impairment test will be performed. The Company's policy is that a quantitative impairment test be performed for each reporting unit at least once during every three-year period.

Under the qualitative assessment, various events and circumstances (or factors) that would affect the estimated fair value of a reporting unit are identified (similar to impairment indicators above). These factors are then classified by the type of impact they would have on the estimated fair value using positive, neutral, and adverse categories based on current business conditions. Additionally, an assessment of the level of impact that a particular factor would have on the estimated fair value is determined using high, medium, and low weighting. Furthermore, management considers the results of the most recent quantitative impairment test completed for a reporting unit and compares the weighted average cost of capital (WACC) between the current and prior years for each reporting unit.

During the 2022 annual review of goodwill for impairment, management performed the qualitative assessment for the Building and Construction Systems reporting unit. Management concluded it was not more likely than not that the estimated fair value of this reporting unit was less than the carrying value. As such, no further analysis was required. Management last proceeded directly to the quantitative impairment test for the Building and Construction Systems reporting unit in 2021. At that time, the estimated fair value of this reporting unit was substantially in excess of its carrying value, resulting in no impairment.

Under the quantitative impairment test, the evaluation of the recoverability of goodwill involves comparing the current fair value of each reporting unit to its carrying value, including goodwill. Arconic uses a DCF model to estimate the current fair value of its reporting units when testing for impairment, as management believes forecasted cash flows are the best indicator of such fair value. Several significant assumptions and estimates are involved in the application of the DCF model to forecast operating cash flows, including sales growth (volumes and pricing), production costs, capital spending, working capital levels, and discount rate. Certain of these assumptions may vary significantly among the reporting units. Cash flow forecasts are generally based on approved business unit five-year operating plans and a terminal value. The WACC rate for the individual reporting units is estimated by management with the assistance of valuation experts. In the event the estimated fair value of a reporting unit per the DCF model is less than the carrying value, the Company would recognize an impairment charge equal to the excess of the reporting unit's carrying value over its fair value without exceeding the total amount of goodwill applicable to that reporting unit.

During the 2022 annual review of goodwill for impairment, management proceeded directly to the quantitative impairment test for the Rolled Products reporting unit. The result of this review indicated the estimated fair value of this reporting unit was substantially in excess of the carrying value, resulting in no impairment.

The respective annual review of goodwill for impairment in 2021 and 2020 indicated that goodwill was not impaired for any of Arconic's reporting units, except for the Extrusions segment (see below), and there were no triggering events since that time that necessitated an interim quantitative impairment test for any of the Company's reporting units. That said, in light of the economic impact of the COVID-19 pandemic (i.e., stock market volatility, customer demand disruption, etc.), Arconic did perform periodic qualitative assessments throughout 2020 in which management concluded that no impairment existed and, therefore, no further analysis was required.

In 2021, the estimated fair value of the Extrusions reporting unit was lower than the associated carrying value by an amount greater than the carrying value of the Extrusions reporting unit's goodwill. Accordingly, the Company recorded an impairment charge of $65 in the fourth quarter of 2021. As a result, there is no goodwill remaining for the Extrusions reporting unit.

The impairment of the Extrusion reporting unit's goodwill resulted from a combination of market-based factors, including continued delays in aerospace market improvement, the expectation that significant cost inflation will extend beyond 2021 resulting in increasingly limited ability for management to drive margin expansion, and a longer than previously anticipated shift in product mix to lower margin industrial products to replace most of the lost aerospace volume. Further, current market factors also resulted in a 150-basis point increase in the WACC compared to the fourth quarter of 2020. Accordingly, given these factors, the estimated fair value of the Extrusions reporting unit was less than the carrying value resulting in a full impairment of its goodwill.

Environmental Matters. Expenditures for current operations are expensed or capitalized, as appropriate. Expenditures relating to existing conditions caused by past operations, which will not contribute to future revenues, are expensed. Liabilities are recorded when remediation costs are probable and can be reasonably estimated. The liability may include costs such as site investigations, consultant fees, feasibility studies, outside contractors, and monitoring expenses. Estimates are generally not reduced by potential claims for recovery, which are recognized when probable and as agreements are reached with third parties. The estimates may also include costs related to other potentially responsible parties to the extent that Arconic has reason to believe such parties will not fully pay their proportionate share. The liability is continuously reviewed and adjusted to reflect current remediation progress, prospective estimates of required activity, and other factors that may be relevant, including changes in technology or regulations. The portion of the liability associated with post-remediation operations, maintenance, and monitoring activities are discounted using a risk-free-rate.

Litigation Matters. For asserted claims and assessments, liabilities are recorded when an unfavorable outcome of a matter is deemed to be probable and the loss is reasonably estimable. Management determines the likelihood of an unfavorable outcome based on many factors such as, among others, the nature of the matter, available defenses and case strategy, progress of the matter, views and opinions of legal counsel and other advisors, applicability and success of appeals processes, and the outcome of similar historical matters. Once an unfavorable outcome is deemed probable, management weighs the probability of estimated losses, and the most reasonable loss estimate is recorded. If an unfavorable outcome of a matter is deemed to be reasonably possible, the matter is disclosed and no liability is recorded. With respect to unasserted claims or assessments, management must first determine the probability an assertion will be made is likely; then a determination as to the likelihood of an unfavorable outcome and the ability to reasonably estimate the potential loss is made. Legal matters are reviewed on a continuous basis to determine if there has been a change in management's judgment regarding the likelihood of an unfavorable outcome or the estimate of a potential loss.

Pension and Other Postretirement Benefits. In all periods prior to January 1, 2020 (see below), certain employees attributable to the Arconic Corporation Businesses participated in defined benefit pension and other postretirement benefit plans sponsored by ParentCo (the "Shared Plans"), which also included participants attributable to non-Arconic Corporation Businesses. Arconic accounted for the portion of the Shared Plans related to its employees as multiemployer benefit plans. Accordingly, the Company did not record an asset or liability to recognize any portion of the funded status of the Shared Plans. However, the related expense recorded by Arconic was based primarily on pensionable compensation and estimated interest costs related to participants attributable to the Arconic Corporation Businesses.

In all periods prior to the Separation Date, certain other ParentCo plans that were entirely attributable to employees of the Arconic Corporation Businesses ("Direct Plans") were accounted for as defined benefit pension and other postretirement benefit plans. Accordingly, the funded or unfunded position of each Direct Plan was recorded in the Consolidated Balance Sheet. Actuarial gains and losses that have not yet been recognized in earnings were recorded in accumulated other comprehensive income, net of taxes, until they were amortized as a component of net periodic benefit cost. The determination of benefit obligations and recognition of expenses related to the Direct Plans is dependent on various assumptions, including, among others, discount rates, long-term expected rates of return on plan assets, and future compensation increases. ParentCo's management developed each assumption using relevant company experience in conjunction with market-related data for each individual location in which such plans exist.

In preparation for the Separation, effective January 1, 2020, certain of the Shared Plans were separated into standalone plans for both Arconic ("New Direct Plans") and ParentCo (see Note H to the Consolidated Financial Statements in Part II Item 8 of this Form 10-K). Additionally, effective April 1, 2020, portions of the other remaining Shared Plans were assumed by Arconic ("Additional New Direct Plans") (see Note H to the Consolidated Financial Statements in Part II Item 8 of this Form 10-K). Accordingly, beginning on the respective effective dates, the New Direct Plans and the Additional New Direct Plans are accounted for as defined benefit pension and other postretirement plans. Additionally, the Direct Plans continue to be accounted for as defined benefit pension and other postretirement plans.

Liabilities and expenses for pension and other postretirement benefits are determined using actuarial methodologies and incorporate significant assumptions, including the interest rate used to discount the future estimated liability, the expected long-term rate of return on plan assets, and several assumptions relating to the employee workforce (compensation increases, health care cost trend rates, retirement age, and mortality).

The interest rate used to discount future estimated liabilities is determined using a Company-specific yield curve model (above-median) developed with the assistance of an external actuary. The cash flows of the projected benefit obligations are discounted using a single equivalent rate derived from yields on high quality corporate bonds, which represent a broad diversification of issuers in various sectors. The yield curve model parallels the projected plan cash flows, which have a weighted average duration of 11 years, and the underlying cash flows of the bonds included in the model exceed the cash flows needed to satisfy the plan obligations multiple times. If a deep market of high quality corporate bonds does not exist in a country, then the yield on government bonds is used. In 2022, the weighted average discount rate used to determine benefit obligations for pension plans was 5.49% and for other postretirement benefit plans was 5.62%. The impact on the combined pension and other postretirement benefit liabilities of a change in the weighted average discount rate of 1/4 of 1% would be approximately $60 and either a charge or credit of approximately $1 to pretax earnings in the following year.

The expected long-term rate of return on plan assets is generally applied to a five-year market-related value of plan assets (the fair value at the plan measurement date is used for certain non-U.S. plans). The process used by management to develop this assumption is one that relies on forward-looking investment returns by asset class. Management incorporates expected future investment returns on current and planned asset allocations using information from various external investment managers and consultants, as well as management's own judgment. For 2022, management used 5.18% as its weighted-average expected long-term rate of return, which was based on the prevailing and planned strategic asset allocations, as well as estimates of future

returns by asset class. For 2023, management anticipates that the weighted-average expected long-term rate of return will be in the range of 5.50% to 6.50%. A change in the assumption for the weighted average expected long-term rate of return on plan assets of 1/4 of 1% would impact pretax earnings by approximately $4 for 2023.

Stock-Based Compensation. In all periods prior to the Separation Date, eligible employees attributable to the Arconic Corporation Businesses participated in ParentCo's stock-based compensation plan. The compensation expense recorded by the Company included the expense associated with these employees, as well as the expense associated with an allocation of stock-based compensation expense for ParentCo's corporate employees. Beginning on the Separation Date, Arconic recorded stock-based compensation expense for all of the Company's employees eligible to participate in Arconic's stock-based compensation plan. The following describes the manner in which stock-based compensation expense was initially determined for both Arconic and ParentCo.

Compensation expense for employee equity grants is recognized using the non-substantive vesting period approach, in which the expense is recognized ratably over the requisite service period based on the grant date fair value. The fair value of stock options is estimated on the date of grant using a lattice-pricing model. The fair value of performance stock units containing a market condition is valued using a Monte Carlo valuation model. Determining the fair value at the grant date requires judgment, including estimates for the average risk-free interest rate, dividend yield, volatility, and exercise behavior. These assumptions may differ significantly between grant dates because of changes in the actual results of these inputs that occur over time.

In 2022, 2021, and 2020, Arconic recognized stock-based compensation expense of $15 ($12 after-tax), $22 ($17 after-tax), and $23 ($18 after-tax), respectively.

Income Taxes. The provision for income taxes is determined using the asset and liability approach of accounting for income taxes. Under this approach, the provision for income taxes represents income taxes paid or payable (or received or receivable) for the current year plus the change in deferred taxes during the year. Deferred taxes represent the future tax consequences expected to occur when the reported amounts of assets and liabilities are recovered or paid, and result from differences between the financial and tax bases of Arconic's assets and liabilities and are adjusted for changes in tax rates and tax laws when enacted.

In all periods prior to the Separation Date, the Arconic Corporation Businesses were included in the income tax filings of ParentCo. The provision for income taxes was determined in the same manner described above, but on a separate return methodology as if the Company was a standalone taxpayer filing hypothetical income tax returns where applicable. Any additional accrued tax liability or refund arising as a result of this approach was assumed to be immediately settled with ParentCo as a component of Parent Company net investment. Deferred taxes were also determined in the same manner described above and were reported in the Consolidated Balance Sheet for net operating losses, credits or other attributes to the extent that such attributes were expected to transfer to Arconic upon the Separation. Any difference from attributes generated in a hypothetical return on a separate return basis was adjusted as a component of Parent Company net investment.

Valuation allowances are recorded to reduce deferred tax assets when it is more likely than not (greater than 50%) that a tax benefit will not be realized. In evaluating the need for a valuation allowance, management considers all potential sources of taxable income, including income available in carryback periods, future reversals of taxable temporary differences, projections of taxable income, and income from tax planning strategies, as well as all available positive and negative evidence. Positive evidence includes factors such as a history of profitable operations, projections of future profitability within the carryforward period, including from tax planning strategies, and the Company's experience with similar operations. Existing favorable contracts and the ability to sell products into established markets are additional positive evidence. Negative evidence includes items such as cumulative losses, projections of future losses, or carryforward periods that are not long enough to allow for the utilization of a deferred tax asset based on existing projections of income. Deferred tax assets for which no valuation allowance is recorded may not be realized upon changes in facts and circumstances, resulting in a future charge to establish a valuation allowance. Existing valuation allowances are re-examined under the same standards of positive and negative evidence. If it is determined that it is more likely than not that a deferred tax asset will be realized, the appropriate amount of the valuation allowance, if any, is released. Deferred tax assets and liabilities are also re-measured to reflect changes in underlying tax rates due to law changes and the grant and lapse of tax holidays.

Arconic applies a tax law ordering approach when considering the need for a valuation allowance on net operating losses expected to offset Global Intangible Low Taxed Income (GILTI) income inclusions. Under this approach, reductions in cash tax savings are not considered as part of the valuation allowance assessment. Instead, future GILTI inclusions are considered a source of taxable income that support the realizability of deferred tax assets.

Tax benefits related to uncertain tax positions taken or expected to be taken on a tax return are recorded when such benefits meet a more likely than not threshold. Otherwise, these tax benefits are recorded when a tax position has been effectively

settled, which means that the statute of limitations has expired or the appropriate taxing authority has completed its examination even though the statute of limitations remains open. Interest and penalties related to uncertain tax positions are recognized as part of the provision for income taxes and are accrued beginning in the period that such interest and penalties would be applicable under relevant tax law until such time that the related tax benefits are recognized.

Recently Adopted and Recently Issued Accounting Guidance

See the respective section of Note B to the Consolidated Financial Statements in Part II Item 8 of this Form 10-K.

Item 7A. Quantitative and Qualitative Disclosures About Market Risk.

Not material.

Item 8. Financial Statements and Supplementary Data.

Management's Reports to Arconic Corporation Stockholders

Management's Report on Financial Statements and Practices

The accompanying Consolidated Financial Statements of Arconic Corporation and its subsidiaries (the "Company") were prepared by management, which is responsible for their integrity and objectivity, in accordance with accounting principles generally accepted in the United States of America (GAAP) and include amounts that are based on management's best judgments and estimates. The other financial information included in the Company's Annual Report on Form 10-K for the year ended December 31, 2022 is consistent with that in the Consolidated Financial Statements.

Management recognizes its responsibility for conducting the Company's affairs according to the highest standards of personal and corporate conduct. This responsibility is characterized and reflected in key policy statements issued from time to time regarding, among other things, conduct of its business activities within the laws of the host countries in which the Company operates and potentially conflicting outside business interests of its employees. The Company maintains a systematic program to assess compliance with these policies.

Management's Report on Internal Control over Financial Reporting

Management is responsible for establishing and maintaining adequate internal control over financial reporting, as defined in Rules 13a-15(f) and 15d-15(f) of the U.S. Securities Exchange Act of 1934 (as amended), for the Company. The Company's internal control over financial reporting is a process designed to provide reasonable assurance regarding the reliability of financial reporting and the preparation of financial statements for external purposes in accordance with GAAP. The Company's internal control over financial reporting includes those policies and procedures that (i) pertain to the maintenance of records that, in reasonable detail, accurately and fairly reflect the transactions and dispositions of the assets of the Company, (ii) provide reasonable assurance that transactions are recorded as necessary to permit preparation of financial statements in accordance with GAAP, and that receipts and expenditures of the Company are being made only in accordance with authorizations of management and directors of the Company, and (iii) provide reasonable assurance regarding prevention or timely detection of unauthorized acquisition, use, or disposition of the Company's assets that could have a material effect on the financial statements.

Because of its inherent limitations, internal control over financial reporting may not prevent or detect misstatements. Also, projections of any evaluation of effectiveness to future periods are subject to the risk that controls may become inadequate because of changes in conditions, or that the degree of compliance with the policies or procedures may deteriorate.

Management conducted an assessment to evaluate the effectiveness of the Company's internal control over financial reporting as of December 31, 2022 using the criteria in *Internal Control—Integrated Framework* (2013) issued by the Committee of Sponsoring Organizations of the Treadway Commission. Based on this assessment, management concluded that the Company maintained effective internal control over financial reporting as of December 31, 2022.

PricewaterhouseCoopers LLP, an independent registered public accounting firm, audited the effectiveness of the Company's internal control over financial reporting as of December 31, 2022, as stated in their report, which is included herein.

/s/ *Timothy D. Myers*

Timothy D. Myers
Chief Executive Officer

/s/ *Erick R. Asmussen*

Erick R. Asmussen
Executive Vice President and
Chief Financial Officer

February 21, 2023

Report of Independent Registered Public Accounting Firm

To the Board of Directors and Stockholders of Arconic Corporation

Opinions on the Financial Statements and Internal Control over Financial Reporting

We have audited the accompanying consolidated balance sheets of Arconic Corporation and its subsidiaries (the "Company") as of December 31, 2022 and 2021, and the related consolidated statements of operations, of comprehensive (loss) income, of changes in equity and of cash flows for each of the three years in the period ended December 31, 2022, including the related notes (collectively referred to as the "consolidated financial statements"). We also have audited the Company's internal control over financial reporting as of December 31, 2022, based on criteria established in *Internal Control - Integrated Framework* (2013) issued by the Committee of Sponsoring Organizations of the Treadway Commission (COSO).

In our opinion, the consolidated financial statements referred to above present fairly, in all material respects, the financial position of the Company as of December 31, 2022 and 2021, and the results of its operations and its cash flows for each of the three years in the period ended December 31, 2022 in conformity with accounting principles generally accepted in the United States of America. Also in our opinion, the Company maintained, in all material respects, effective internal control over financial reporting as of December 31, 2022, based on criteria established in *Internal Control - Integrated Framework* (2013) issued by the COSO.

Basis for Opinions

The Company's management is responsible for these consolidated financial statements, for maintaining effective internal control over financial reporting, and for its assessment of the effectiveness of internal control over financial reporting, included in the accompanying Management's Report on Internal Control over Financial Reporting. Our responsibility is to express opinions on the Company's consolidated financial statements and on the Company's internal control over financial reporting based on our audits. We are a public accounting firm registered with the Public Company Accounting Oversight Board (United States) (PCAOB) and are required to be independent with respect to the Company in accordance with the U.S. federal securities laws and the applicable rules and regulations of the Securities and Exchange Commission and the PCAOB.

We conducted our audits in accordance with the standards of the PCAOB. Those standards require that we plan and perform the audits to obtain reasonable assurance about whether the consolidated financial statements are free of material misstatement, whether due to error or fraud, and whether effective internal control over financial reporting was maintained in all material respects.

Our audits of the consolidated financial statements included performing procedures to assess the risks of material misstatement of the consolidated financial statements, whether due to error or fraud, and performing procedures that respond to those risks. Such procedures included examining, on a test basis, evidence regarding the amounts and disclosures in the consolidated financial statements. Our audits also included evaluating the accounting principles used and significant estimates made by management, as well as evaluating the overall presentation of the consolidated financial statements. Our audit of internal control over financial reporting included obtaining an understanding of internal control over financial reporting, assessing the risk that a material weakness exists, and testing and evaluating the design and operating effectiveness of internal control based on the assessed risk. Our audits also included performing such other procedures as we considered necessary in the circumstances. We believe that our audits provide a reasonable basis for our opinions.

Definition and Limitations of Internal Control over Financial Reporting

A company's internal control over financial reporting is a process designed to provide reasonable assurance regarding the reliability of financial reporting and the preparation of financial statements for external purposes in accordance with generally accepted accounting principles. A company's internal control over financial reporting includes those policies and procedures that (i) pertain to the maintenance of records that, in reasonable detail, accurately and fairly reflect the transactions and dispositions of the assets of the company; (ii) provide reasonable assurance that transactions are recorded as necessary to permit preparation of financial statements in accordance with generally accepted accounting principles, and that receipts and expenditures of the company are being made only in accordance with authorizations of management and directors of the company; and (iii) provide reasonable assurance regarding prevention or timely detection of unauthorized acquisition, use, or disposition of the company's assets that could have a material effect on the financial statements.

Because of its inherent limitations, internal control over financial reporting may not prevent or detect misstatements. Also, projections of any evaluation of effectiveness to future periods are subject to the risk that controls may become inadequate because of changes in conditions, or that the degree of compliance with the policies or procedures may deteriorate.

Critical Audit Matters

The critical audit matter communicated below is a matter arising from the current period audit of the consolidated financial statements that was communicated or required to be communicated to the audit committee and that (i) relates to accounts or disclosures that are material to the consolidated financial statements and (ii) involved our especially challenging, subjective, or complex judgments. The communication of critical audit matters does not alter in any way our opinion on the consolidated financial statements, taken as a whole, and we are not, by communicating the critical audit matter below, providing a separate opinion on the critical audit matter or on the accounts or disclosures to which it relates.

Valuation of Long-Lived Assets – Extrusions Asset Group

As described in Notes B and E to the consolidated financial statements, the long-lived assets of the Extrusions asset group include properties, plants and equipment, and intangible assets. Management reviews the properties, plants, and equipment for impairment whenever events or changes in circumstances indicate that the carrying value of those assets may not be recoverable. In September 2022, due to an expected decline in the forecasted financial performance for the Extrusions segment (and asset group), including continued forecasted losses, management evaluated the recoverability of the long-lived assets of the Extrusions asset group by comparing the aggregate carrying value to the undiscounted future cash flows. The result was that the long-lived assets were deemed by management to be impaired and the amount of the impairment charge was measured as the difference between the aggregate carrying value and the aggregate fair value of the long-lived assets. Fair value was determined by management using an orderly liquidation methodology for machinery and equipment and a sales comparison approach for land and structures. Significant judgments and assumptions were applied in estimating the fair value of the long-lived assets, including the use of market-based information specific to the machinery and equipment and information from recent sales or current listings of comparable properties for the land and structures. As a result of the evaluation, management recognized an impairment charge of $90 million for properties, plants, and equipment of the Extrusions asset group.

The principal considerations for our determination that performing procedures relating to the valuation of long-lived assets of the Extrusions asset group is a critical audit matter are (i) the significant judgment by management when developing the fair value estimates of the machinery, equipment, land and structures of the Extrusions asset group; (ii) a high degree of auditor judgment, subjectivity and effort in performing procedures and evaluating audit evidence related to management's valuation methods and significant assumptions related to market based information for machinery and equipment and information from recent sales or current listings of comparable properties for land and structures; and (iii) the audit effort involved the use of professionals with specialized skill and knowledge.

Addressing the matter involved performing procedures and evaluating audit evidence in connection with forming our overall opinion on the consolidated financial statements. These procedures included testing the effectiveness of controls relating to the valuation of the long-lived assets of the Extrusion asset group, including controls over valuation of the machinery, equipment, land and structures. These procedures also included, among others (i) testing management's process for developing the fair value estimates of the machinery, equipment, land and structures; (ii) evaluating the appropriateness of the valuation methods; (iii) testing the completeness and accuracy of underlying data used in the valuation methods; and (iv) evaluating the reasonableness of the significant assumptions used by management related to market based information for machinery and equipment and information from recent sales or current listings of comparable properties for the land and structures. Evaluating management's significant assumptions related to market based information for machinery and equipment and information from recent sales or current listings of comparable properties for the land and structures involved evaluating whether the assumptions used by management were reasonable considering (i) historical cost and salvage information of assets within the asset group; (ii) the consistency with comparable salvage values, and sales or listings of comparable properties; and (iii) whether the assumptions were consistent with evidence obtained in other areas of the audit. Professionals with specialized skill and knowledge were used to assist in the evaluation of the estimates of fair value of the machinery, equipment, land and structures of the Extrusions asset group.

/s/ PricewaterhouseCoopers LLP

Pittsburgh, Pennsylvania
February 21, 2023

We have served as the Company's auditor since 2019.

Arconic Corporation and subsidiaries
Statement of Consolidated Operations
(in millions, except per-share amounts)

For the year ended December 31,	2022	2021	2020
Sales (C and D)	$ 8,961	$ 7,504	$ 5,675
Cost of goods sold (exclusive of expenses below)	8,032	6,573	4,862
Selling, general administrative, and other expenses	246	247	258
Research and development expenses	37	34	36
Provision for depreciation and amortization	237	253	251
Impairment of goodwill (B & O)	—	65	—
Restructuring and other charges (E)	456	624	188
Operating (loss) income	(47)	(292)	80
Interest expense (F)	104	100	118
Other expenses, net (G)	41	67	70
Loss before income taxes	(192)	(459)	(108)
(Benefit) Provision for income taxes (I)	(11)	(62)	1
Net loss	(181)	(397)	(109)
Less: Net income attributable to noncontrolling interest	1	—	—
Net loss attributable to Arconic Corporation	$ (182)	$ (397)	$ (109)
Earnings Per Share Attributable to Arconic Corporation Common Stockholders (J):			
Basic	$ (1.75)	$ (3.65)	$ (1.00)
Diluted	$ (1.75)	$ (3.65)	$ (1.00)

The accompanying notes are an integral part of the consolidated financial statements.

Arconic Corporation and subsidiaries
Statement of Consolidated Comprehensive (Loss) Income
(in millions)

	Arconic Corporation			Noncontrolling interest			Total		
For the year ended December 31,	**2022**	**2021**	**2020**	**2022**	**2021**	**2020**	**2022**	**2021**	**2020**
Net loss	$ (182)	$ (397)	$ (109)	$ 1	$ —	$ —	$ (181)	$ (397)	$ (109)
Other comprehensive income, net of tax (L):									
Change in unrecognized net actuarial loss and prior service cost/benefit related to pension and other postretirement benefits	229	670	54	—	—	—	229	670	54
Foreign currency translation adjustments	(76)	(4)	87	—	—	—	(76)	(4)	87
Net change in unrecognized losses on cash flow hedges	21	(16)	5	—	—	—	21	(16)	5
Total Other comprehensive income, net of tax	174	650	146	—	—	—	174	650	146
Comprehensive (loss) income	$ (8)	$ 253	$ 37	$ 1	$ —	$ —	$ (7)	$ 253	$ 37

The accompanying notes are an integral part of the consolidated financial statements.

Arconic Corporation and subsidiaries
Consolidated Balance Sheet
(in millions)

December 31,	2022	2021
Assets		
Current assets:		
Cash and cash equivalents	$ 261	$ 335
Receivables from customers, less allowances of $1 in both 2022 and 2021 (V)	791	922
Other receivables	183	226
Inventories (M)	1,622	1,630
Fair value of hedging instruments and derivatives (U)	21	1
Prepaid expenses and other current assets (T)	124	54
Total current assets	3,002	3,168
Properties, plants, and equipment, net (N)	2,361	2,651
Goodwill (B & O)	292	322
Operating lease right-of-use assets (P)	115	122
Deferred income taxes (I)	188	229
Other noncurrent assets	57	88
Total assets	$ 6,015	$ 6,580
Liabilities		
Current liabilities:		
Accounts payable, trade	$ 1,578	$ 1,718
Accrued compensation and retirement costs	119	116
Taxes, including income taxes	43	61
Environmental remediation (T)	40	15
Operating lease liabilities (P)	34	35
Fair value of hedging instruments and derivatives (U)	7	23
Other current liabilities (T)	150	95
Total current liabilities	1,971	2,063
Long-term debt (Q)	1,597	1,594
Accrued pension benefits (H)	586	717
Accrued other postretirement benefits (H)	302	411
Environmental remediation (T)	45	49
Operating lease liabilities (P)	83	90
Deferred income taxes (I)	3	12
Other noncurrent liabilities	71	85
Total liabilities	4,658	5,021
Contingencies and commitments (T)		
Equity		
Arconic Corporation stockholders' equity:		
Common stock (K)	1	1
Additional capital	3,373	3,368
Accumulated deficit	(734)	(552)
Treasury stock (K)	(346)	(161)
Accumulated other comprehensive loss (L)	(937)	(1,111)
Total Arconic Corporation stockholders' equity	1,357	1,545
Noncontrolling interest (S)	—	14
Total equity	1,357	1,559
Total liabilities and equity	$ 6,015	$ 6,580

The accompanying notes are an integral part of the consolidated financial statements.

Arconic Corporation and subsidiaries
Statement of Consolidated Cash Flows
(in millions)

For the year ended December 31,	2022	2021	2020
Operating Activities			
Net loss	$ (181)	$ (397)	$ (109)
Adjustments to reconcile net loss to cash provided from (used for) operations:			
Depreciation and amortization	237	253	251
Impairment of goodwill (B & O)	—	65	—
Deferred income taxes (I)	(45)	(100)	(16)
Restructuring and other charges (E)	456	624	188
Net periodic pension benefit cost (H)	81	68	82
Stock-based compensation (K)	15	22	23
Amortization of debt issuance costs (Q)	5	5	25
Other	4	16	(1)
Changes in assets and liabilities, excluding effects of acquisitions, divestitures, and foreign currency translation adjustments:			
Decrease (Increase) in receivables (V)	34	(381)	(235)
(Increase) Decrease in inventories	(149)	(596)	65
(Increase) in prepaid expenses and other current assets	(80)	(1)	(16)
(Decrease) Increase in accounts payable, trade	(67)	581	82
Increase (Decrease) in accrued expenses	38	(129)	(217)
Increase in taxes, including income taxes	2	21	99
Pension contributions (H)	(31)	(458)	(271)
(Increase) Decrease in noncurrent assets	(6)	(8)	35
Increase in noncurrent liabilities	25	8	21
Cash provided from (used for) operations	338	(407)	6
Financing Activities			
Net transfers from former parent company	—	—	216
Separation payment to former parent company (A)	—	—	(728)
Additions to debt (original maturities greater than three months) (Q)	—	319	2,400
Debt issuance costs (Q)	(1)	(5)	(57)
Payments on debt (original maturities greater than three months) (Q)	—	—	(1,100)
Repurchases of common stock (K)	(185)	(161)	—
Other	(10)	(18)	13
Cash (used for) provided from financing activities	(196)	135	744
Investing Activities			
Capital expenditures	(245)	(184)	(163)
Proceeds from the sale of assets and businesses (S)	30	(1)	125
Other	1	4	—
Cash used for investing activities	(214)	(181)	(38)
Effect of exchange rate changes on cash and cash equivalents and restricted cash	(2)	1	3
Net change in cash and cash equivalents and restricted cash	(74)	(452)	715
Cash and cash equivalents and restricted cash at beginning of year (R)	335	787	72
Cash and cash equivalents and restricted cash at end of year (R)	$ 261	$ 335	$ 787

The accompanying notes are an integral part of the consolidated financial statements.

Arconic Corporation and subsidiaries
Statement of Changes in Consolidated Equity
(dollars in millions)

	Common shares outstanding	Common stock	Additional capital	Accumulated deficit	Treasury stock	Parent Company net investment	Accumulated other comprehensive income (loss)	Noncontrolling interest	Total equity
Balance at December 31, 2019	—	$ —	$ —	$ —	$ —	$ 2,664	$ 295	$ 14	$ 2,973
Net (loss) income	—	—	—	(155)	—	46	—	—	(109)
Other comprehensive income (L)	—	—	—	—	—	—	146	—	146
Establishment of additional defined benefit plans (H)	—	—	—	—	—	349	(1,752)	—	(1,403)
Change in ParentCo contribution	—	—	—	—	—	217	—	—	217
Separation payment to former parent company (A)	—	—	—	—	—	(728)	—	—	(728)
Separation-related adjustments	—	—	3,334	—	—	(2,548)	(450)	—	336
Issuance of common stock (K)	109,021,376	1	(1)	—	—	—	—	—	—
Stock-based compensation (K)	183,850	—	15	—	—	—	—	—	15
Balance at December 31, 2020	109,205,226	$ 1	$ 3,348	$ (155)	$ —	$ —	$ (1,761)	$ 14	$ 1,447
Net loss	—	—	—	(397)	—	—	—	—	(397)
Other comprehensive income (L)	—	—	—	—	—	—	650	—	650
Repurchases of common stock (K)	(4,912,505)	—	—	—	(161)	—	—	—	(161)
Stock-based compensation (K)	1,034,164	—	22	—	—	—	—	—	22
Other	—	—	(2)	—	—	—	—	—	(2)
Balance at December 31, 2021	105,326,885	$ 1	$ 3,368	$ (552)	$ (161)	$ —	$ (1,111)	$ 14	$ 1,559
Net (loss) income	—	—	—	(182)	—	—	—	1	(181)
Other comprehensive income (L)	—	—	—	—	—	—	174	—	174
Repurchases of common stock (K)	(6,935,507)	—	—	—	(185)	—	—	—	(185)
Stock-based compensation (K)	1,040,816	—	15	—	—	—	—	—	15
Divestiture (S)	—	—	—	—	—	—	—	(15)	(15)
Other	—	—	(10)	—	—	—	—	—	(10)
Balance at December 31, 2022	99,432,194	$ 1	$ 3,373	$ (734)	$ (346)	$ —	$ (937)	$ —	$ 1,357

The accompanying notes are an integral part of the consolidated financial statements.

Arconic Corporation and subsidiaries
Notes to the Consolidated Financial Statements
(dollars in millions, except per-share amounts)

A. Basis of Presentation

Arconic Corporation ("Arconic" or the "Company") is a manufacturer of fabricated aluminum products, including sheet and plate, extrusions, and architectural products and systems, with a primary focus on the ground transportation, aerospace, building and construction, industrial products, and packaging end markets. The Company has 20 primary operating locations in 7 countries around the world, situated in the United States, Canada, China, France, Germany, Hungary, and the United Kingdom. Arconic's previous operations in Russia were divested in November 2022 (see Note S).

In the 2022 fourth quarter, Arconic recorded an adjustment of $9 in Cost of goods sold to increase its environmental reserves to correct the accrual related to anticipated costs associated with the Company's obligations to perform future operations, maintenance, and monitoring (OM&M) activities, as required by federal, state, and/or local environmental agencies, at several environmental remediation sites. The adjustment was derived from a site-by-site analysis based upon OM&M plans submitted to respective environmental regulatory agencies, and resulted in extending the time horizon for OM&M costs. Management has concluded that the impact was not material to any previously reported periods.

In the 2022 first quarter, the Company recorded a net gain of $3 in Cost of goods sold related to the unrealized impact associated with the change in the estimated fair value of natural gas supply contracts now determined to be derivatives (see Note U). This amount was comprised of an unrealized loss of $5 for the 2022 first quarter, an unrealized gain of $6 for the 2021 annual period, and an unrealized gain of $2 for the 2020 fourth quarter. The out-of-period amounts were not material to any interim or annual period.

References in these Notes to (i) "ParentCo" refer to Arconic Inc., a Delaware corporation, and its consolidated subsidiaries (through March 31, 2020, at which time it was renamed Howmet Aerospace Inc. ("Howmet")), and (ii) "2016 Separation Transaction" refer to the November 1, 2016 separation of Alcoa Inc., a Pennsylvania corporation, into two standalone, publicly-traded companies, Arconic Inc. and Alcoa Corporation.

The Separation. On April 1, 2020 (the "Separation Date"), ParentCo separated into two standalone, publicly-traded companies, Arconic and Howmet, effective at 12:01 a.m. Eastern Daylight Time (the "Separation"). The spin-off company, Arconic, comprised the rolled aluminum products, aluminum extrusions, and architectural products operations of ParentCo, as well as the Latin America extrusions operations sold in April 2018, (collectively, the "Arconic Corporation Businesses"). The existing publicly-traded company, Howmet, continued to own the engine products, engineered structures, fastening systems, and forged wheels operations (collectively, the "Howmet Aerospace Businesses"). ParentCo common stockholders of record as of the close of business on March 19, 2020 (the "Record Date") received one share of Arconic common stock for every four shares of ParentCo common stock (the "Separation Ratio") held as of the Record Date (ParentCo paid cash to its common stockholders in lieu of fractional shares).

To effect the Separation, ParentCo undertook a series of transactions to separate the net assets and certain legal entities of ParentCo, resulting in a cash payment of $728 to ParentCo by Arconic from a portion of the aggregate net proceeds of previously executed financing arrangements (see Note Q). In connection with the Separation, 109,021,376 shares of Arconic common stock were distributed to ParentCo stockholders. This was determined by applying the Separation Ratio to the 436,085,504 shares of ParentCo's outstanding common stock as of the Record Date. "Regular-way" trading of Arconic's common stock began with the opening of the New York Stock Exchange on April 1, 2020 under the ticker symbol "ARNC." Arconic's common stock has a par value of $0.01 per share.

In connection with the Separation, Arconic and Howmet entered into several agreements to implement the legal and structural separation between the two companies; govern the relationship between Arconic and Howmet after the completion of the Separation; and allocate between Arconic and Howmet various assets, liabilities, and obligations, including, among other things, employee benefits, environmental liabilities, intellectual property, and tax-related assets and liabilities. These agreements included a Separation and Distribution Agreement, Tax Matters Agreement, Employee Matters Agreement, and certain Patent, Know-How, Trade Secret License and Trademark License Agreements. The Separation and Distribution Agreement identified the assets to be transferred, the liabilities to be assumed, and the contracts to be transferred to each of Arconic and Howmet as part of the Separation, and provided for when and how these transfers and assumptions were to occur.

ParentCo incurred costs to evaluate, plan, and execute the Separation, and Arconic was allocated a pro rata portion of these costs based on segment revenue (see Cost Allocations below). ParentCo recognized $38 from January 1, 2020 through March 31, 2020 (the "2020 Pre-Separation Period") for such costs, of which $18 was allocated to Arconic. The allocated amounts were included in Selling, general administrative, and other expenses on the accompanying Statement of Consolidated Operations.

Basis of Presentation. The Consolidated Financial Statements of Arconic are prepared in conformity with accounting principles generally accepted in the United States of America (GAAP). In accordance with GAAP, certain situations require management to make estimates based on judgments and assumptions, which may affect the reported amounts of assets and liabilities and the disclosure of contingent assets and liabilities at the date of the financial statements. They also may affect the reported amounts of revenues and expenses during the reporting periods. Actual results could differ from those estimates upon subsequent resolution of identified matters. These estimates are based on historical experience and, in some cases, assumptions based on current and future market experience. Management has made its best estimates using all relevant information available at the time.

Principles of Consolidation. The Consolidated Financial Statements of the Company include the accounts of Arconic and companies in which Arconic has a controlling interest. Intercompany transactions have been eliminated.

Management evaluates whether an Arconic entity or interest is a variable interest entity and whether the Company is the primary beneficiary. Consolidation is required if both of these criteria are met. Arconic does not have any variable interest entities requiring consolidation.

Prior to the Separation Date, the Company did not operate as a separate, standalone entity. Arconic's operations were included in ParentCo's financial results. Accordingly, in all periods prior to the Separation Date, the accompanying Consolidated Financial Statements were prepared from ParentCo's historical accounting records and were presented on a standalone basis as if the Arconic Corporation Businesses had been conducted independently from ParentCo. Such Consolidated Financial Statements include the historical operations that were considered to comprise the Arconic Corporation Businesses, as well as certain assets and liabilities that were historically held at ParentCo's corporate level but were specifically identifiable or otherwise attributable to the Arconic Corporation Businesses. ParentCo's net investment in these operations was reflected as Parent Company net investment on the accompanying Consolidated Financial Statements. All significant transactions and accounts within the Arconic Corporation Businesses were eliminated. All significant intercompany transactions between ParentCo and the Arconic Corporation Businesses were included within Parent Company net investment on the accompanying Consolidated Financial Statements.

Cost Allocations. The description and information on cost allocations is applicable for all periods prior to the Separation Date presented in the accompanying Consolidated Financial Statements.

The Consolidated Financial Statements of Arconic include general corporate expenses of ParentCo that were not historically charged to the Arconic Corporation Businesses for certain support functions that were provided on a centralized basis, such as expenses related to finance, audit, legal, information technology, human resources, communications, compliance, facilities, employee benefits and compensation, and research and development activities. These general corporate expenses were reported on the accompanying Statement of Consolidated Operations within Cost of goods sold, Selling, general administrative and other expenses, and Research and development expenses. These expenses were allocated to the Arconic Corporation Businesses on the basis of direct usage when identifiable, with the remainder allocated based on the Arconic Corporation Businesses' segment revenue as a percentage of ParentCo's total segment revenue, as reported in the respective periods.

All external debt not directly attributable to the Arconic Corporation Businesses was excluded from the Company's Consolidated Balance Sheet. Financing costs related to these debt obligations were allocated to the Arconic Corporation Businesses based on the ratio of capital invested by ParentCo in the Arconic Corporation Businesses to the total capital invested by ParentCo in both the Arconic Corporation Businesses and the Howmet Aerospace Businesses and were included on the accompanying Statement of Consolidated Operations within Interest expense.

The following table reflects the allocations described above:

	2020
Selling, general administrative, and other expenses*	$ 25
Provision for depreciation and amortization	1
Restructuring and other charges (E)	2
Interest expense (F)	28
Other (income), net	(5)

* In the 2020 Pre-Separation Period, amount includes an allocation of $18 for costs incurred by ParentCo associated with the Separation (see The Separation above).

Management believes the assumptions regarding the allocation of ParentCo's general corporate expenses and financing costs were reasonable.

Nevertheless, the Company's Consolidated Financial Statements may not include all of the actual expenses that would have been incurred and may not reflect Arconic's consolidated results of operations, financial position, and cash flows had it been a standalone company during the periods prior to the Separation Date. Actual costs that would have been incurred if Arconic had been a standalone company would depend on multiple factors, including organizational structure, capital structure, and strategic decisions made in various areas, including information technology and infrastructure. Transactions between the Arconic Corporation Businesses and ParentCo, including sales to the Howmet Aerospace Businesses, were considered to be effectively settled for cash at the time the transaction was recorded. The total net effect of the settlement of these transactions was reported on the accompanying Statement of Consolidated Cash Flows as a financing activity and on Arconic's Consolidated Balance Sheet as Parent Company net investment.

Cash Management. Prior to the Separation Date, cash was managed centrally with certain net earnings reinvested locally and working capital requirements met from existing liquid funds. Accordingly, the cash and cash equivalents held by ParentCo at the corporate level were not attributed to the Arconic Corporation Businesses in any period presented in the accompanying Consolidated Financial Statements that was prior to the Separation Date. Only cash amounts specifically attributable to the Arconic Corporation Businesses were reported in the accompanying Consolidated Financial Statements for any period presented that was prior to the Separation Date. Transfers of cash, both to and from ParentCo's centralized cash management system, were reported as a component of Parent Company net investment on Arconic's Consolidated Balance Sheet and as a financing activity on the accompanying Statement of Consolidated Cash Flows.

B. Summary of Significant Accounting Policies

Cash Equivalents. Cash equivalents are highly liquid investments purchased with an original maturity of three months or less. In all periods prior to the Separation Date, the cash and cash equivalents held by ParentCo at the corporate level were not attributed to the Arconic Corporation Businesses. Only cash amounts specifically attributable to the Arconic Corporation Businesses were reported on the Company's Consolidated Financial Statements.

Inventory Valuation. Inventories are carried at the lower of cost and net realizable value, with cost for most inventories determined under the average cost method. The cost of certain non-U.S. inventories is determined under the first in, first out (FIFO) method.

Properties, Plants, and Equipment. Properties, plants, and equipment are recorded at cost. Also, interest related to the construction of qualifying assets is capitalized as part of the construction costs. Depreciation is recorded on the straight-line method at rates based on the estimated useful lives of the assets. The following table details the weighted-average useful lives of structures and machinery and equipment by reporting segment (numbers in years):

	Structures	Machinery and equipment
Rolled Products	32	22
Building and Construction Systems	25	18
Extrusions	32	20

Repairs and maintenance are charged to expense as incurred. Generally, gains or losses from the sale of asset groups are recorded in Restructuring and other charges while gains and losses from the sale of individual assets are recorded in Other expenses (income), net.

Properties, plants, and equipment are reviewed for impairment whenever events or changes in circumstances indicate that the carrying value of such assets may not be recoverable. Recoverability of assets is determined by comparing the estimated undiscounted net cash flows of the related operations (asset group) to the carrying value of the associated assets. An impairment loss would be recognized when the carrying value of the assets exceeds the estimated undiscounted net cash flows of the asset group. The amount of the impairment loss to be recorded is calculated as the excess of the carrying value of the assets over their fair value, with fair value determined using the best information available, which generally is a discounted cash flow (DCF) model. In certain circumstances, Arconic may use the services of a third-party to assist with other valuation techniques. The determination of what constitutes an asset group, the associated estimated undiscounted net cash flows, and the estimated useful lives of the assets also require significant judgments.

Goodwill. Goodwill is not amortized; it is reviewed for impairment annually (in the fourth quarter) or more frequently if indicators of impairment exist or if a decision is made to sell, exit, or realign a business. A significant amount of judgment is involved in determining if an indicator of impairment has occurred. Such indicators may include, among others, deterioration in general economic conditions, negative developments in equity and credit markets, adverse changes in the markets in which an entity operates, increases in input costs that have a negative effect on earnings and cash flows, or a trend of negative or declining cash flows over multiple periods. The fair value that could be realized in an actual transaction may differ from that used to evaluate goodwill for impairment.

Goodwill is allocated among and evaluated for impairment at the reporting unit level, which is defined as an operating segment or one level below an operating segment. Arconic has three reporting units, the Rolled Products segment, the Building and Construction Systems segment, and the Extrusions segment (full impairment of goodwill in 2021 – see below). At the time of the Company's annual 2022 review of goodwill for impairment, the carrying value of goodwill for Rolled Products and Building and Construction Systems was $232 and $67, respectively.

In reviewing goodwill for impairment, an entity has the option to first assess qualitative factors to determine whether the existence of events or circumstances leads to a determination that it is more likely than not (greater than 50%) that the estimated fair value of a reporting unit is less than its carrying amount. If an entity elects to perform a qualitative assessment and determines that an impairment is more likely than not, the entity is then required to perform a quantitative impairment test (described below), otherwise no further analysis is required. An entity also may elect not to perform the qualitative assessment and, instead, proceed directly to the quantitative impairment test. The ultimate outcome of the goodwill impairment review for a reporting unit should be the same whether an entity chooses to perform the qualitative assessment or proceeds directly to the quantitative impairment test.

Arconic determines annually, based on facts and circumstances, which of its reporting units will be subject to the qualitative assessment. For those reporting units where a qualitative assessment is either not performed or for which the conclusion is that an impairment is more likely than not, a quantitative impairment test will be performed. The Company's policy is that a quantitative impairment test be performed for each reporting unit at least once during every three-year period.

Under the qualitative assessment, various events and circumstances (or factors) that would affect the estimated fair value of a reporting unit are identified (similar to impairment indicators above). These factors are then classified by the type of impact they would have on the estimated fair value using positive, neutral, and adverse categories based on current business conditions. Additionally, an assessment of the level of impact that a particular factor would have on the estimated fair value is determined using high, medium, and low weighting. Furthermore, management considers the results of the most recent quantitative impairment test completed for a reporting unit and compares the weighted average cost of capital (WACC) between the current and prior years for each reporting unit.

During the 2022 annual review of goodwill for impairment, management performed the qualitative assessment for the Building and Construction Systems reporting unit. Management concluded it was not more likely than not that the estimated fair value of this reporting unit was less than the carrying value. As such, no further analysis was required. Management last proceeded directly to the quantitative impairment test for the Building and Construction Systems reporting unit in 2021. At that time, the estimated fair value of this reporting unit was substantially in excess of its carrying value, resulting in no impairment.

Under the quantitative impairment test, the evaluation of the recoverability of goodwill involves comparing the current fair value of each reporting unit to its carrying value, including goodwill. Arconic uses a DCF model to estimate the current fair value of its reporting units when testing for impairment, as management believes forecasted cash flows are the best indicator of such fair value. Several significant assumptions and estimates are involved in the application of the DCF model to forecast operating cash flows, including sales growth (volumes and pricing), production costs, capital spending, working capital levels, and discount rate. Certain of these assumptions may vary significantly among the reporting units. Cash flow forecasts are generally based on approved business unit five-year operating plans and a terminal value. The WACC rate for the individual reporting units is estimated by management with the assistance of valuation experts. In the event the estimated fair value of a reporting unit per the DCF model is less than the carrying value, the Company would recognize an impairment charge equal to the excess of the reporting unit's carrying value over its fair value without exceeding the total amount of goodwill applicable to that reporting unit.

During the 2022 annual review of goodwill for impairment, management proceeded directly to the quantitative impairment test for the Rolled Products reporting unit. The result of this review indicated the estimated fair value of this reporting unit was substantially in excess of the carrying value, resulting in no impairment.

The respective annual review of goodwill for impairment in 2021 and 2020 indicated that goodwill was not impaired for any of Arconic's reporting units, except for the Extrusions segment (see below), and there were no triggering events since that time that necessitated an interim quantitative impairment test for any of the Company's reporting units. That said, in light of the

economic impact of the COVID-19 pandemic (i.e., stock market volatility, customer demand disruption, etc.), Arconic did perform periodic qualitative assessments throughout 2020 in which management concluded that no impairment existed and, therefore, no further analysis was required.

In 2021, the estimated fair value of the Extrusions reporting unit was lower than the associated carrying value by an amount greater than the carrying value of the Extrusions reporting unit's goodwill. Accordingly, the Company recorded an impairment charge of $65 in the fourth quarter of 2021. As a result, there isno goodwill remaining for the Extrusions reporting unit.

The impairment of the Extrusion reporting unit's goodwill resulted from a combination of market-based factors, including continued delays in aerospace market improvement, the expectation that significant cost inflation will extend beyond 2021 resulting in increasingly limited ability for management to drive margin expansion, and a longer than previously anticipated shift in product mix to lower margin industrial products to replace most of the lost aerospace volume. Further, current market factors also resulted in a 150-basis point increase in the WACC compared to the fourth quarter of 2020. Accordingly, given these factors, the estimated fair value of the Extrusions reporting unit was less than the carrying value resulting in a full impairment of its goodwill.

Other Intangible Assets. Intangible assets with finite useful lives are amortized on a straight-line basis over the periods benefited.The following table details the weighted-average useful lives of software and other intangible assets by reporting segment (numbers in years):

	Software	Other intangible assets
Rolled Products	8	—
Building and Construction Systems	4	20
Extrusions	3	10

Leases. Arconic determines whether a contract contains a lease at inception. The Company leases certain land and buildings, plant equipment, vehicles, and computer equipment, which have been classified as operating leases. Certain real estate leases include one or more options to renew; the exercise of lease renewal options is at Arconic's discretion. The Company includes renewal option periods in the lease term when it is determined that the options are reasonably certain to be exercised. Certain of Arconic's real estate lease agreements include rental payments that either have fixed contractual increases over time or adjust periodically for inflation. Also, certain of the Company's lease agreements include variable lease payments. The variable portion of payments is not included in the initial measurement of the right-of-use asset or lease liability due to the uncertainty of the payment amount and is recorded as lease cost in the period incurred.

Operating lease right-of-use assets and lease liabilities with an initial term greater than 12 months are recorded on the balance sheet at the present value of the future minimum lease payments over the lease term calculated at the lease commencement date and are recognized as lease expense on a straight-line basis over the lease term. Arconic uses an incremental collateralized borrowing rate based on the information available at the lease commencement date in determining the present value of future payments, as most of the Company's leases do not provide an implicit rate. The operating lease right-of-use assets also include any lease prepayments made and are reduced by lease incentives and accrued exit costs.

Environmental Matters. Expenditures for current operations are expensed or capitalized, as appropriate. Expenditures relating to existing conditions caused by past operations, which will not contribute to future revenues, are expensed. Liabilities are recorded when remediation costs are probable and can be reasonably estimated. The liability may include costs such as site investigations, consultant fees, feasibility studies, outside contractors, and monitoring expenses. Estimates are generally not reduced by potential claims for recovery, which are recognized when probable and as agreements are reached with third parties. The estimates may also include costs related to other potentially responsible parties to the extent that Arconic has reason to believe such parties will not fully pay their proportionate share. The liability is continuously reviewed and adjusted to reflect current remediation progress, prospective estimates of required activity, and other factors that may be relevant, including changes in technology or regulations. The portion of the liability associated with post-remediation operations, maintenance, and monitoring activities are discounted using a risk-free-rate.

Litigation Matters. For asserted claims and assessments, liabilities are recorded when an unfavorable outcome of a matter is deemed to be probable and the loss is reasonably estimable. Management determines the likelihood of an unfavorable outcome based on many factors such as, among others, the nature of the matter, available defenses and case strategy, progress of the matter, views and opinions of legal counsel and other advisors, applicability and success of appeals processes, and the outcome of similar historical matters. Once an unfavorable outcome is deemed probable, management weighs the probability of

estimated losses, and the most reasonable loss estimate is recorded. If an unfavorable outcome of a matter is deemed to be reasonably possible, the matter is disclosed and no liability is recorded. With respect to unasserted claims or assessments, management must first determine the probability an assertion will be made is likely; then a determination as to the likelihood of an unfavorable outcome and the ability to reasonably estimate the potential loss is made. Legal matters are reviewed on a continuous basis to determine if there has been a change in management's judgment regarding the likelihood of an unfavorable outcome or the estimate of a potential loss.

Revenue Recognition. The Company's contracts with customers are generally comprised of acknowledged purchase orders incorporating Arconic's standard terms and conditions, or for larger customers, may also generally include terms under negotiated multi-year agreements. These customer contracts typically consist of the manufacture of products, which represent single performance obligations that are satisfied upon transfer of control of the product to the customer. The Company produces aluminum sheet and plate; extruded and machined parts; integrated aluminum architectural systems; and architectural extrusions. Transfer of control is assessed based on alternative use of the products produced and Arconic's enforceable right to payment for performance to date under the contract terms. Transfer of control and revenue recognition generally occur upon shipment or delivery of the product, which is when title, ownership, and risk of loss pass to the customer and is based on the applicable shipping terms. The shipping terms vary across all businesses and depend on the product, the country of origin, and the type of transportation (truck, train, or vessel).

In certain circumstances, the Company receives advanced payments from its customers for product to be delivered in future periods. These advanced payments are recorded as deferred revenue until the product is delivered and title and risk of loss have passed to the customer in accordance with the terms of the contract. Deferred revenue is reported in Other current liabilities and Other noncurrent liabilities on the Consolidated Balance Sheet.

Pension and Other Postretirement Benefits. In all periods prior to January 1, 2020 (see below), certain employees attributable to the Arconic Corporation Businesses participated in defined benefit pension and other postretirement benefit plans sponsored by ParentCo (the "Shared Plans"), which also included participants attributable to non-Arconic Corporation Businesses. Arconic accounted for the portion of the Shared Plans related to its employees as multiemployer benefit plans. Accordingly, the Company did not record an asset or liability to recognize any portion of the funded status of the Shared Plans. However, the related expense recorded by Arconic was based primarily on pensionable compensation and estimated interest costs related to participants attributable to the Arconic Corporation Businesses.

In all periods prior to the Separation Date, certain other ParentCo plans that were entirely attributable to employees of the Arconic Corporation Businesses ("Direct Plans") were accounted for as defined benefit pension and other postretirement benefit plans. Accordingly, the funded or unfunded position of each Direct Plan was recorded in the Consolidated Balance Sheet. Actuarial gains and losses that have not yet been recognized in earnings were recorded in accumulated other comprehensive income, net of taxes, until they were amortized as a component of net periodic benefit cost. The determination of benefit obligations and recognition of expenses related to the Direct Plans is dependent on various assumptions, including, among others, discount rates, long-term expected rates of return on plan assets, and future compensation increases. ParentCo's management developed each assumption using relevant company experience in conjunction with market-related data for each individual location in which such plans exist.

In preparation for the Separation, effective January 1, 2020, certain of the Shared Plans were separated into standalone plans for both Arconic ("New Direct Plans") and ParentCo (see Note H). Additionally, effective April 1, 2020, portions of the other remaining Shared Plans were assumed by Arconic ("Additional New Direct Plans") (see Note H). Accordingly, beginning on the respective effective dates, the New Direct Plans and the Additional New Direct Plans are accounted for as defined benefit pension and other postretirement plans. Additionally, the Direct Plans continue to be accounted for as defined benefit pension and other postretirement plans.

Stock-Based Compensation. In all periods prior to the Separation Date, eligible employees attributable to the Arconic Corporation Businesses participated in ParentCo's stock-based compensation plan. The compensation expense recorded by the Company included the expense associated with these employees, as well as the expense associated with an allocation of stock-based compensation expense for ParentCo's corporate employees (see Cost Allocations in Note A). Beginning on the Separation Date, Arconic recorded stock-based compensation expense for all of the Company's employees eligible to participate in Arconic's stock-based compensation plan. The following describes the manner in which stock-based compensation expense was initially determined for both Arconic and ParentCo.

Compensation expense for employee equity grants is recognized using the non-substantive vesting period approach, in which the expense is recognized ratably over the requisite service period based on the grant date fair value. The fair value of stock options is estimated on the date of grant using a lattice-pricing model. The fair value of performance stock units containing a market condition is valued using a Monte Carlo valuation model. Determining the fair value at the grant date

requires judgment, including estimates for the average risk-free interest rate, dividend yield, volatility, and exercise behavior. These assumptions may differ significantly between grant dates because of changes in the actual results of these inputs that occur over time.

Income Taxes. The provision for income taxes is determined using the asset and liability approach of accounting for income taxes. Under this approach, the provision for income taxes represents income taxes paid or payable (or received or receivable) for the current year plus the change in deferred taxes during the year. Deferred taxes represent the future tax consequences expected to occur when the reported amounts of assets and liabilities are recovered or paid, and result from differences between the financial and tax bases of Arconic's assets and liabilities and are adjusted for changes in tax rates and tax laws when enacted.

In all periods prior to the Separation Date, the Arconic Corporation Businesses were included in the income tax filings of ParentCo. The provision for income taxes was determined in the same manner described above, but on a separate return methodology as if the Company was a standalone taxpayer filing hypothetical income tax returns where applicable. Any additional accrued tax liability or refund arising as a result of this approach was assumed to be immediately settled with ParentCo as a component of Parent Company net investment. Deferred taxes were also determined in the same manner described above and were reported in the Consolidated Balance Sheet for net operating losses, credits or other attributes to the extent that such attributes were expected to transfer to Arconic upon the Separation. Any difference from attributes generated in a hypothetical return on a separate return basis was adjusted as a component of Parent Company net investment.

Valuation allowances are recorded to reduce deferred tax assets when it is more likely than not (greater than 50%) that a tax benefit will not be realized. In evaluating the need for a valuation allowance, management considers all potential sources of taxable income, including income available in carryback periods, future reversals of taxable temporary differences, projections of taxable income, and income from tax planning strategies, as well as all available positive and negative evidence. Positive evidence includes factors such as a history of profitable operations, projections of future profitability within the carryforward period, including from tax planning strategies, and the Company's experience with similar operations. Existing favorable contracts and the ability to sell products into established markets are additional positive evidence. Negative evidence includes items such as cumulative losses, projections of future losses, or carryforward periods that are not long enough to allow for the utilization of a deferred tax asset based on existing projections of income. Deferred tax assets for which no valuation allowance is recorded may not be realized upon changes in facts and circumstances, resulting in a future charge to establish a valuation allowance. Existing valuation allowances are re-examined under the same standards of positive and negative evidence. If it is determined that it is more likely than not that a deferred tax asset will be realized, the appropriate amount of the valuation allowance, if any, is released. Deferred tax assets and liabilities are also re-measured to reflect changes in underlying tax rates due to law changes and the grant and lapse of tax holidays.

Arconic applies a tax law ordering approach when considering the need for a valuation allowance on net operating losses expected to offset Global Intangible Low Taxed Income (GILTI) income inclusions. Under this approach, reductions in cash tax savings are not considered as part of the valuation allowance assessment. Instead, future GILTI inclusions are considered a source of taxable income that support the realizability of deferred tax assets.

Tax benefits related to uncertain tax positions taken or expected to be taken on a tax return are recorded when such benefits meet a more likely than not threshold. Otherwise, these tax benefits are recorded when a tax position has been effectively settled, which means that the statute of limitations has expired or the appropriate taxing authority has completed its examination even though the statute of limitations remains open. Interest and penalties related to uncertain tax positions are recognized as part of the provision for income taxes and are accrued beginning in the period that such interest and penalties would be applicable under relevant tax law until such time that the related tax benefits are recognized.

Foreign Currency. The local currency is the functional currency for the Company's operations outside the United States with the following exception. The U.S. dollar was the functional currency for Arconic's former operations in Russia (divested in November 2022 - see Note S). The determination of the functional currency for the Company's operations is made based on the appropriate economic and management indicators.

Recently Adopted Accounting Guidance. There was no relevant guidance previously issued by the Financial Accounting Standards Board (FASB) requiring adoption in 2022 by Arconic. FASB guidance adopted in 2021 and 2020 by the Company related to accounting and/or reporting for income taxes, credit losses, and employee defined benefit plans did not have a material impact as previously disclosed in Arconic's Consolidated Financial Statements issued in the respective periods.

Recently Issued Accounting Guidance. In March 2020, the FASB issued guidance that provides optional expedients and exceptions for a limited period of time to ease the potential burden in accounting for (or recognizing the effects of) reference rate reform on financial reporting. These expedients and exceptions may be used when applying GAAP, if certain criteria are met, to contracts, hedging relationships, and other transactions that reference London Inter-Bank Offered Rate (LIBOR) or

another reference rate expected to be discontinued because of such reform. The purpose of this guidance is to provide relief to entities from experiencing unintended accounting and/or financial reporting outcomes or consequences due to reference rate reform. This guidance became effective immediately on March 12, 2020 and may be applied prospectively to contract modifications made and hedging relationships entered into or evaluated on or before December 31, 2022, after which time the expedients and exceptions expire (see below). In January 2021, the FASB issued clarifying guidance to specify that certain of the optional expedients and exceptions apply to derivatives that use an interest rate for margining, discounting, or contract price alignment that is modified as a result of reference rate reform. This additional guidance may be applied on a full retrospective basis as of any date from the beginning of an interim period that includes or is subsequent to March 12, 2020, or prospectively in the manner previously described for the guidance issued on March 12, 2020. In December 2022, the FASB extended the expiration of the optional expedients and exceptions to June 30, 2023. Through December 31, 2022, the Company has not experienced any unintended outcomes or consequences of reference rate reform that would necessitate the adoption of this guidance. Additionally, Arconic's credit agreement, which previously provided a credit facility that was referenced to LIBOR in certain borrowing situations, was amended in February 2022 to replace LIBOR with the Secured Oversight Financing Rate (SOFR) (see Note Q). Management will continue to closely monitor all potential instances of reference rate reform to determine if adoption of this guidance becomes necessary in the future.

In September 2022, the FASB issued guidance to enhance the transparency of supplier finance programs. Under the new guidance, a buyer in a supplier finance program must disclose at least annually qualitative and quantitative information about its supplier finance programs to allow a user of financial statements to understand the program's nature, activity during the period, changes from period to period, and potential magnitude. These changes become effective for Arconic on January 1, 2023. The Company has an existing arrangement with a financial institution that provides for a capacity of up to $250. Other than complying with the disclosure requirements, management has determined that the adoption of this guidance will not have an impact on the Consolidated Financial Statements.

C. Revenue from Contracts with Customers

The following table disaggregates revenue by major end market served. Differences between segment totals and consolidated Arconic are in Corporate.

For the year ended December 31,	Rolled Products	Building and Construction Systems	Extrusions	Total
2022				
Ground Transportation	$ 3,010	$ —	$ 109	$ 3,119
Packaging	1,603	—	—	1,603
Building and Construction	277	1,245	—	1,522
Aerospace	764	—	195	959
Industrial Products and Other	1,659	—	105	1,764
Total end-market revenue	$ 7,313	$ 1,245	$ 409	$ 8,967
2021				
Ground Transportation	$ 2,697	$ —	$ 102	$ 2,799
Packaging	1,217	—	—	1,217
Building and Construction	244	1,011	—	1,255
Aerospace	512	—	119	631
Industrial Products and Other	1,517	—	85	1,602
Total end-market revenue	$ 6,187	$ 1,011	$ 306	$ 7,504
2020				
Ground Transportation	$ 1,761	$ —	$ 88	$ 1,849
Packaging	773	—	—	773
Building and Construction	154	963	—	1,117
Aerospace	598	—	222	820
Industrial Products and Other	1,049	—	71	1,120
Total end-market revenue	$ 4,335	$ 963	$ 381	$ 5,679

D. Segment and Related Information

Segment Information

Arconic has three operating and reportable segments, which are organized by product on a global basis: Rolled Products, Building and Construction Systems, and Extrusions (see segment descriptions below). The Company determined the chief operating decision maker to be the CEO, who regularly reviews the financial information of these three segments to assess performance and allocate resources.

Arconic's profit or loss measure for its reportable segments is Segment Adjusted EBITDA (Earnings before interest, taxes, depreciation, and amortization). The Company calculates Segment Adjusted EBITDA as Total sales (third-party and intersegment) minus each of (i) Cost of goods sold, (ii) Selling, general administrative, and other expenses, and (iii) Research and development expenses, plus each of (i) Stock-based compensation expense, (ii) Metal price lag, and (iii) Unrealized (gains) losses on mark-to-market hedging instruments and derivatives (see below). Arconic's Segment Adjusted EBITDA may not be comparable to similarly titled measures of other companies' reportable segments.

Effective in the first quarter of 2022, management modified the Company's definition of Segment Adjusted EBITDA to exclude the impact of unrealized gains and losses on mark-to-market hedging instruments and derivatives. This modification was deemed appropriate as Arconic is considering entering in additional hedging instruments in future reporting periods if favorable conditions exist to mitigate cost inflation. Certain of these instruments may not qualify for hedge accounting resulting in unrealized gains and losses being recorded directly to Sales or Cost of goods sold, as appropriate (i.e., mark-to-market). Additionally, this change was also applied to derivatives that do not qualify for hedge accounting for consistency purposes. The Company does not have a regular practice of entering into contracts that are treated as derivatives for accounting purposes. Ultimately, this change was made to maintain the transparency and visibility of the underlying operating performance of Arconic's reportable segments. Prior to this change, the Company had a limited number of hedging instruments and derivatives that did not qualify for hedge accounting, the unrealized impact of which was not material to Arconic's Segment Adjusted EBITDA performance measure. Accordingly, prior period information presented was not recast to reflect this change.

Segment assets are comprised of customer receivables; inventories; properties, plants, and equipment, net; and goodwill.

The accounting policies of the segments are the same as those described in the Summary of Significant Accounting Policies (seeNote B). Transactions among segments are established based on negotiation among the parties.

The following are detailed descriptions of Arconic's reportable segments:

Rolled Products. This segment produces aluminum sheet and plate for a variety of end markets. Sheet and plate are sold directly to customers and through distributors related to the aerospace, ground transportation, packaging, building and construction, and industrial products (mainly used in the production of machinery and equipment and consumer durables) end markets. A small portion of this segment also produced aseptic foil for the packaging end market prior to February 1, 2020 (see Note S). While the customer base for flat-rolled products is large, a significant amount of sales of sheet and plate is to a relatively small number of customers. Prices for these products are generally based on the price of metal plus a premium for adding value to the aluminum to produce a semi-finished product, resulting in a business model in which the underlying price of metal is contractually passed-through to customers.

Building and Construction Systems. This segment manufactures products that are used primarily in the non-residential building and construction end market. These products include integrated aluminum architectural systems and architectural extrusions, which are sold directly to customers and through distributors.

Extrusions. This segment produces a range of extruded and machined parts for the aerospace, ground transportation, and industrial products end markets. These products are sold directly to customers and through distributors. Prices for these products are generally based on the price of metal plus a premium for adding value to the aluminum to produce a semi-finished product, resulting in a business model in which the underlying price of metal is contractually passed-through to customers.

The operating results and assets of Arconic's reportable segments were as follows (differences between segment totals and Arconic's consolidated totals for line items not reconciled are in Corporate):

	Rolled Products	Building and Construction Systems	Extrusions	Total
2022				
Sales:				
Third-party sales	$ 7,313	$ 1,245	$ 409	$ 8,9
Intersegment sales	44	—	2	
Total sales	$ 7,357	$ 1,245	$ 411	$ 9,0
Segment Adjusted EBITDA	$ 581	$ 195	$ (47)	$ 7
Provision for depreciation and amortization	$ 190	$ 17	$ 18	$ 2
2021				
Sales:				
Third-party sales	$ 6,187	$ 1,011	$ 306	$ 7,5
Intersegment sales	33	—	1	
Total sales	$ 6,220	$ 1,011	$ 307	$ 7,5
Segment Adjusted EBITDA	$ 655	$ 130	$ (28)	$ 7
Provision for depreciation and amortization	$ 197	$ 17	$ 23	$ 2
2020				
Sales:				
Third-party sales	$ 4,335	$ 963	$ 381	$ 5,6
Intersegment sales	19	—	2	
Total sales	$ 4,354	$ 963	$ 383	$ 5,7
Segment Adjusted EBITDA	$ 527	$ 137	$ (16)	$ 6
Provision for depreciation and amortization	$ 192	$ 18	$ 25	$ 2

2022				
Assets:				
Segment assets[1]	$ 4,308	$ 474	$ 312	$ 5,094
Supplemental information:				
Capital expenditures	194	18	16	228
Goodwill	224	68	—	292
2021				
Assets:				
Segment assets[2]	$ 4,766	$ 416	$ 381	$ 5,563
Supplemental information:				
Capital expenditures	147	11	14	172
Goodwill	253	69	—	322

(1) In the 2022 third quarter, the Extrusions segment recorded a $92 asset impairment charge comprised of $90 for Properties, plants, and equipment and $2 for intangible assets (included within Other noncurrent assets) (see Note E). Also, in November 2022, Arconic completed the sale of all of its operations in Russia (seeNote S), which was previously reported in Rolled Products.

(2) In the 2021 fourth quarter, the Rolled Products segment recorded a net adjustment of $10 (approximately $7 of which relates to prior quarters in 2021) related to write-downs of scrap inventory. The out-of-period amounts were not material to any interim or annual period.

The following tables reconcile certain segment information to consolidated totals:

For the year ended December 31,	2022	2021	2020
Sales:			
Total segment sales	$ 9,013	$ 7,538	$ 5,700
Elimination of intersegment sales	(46)	(34)	(21)
Other	(6)	—	(4)
Consolidated sales	$ 8,961	$ 7,504	$ 5,675

For the year ended December 31,	2022	2021	2020
Total Segment Adjusted EBITDA	$ 729	$ 757	$ 648
Unallocated amounts:			
Corporate expenses[1]	(29)	(33)	(24)
Stock-based compensation expense (K)	(15)	(22)	(23)
Metal price lag[2]	17	(16)	(27)
Unrealized gains on mark-to-market hedging instruments and derivatives (U)	6	—	—
Provision for depreciation and amortization	(237)	(253)	(251)
Impairment of goodwill (B & O)	—	(65)	—
Restructuring and other charges[3],[4] (E)	(456)	(624)	(188)
Other[5]	(62)	(36)	(55)
Operating (loss) income	(47)	(292)	80
Interest expense (F)	(104)	(100)	(118)
Other expenses, net (G)	(41)	(67)	(70)
Benefit (Provision) for income taxes(I)	11	62	(1)
Net income attributable to noncontrolling interest	(1)	—	—
Consolidated net loss attributable to Arconic[1]	$ (182)	$ (397)	$ (109)

(1) Corporate expenses are composed of general administrative and other expenses of operating the corporate headquarters and other global administrative facilities. The amounts presented for all periods prior to second quarter 2020 include an allocation of ParentCo's corporate expenses, including research and development expenses, for the portion of the period prior to the Separation Date (see Cost Allocations in Note A).

(2) Metal price lag represents the financial impact of the timing difference between when aluminum prices included in Sales are recognized and when aluminum purchase prices included in Cost of goods sold are realized. This adjustment aims to remove the effect of the volatility in metal prices and the calculation of this impact considers applicable metal hedging transactions.

(3) In 2022, Restructuring and other charges includes a loss of $306 related to the sale of the Company's operations in Russia and a $92 asset impairment charge related to the Extrusions segment (see Note E).

(4) In 2022, 2021, and 2020, Restructuring and other charges include a $47, $584, and $199, respectively, charge for the settlement of certain employee retirement benefits virtually all of which were within the United States and the United Kingdom (see Note H).

(5) Other includes certain items that impact Cost of goods sold and Selling, general administrative, and other expenses on the Company's Statement of Consolidated Operations that are not included in Segment Adjusted EBITDA. In 2022, Other includes costs related to environmental remediation charges of $27 (see Environmental Matters in Note T) and costs related to the new labor agreement of $19 (see Note H). These charges were recorded in Cost of goods sold on the accompanying Statement of Consolidated Operations.

December 31,	2022	2021
Assets:		
Total segment assets	$ 5,094	$ 5,563
Unallocated amounts:		
Cash and cash equivalents	261	335
Prepaid expenses and other current assets	124	54
Fair value of hedging instruments and derivatives	21	1
Corporate fixed assets, net	135	153
Operating lease right-of-use assets	115	122
Deferred income taxes (I)	188	229
Other noncurrent assets	57	88
Other	20	35
Consolidated assets	$ 6,015	$ 6,580

Customer Information

In 2021 and 2020 Arconic generated more than 10% of its consolidated sales from one customer, Ford Motor Company. These sales amounted to $761 and $647 in 2021, and 2020 respectively, and were included in the Rolled Products segment.

Geographic Area Information

Geographic information for sales was as follows (based upon the country where the point of sale occurred):

For the year ended December 31,	2022	2021	2020
Sales:			
United States	$ 5,972	$ 4,753	$ 3,697
Russia*	828	793	535
China	807	696	429
Hungary*	608	625	462
France	247	260	207
United Kingdom	245	169	144
Other	254	208	201
	$ 8,961	$ 7,504	$ 5,675

* In all periods presented, sales of a portion of aluminum products from Arconic's plant in Russia were completed through the Company's international selling company located in Hungary. Also, in November 2022, Arconic completed the sale of all of its operations in Russia (see Note S).

Geographic information for long-lived assets was as follows (based upon the physical location of the assets):

December 31,	2022	2021
Long-lived assets:		
United States	$ 1,943	$ 1,998
China	208	242
Russia*	—	200
Hungary	99	98
United Kingdom	78	79
France	18	15
Other	15	19
	$ 2,361	$ 2,651

* In November 2022, Arconic completed the sale of all of its operations in Russia (seeNote S).

E. Restructuring and Other Charges

Restructuring and other charges for each year in the three-year period ended December 31, 2022 were comprised of the following:

	2022	2021	2020
Net loss (gain) on divestitures of assets and businesses (S)	$ 306	$ 1	$ (49)
Asset impairments	92	34	15
Settlements related to employee retirement benefit plans (H)	47	584	199
Layoff costs	2	3	23
Other*	10	6	14
Reversals of previously recorded layoff and other costs	(1)	(4)	(14)
Restructuring and other charges	$ 456	$ 624	$ 188

* In 2020, Other includes $2 related to the allocation of ParentCo's corporate restructuring activity to Arconic (see Cost Allocations inNote A).

Layoff costs were recorded based on approved detailed action plans submitted by the operating locations that specified positions to be eliminated, benefits to be paid under existing severance plans, union contracts or statutory requirements, and the expected timetable for completion of the plans.

2022 Actions. In 2022, Arconic recorded Restructuring and other charges of $456, which were comprised of the following components: a $306 net loss related to the sale of the Company's operations in Russia (See Russia in Note S); a $92 asset impairment charge related to the Extrusions segment (see below); a $46 charge for the settlement of certain employee retirement benefits related to U.S. pension plans (see Note H); a $5 charge related to several legacy non-U.S. matters, including $1 for an environmental remediation obligation related to Italy (see Environmental Matters in Note T) and $1 for the full settlement of certain employee retirement benefits related to Brazil (see Note H); a $2 charge related to idling certain operations in the Extrusions segment (actions initiated in 2021); and a net $5 charge for other items.

In September 2022, management initiated a business review of the Extrusions segment aimed at identifying alternatives to improve financial performance of this segment in future periods. Management continues to assess alternatives and no decisions or commitments were made as of December 31, 2022. In connection with this review, the Company updated its five-year strategic plan, the results of which indicated that there is an expected decline in the forecasted financial performance for the Extrusions segment (and asset group), including continued forecasted losses. As such, management evaluated the recoverability of the long-lived assets of the Extrusions asset group by comparing the aggregate carrying value to the undiscounted future cash flows. The result of this evaluation was that the long-lived assets were deemed to be impaired as the aggregate carrying value exceeded the undiscounted future cash flows. The impairment charge was measured as the difference between the aggregate carrying value and aggregate fair value of the long-lived assets. Fair value was determined using an orderly liquidation methodology for the machinery and equipment and a sales comparison approach for the land and structures. Significant judgments and assumptions were applied in estimating the fair value of the long-lived assets, including the use of market-based information specific to the machinery and equipment and information from recent sales or current listings of comparable

properties for the land and structures. As a result, the Company recorded an impairment charge of $90 for the Properties, plants, and equipment and $2 for the intangible assets (included within Other noncurrent assets).

2021 Actions. In 2021, Arconic recorded Restructuring and other charges of $624, which were comprised of the following components: a $584 charge for the settlement of certain employee retirement benefits (see Note H); a $34 charge for the impairment of several buildings and equipment due to management's decision to abandon these assets located at the Company's primary research and development facility; a $7 charge related to idling certain operations in the Extrusions segment, including layoff costs associated with approximately 115 employees; a $4 net benefit for legacy tax and legal matters related to Brazil; a $4 charge related to several legal matters, including the assumption of a related environmental remediation obligation (see Environmental Matters in Note T); a $4 credit for the reversal of reserves established in prior periods (see 2020 Actions below); a $1 additional loss on the sale of an aluminum rolling mill in Brazil (see Itapissuma in Note S); and a $2 net charge for other items.

As of September 30, 2021, the employee separations associated with 2021 restructuring programs were essentially complete. In 2021, the Company made cash payments of $2 against layoff reserves related to the 2021 restructuring programs.

2020 Actions. In 2020, Arconic recorded a net charge of $188 in Restructuring and other charges, which were comprised of the following components: a $199 charge for the settlement of certain employee retirement benefits, virtually all within the United States and the United Kingdom (see Note H); a $25 benefit for contingent consideration received related to the 2018 sale of the Texarkana (Texas) rolling mill (see Note S); a $25 net gain related to the sales of an extrusions plant in South Korea and an aluminum rolling mill in Brazil (see Note S); a $21 charge for costs, of which $5 is for layoff costs associated with approximately 90 employees, related to the planned closure and related reorganizations of several small facilities in the Building and Construction Systems and Extrusions segments; an $18 charge for layoff costs associated with the separation of approximately 460 employees across the Company in response to the impact of the COVID 19 pandemic; a $14 credit for the reversal of reserves established in prior periods, including $5 related to an environmental matter (see Note T); a $4 charge for legacy non-income tax matters in Brazil; a $2 charge for an allocation of ParentCo's corporate restructuring activity (see Cost Allocations in Note A); and an $8 net charge for other items.

In 2021, the total number of employees to be separated was updated to 500 from 550 to reflect employees initially identified for separation accepting other positions within the Company and natural attrition. As of December 31, 2021, the employee separations associated with 2020 restructuring programs were essentially complete. In 2021 and 2020, the Company made cash payments of $5 and $15, respectively, against layoff reserves related to the 2020 restructuring programs.

Segment Information. The Company does not include Restructuring and other charges in the results of its reportable segments. The impact of allocating such charges to segment results would have been as follows:

For the year ended December 31,	2022	2021	2020
Rolled Products	$ 302	$ 1	$ (15)
Building and Construction Systems	2	(2)	5
Extrusions	90	7	(14)
Segment total	394	6	(24)
Corporate	62	618	212
	$ 456	$ 624	$ 188

Reserve Activity. Activity and reserve balances for restructuring charges were as follows:

	Layoff costs	Other costs	Total
Reserve balances at December 31, 2019	$ 20	$ 1	$ 21
Separation-related adjustments[1]	2	—	2
Cash payments	(24)	(3)	(27)
Restructuring charges	23	4	27
Other[2]	(8)	(1)	(9)
Reserve balances at December 31, 2020	13	1	14
Cash payments	(10)	(5)	(15)
Restructuring charges	3	6	9
Other[2]	(4)	(1)	(5)
Reserve balances at December 31, 2021	2	1	3
Cash payments	(3)	(3)	(6)
Restructuring charges	2	3	5
Other[2]	—	(1)	(1)
Reserve balances at December 31, 2022[3]	$ 1	$ —	$ 1

(1) Represents liabilities transferred from ParentCo on the Separation Date (see Note A).

(2) Other includes reversals of previously recorded restructuring charges and the effects of foreign currency translation.

(3) The remaining reserves are expected to be paid in cash during 2023.

F. Interest Cost Components

For the year ended December 31,	2022	2021	2020
Amount charged to expense	$ 104	$ 100	$ 118
Amount capitalized	7	4	6
	$ 111	$ 104	$ 124

In 2020 (January through March), total interest costs include an allocation of ParentCo's financing costs of $28 (see Cost Allocations in Note A). Also, in 2020, total interest costs include $19 for the write-off and immediate expensing of certain debt issuance costs related to a debt refinancing (see Note Q). Typically, such costs are deferred and amortized to interest expense over the term of the related financing arrangement.

G. Other Expenses, Net

For the year ended December 31,	2022	2021	2020
Non-service costs — Pension and OPEB (H)	$ 75	$ 60	$ 78
Foreign currency (gains) losses, net	(26)	5	11
Net loss from asset sales	3	—	—
Interest income	(5)	(1)	(4)
Other, net	(6)	3	(15)
	$ 41	$ 67	$ 70

In 2022, Foreign currency (gains) losses, net includes a $39 gain for the remeasurement of monetary balances, primarily cash, related to the Company's former operations in Russia (see Note S) from rubles to the U.S. dollar. This gain was the result of a significant strengthening of the Russian ruble against the U.S. dollar in the period.

In 2020, Other, net includes a $20 benefit for the reversal of a liability previously established at the Separation Date related to a potential indemnification to Howmet by Arconic for an outstanding income tax matter in Spain. Under the terms of the Tax Matters Agreement (see Note A) related to the Separation, Arconic was responsible for 34% of the potential loss related to this matter should Howmet receive an unfavorable ruling from Spain's Supreme Court. At the time of Separation, Arconic

management believed that the likelihood of the Company performing under the indemnification was probable resulting in a liability being established on Arconic's opening balance sheet at the Separation Date. In November 2020, a favorable ruling was received from Spain's Supreme Court bringing a final conclusion to this matter as this decision may not be appealed any further. As no further income tax payment was required of Howmet likewise Arconic no longer has a requirement to perform under the indemnification.

H. Pension and Other Postretirement Benefits

Defined Benefit Pension and Other Postretirement Benefit Plans

Arconic sponsors several defined benefit pension and other postretirement plans covering eligible employees and retirees in U.S. and foreign locations. Pension benefits generally depend on length of service and job grade. Substantially all benefits are paid through pension trusts that are sufficiently funded to ensure that the related plans can pay benefits to retirees as they become due. Most salaried and non-bargaining hourly U.S. employees hired after March 1, 2006, participate in a defined contribution plan instead of a defined benefit plan. The Company has one health care and life insurance postretirement benefit plan covering eligible U.S. retirees. The plan is unfunded and pays a percentage of medical expenses, reduced by deductibles and other coverage. Life benefits are provided by insurance contracts. Arconic retains the right, subject to existing agreements, to change or eliminate these benefits. Certain bargaining hourly U.S. employees hired on or after July 1, 2010 and all salaried and non-bargaining hourly U.S. employees are not eligible for postretirement health care benefits. All salaried and most hourly U.S. employees are not eligible for postretirement life insurance benefits.

United Steelworkers Labor Agreement—On May 14, 2022, the Company and the United Steelworkers reached a tentative four-year labor agreement covering approximately 3,300 employees at four U.S. locations; the previous labor agreement expired on May 15, 2022. The tentative agreement was ratified by the union employees on June 1, 2022. In 2022, Arconic recognized $19 in Cost of goods sold on the accompanying Statement of Consolidated Operations primarily for a one-time signing bonus for the covered employees. Additionally, the new labor agreement provides for, among other items, established annual wage increases and higher multipliers used to calculate the union employees' future pension benefits.

The change to the pension benefits qualifies as a significant plan amendment to the Company's U.S. hourly defined benefit pension plan, and, as a result, Arconic was required to complete a remeasurement of this plan (generally completed on an annual basis as of December 31st), including an interim actuarial valuation of the plan obligations. Communication of the benefit change to the union employees occurred on May 15, 2022, and the effective date of this amendment was May 16, 2022. For purposes of performing an interim remeasurement of the plan, the Company applied a practical expedient to the remeasurement date and selected May 31, 2022. Accordingly, the discount rate used in calculating the plan obligations increased to 4.66% at May 31, 2022 from 2.96% at December 31, 2021. The remeasurement of this plan, together with the amendment for increased benefits, resulted in a $13 net decrease to Accrued pension benefits and a $10 (after-tax) net decrease to Accumulated other comprehensive loss (see Note L) on the accompanying Consolidated Balance Sheet. Additionally, annual net periodic benefit cost to be recognized for this plan in 2022 increased by $8, comprised of a $2 decrease in service cost and a $10 increase in non-service costs.

U.S. Pension Plan Interim Settlement—In September 2022, management concluded that it was probable that lump-sum benefit payments expected to be paid in 2022 under Arconic's U.S. salary defined benefit pension plan will exceed the pre-determined threshold (sum of annual service cost and interest cost) requiring settlement accounting. As a result, the Company was required to complete a remeasurement of this plan (generally completed on an annual basis as of December 31st), including an interim actuarial valuation of the plan obligations. For purposes of performing an interim remeasurement of the plan, Arconic applied a practical expedient to the remeasurement date and selected September 30, 2022. Accordingly, the discount rate used in calculating the plan obligations increased to 5.71% at September 30, 2022 from 2.82% at December 31, 2021. The remeasurement of this plan, together with the settlement of benefits, resulted in a $34 net decrease to Accrued pension benefits and a $26 (after-tax) net decrease to Accumulated other comprehensive loss (see Note L) on the accompanying Consolidated Balance Sheet. Unfavorable plan asset performance offset most of the impact of the increase in the discount rate. Also, the settlement resulted in the accelerated amortization of a portion of the existing net actuarial loss associated with this plan in the amount of $15 ($12 after-tax). This amount was reclassified to earnings through Restructuring and other charges (see Note E) from Accumulated other comprehensive loss (see Note L). Additionally, annual net periodic benefit cost to be recognized for this plan in 2022 increased by $8, all of which relates to non-service costs.

U.S. Pension Plan Annuitizations—In April 2021, Arconic purchased a group annuity contract to transfer the obligation to pay the remaining retirement benefits of approximately 8,400 participants in two U.S. defined benefit pension plans to an insurance company. In connection with this transaction, the Company contributed a total of $250 to the two plans to maintain the funding level of the remaining plan obligations not transferred. This contribution was funded with the net proceeds from a

March 2021 debt offering (see 2021 Activity in Note Q). Prior to this action, these two plans had an aggregate of approximately 23,000 participants.

This transaction represents a significant settlement event, and, as a result, the Company was required to complete a remeasurement of these two plans (generally completed on an annual basis as of December 31st), including an interim actuarial valuation of the plan obligations. Accordingly, the weighted-average discount rate used in calculating the plan obligations increased to 3.10% as of April 30, 2021 from 2.54% as of December 31, 2020. The remeasurement resulted in a combined projected benefit obligation and fair value of plan assets of $3,337 and $2,790, respectively, as of April 30, 2021. From these amounts, the group annuity transaction resulted in the settlement of $995 in plan obligations and the transfer of $1,007 in plan assets. The remeasurement of these two plans, together with the annuitization, resulted in a $152 net decrease to Accrued pension benefits and a $117 (after-tax) net decrease to Accumulated other comprehensive loss (see Note L). Additionally, the annuitization resulted in the accelerated amortization of a portion of the existing net actuarial loss associated with these two plans in the amount of $549 ($423 after-tax). This amount was reclassified to earnings through Restructuring and other charges (see Note E) from Accumulated other comprehensive loss (see Note L).

In December 2020, Arconic purchased a group annuity contract to transfer the obligation to pay the remaining retirement benefits of approximately 7,000 participants from two U.S. defined benefit pension plans to an insurance company. On a combined basis, this transaction resulted in the settlement of approximately $340 in plan obligations and the transfer of approximately $245 in plan assets. Prior to this action, these two plans had approximately 30,000 participants combined. The Company recognized a $140 ($108 after-tax) settlement charge, which represents the accelerated amortization of a portion of the existing net actuarial loss associated with these plans. This amount was reclassified to earnings through Restructuring and other charges (see Note E) from Accumulated other comprehensive loss (see Note L).

U.S. OPEB Plan Amendments—In August 2021, Arconic modified the medical benefit coverage offered to certain Medicare-eligible participants under the Company's U.S. other postretirement benefit plan. Effective January 1, 2022, this modification results in lower premiums and increased benefits to the participants. This change qualifies as a significant plan amendment to the Company's U.S. other postretirement benefit plan. Accordingly, this plan was required to be remeasured, and through this process, the discount rate was updated to 2.78% at August 31, 2021 from 2.61% at December 31, 2020. The amendment, together with the remeasurement of this plan, resulted in a $34 net decrease to the Company's other postretirement benefits liability and a $26 (after-tax) net decrease to Accumulated other comprehensive loss (see Note L) on the accompanying Consolidated Balance Sheet. The impact of this change on the Company's annual net periodic benefit cost is not material. The Company's estimated annual benefit payments will decrease by approximately $4 beginning in 2022.

In July 2020, Arconic and the United Steelworkers agreed to modify the medical benefit coverage offered to certain Medicare-eligible participants under the Company's U.S. other postretirement benefit plan, as provided for in the current master collective bargaining agreement between the parties. Effective January 1, 2021, this modification results in lower premiums and increased benefits to the participants. This change qualifies as a significant plan amendment to the Company's U.S. other postretirement benefit plan. Accordingly, this plan was required to be remeasured, and through this process, the discount rate was updated to 2.54% at July 31, 2020 from 3.17% at December 31, 2019. The amendment, together with the remeasurement of this plan, resulted in a net decrease to both the Company's other postretirement benefits liability of $7 and Accumulated other comprehensive loss of $5 (after-tax). The impact of this change on the Company's annual net periodic benefit cost is not material. The Company's estimated annual benefit payments decreased by approximately $20 beginning in 2021.

U.K. Pension Plan Annuitization—In June 2020, Arconic and Howmet, together, executed several liability management actions related to approximately 1,800 participants in a U.K. defined benefit pension plan. The primary action was the purchase of a group annuity contract to transfer the obligation to pay the remaining retirement benefits of certain plan participants to an insurance company. On a combined basis, these actions resulted in the settlement of approximately $400 in plan obligations and the transfer of approximately $460 in plan assets. In the 2020 second quarter, the Company contributed $10 to the plan to facilitate these actions and maintain the funding level of the remaining plan obligations. Prior to these actions, this plan had approximately 3,350 participants combined.

Accordingly, this plan was required to be remeasured, and through this process, the discount rate was updated to 1.55% at June 30, 2020 from 2.05% at December 31, 2019. The settlement events, together with the remeasurement of the plan, resulted in an approximately $250 net reduction to the Company’s remaining plan obligation and both a decrease to the Company’s pension benefit asset and a settlement charge of $58 ($48 after-tax) in 2020. The settlement charge represents the accelerated amortization of a portion of the existing net actuarial loss associated with this plan. This amount was reclassified to earnings through Restructuring and other charges (see Note E) from Accumulated other comprehensive loss. Subsequent to this remeasurement, the remaining respective plan obligations and plan assets attributable to Arconic and Howmet were transferred into separate plans and the existing U.K. plan was terminated. Immediately following the completion of the transfer, the

Company's remaining plan obligation was approximately $240 and the plan assets were approximately $260 related to 1,050 plan participants.

The Separation—The above descriptions of retirement benefits for Arconic participants also describe the retirement benefits provided to the employees and retirees of ParentCo prior to the Separation Date.

Prior to the Separation Date for certain non-U.S. plans, eligible employees and retirees related to the Arconic Corporation Businesses participated in ParentCo-sponsored defined benefit pension and other postretirement plans (the "Shared Plans"), which included participants related to the Howmet Aerospace Businesses and ParentCo corporate participants, as well as eligible retirees from previously closed or sold operations. Also, prior to the Separation Date, other eligible employees and retirees related to the Arconic Corporation Businesses participated in certain non-U.S. defined benefit pension and other postretirement plans (the "Direct Plans").

The Company accounted for the portion of the Shared Plans related to its employees as multiemployer benefit plans. Accordingly, Arconic did not record an asset or liability to recognize the funded status of the Shared Plans. However, the related pension and other postretirement benefit expenses attributable to Arconic were based primarily on pensionable compensation of active Arconic participants and estimated interest costs, respectively. The Company also recorded an allocation of pension and other postretirement benefit expenses for the Shared Plans attributable to ParentCo corporate participants, as well as to participants related to closed and sold operations (see Cost Allocations in Note A).

The Direct Plans were accounted for as defined benefit pension and other postretirement plans. Accordingly, the funded status of each Direct Plan was recorded in the Company's Consolidated Balance Sheet. Actuarial gains and losses that had not yet been recognized in earnings were recorded in Accumulated other comprehensive loss until they were amortized as a component of net periodic benefit cost. The determination of benefit obligations and recognition of expenses related to Direct Plans were dependent on various assumptions, including, among others, discount rates, long-term expected rates of return on plan assets, and future compensation increases. Management developed each assumption using relevant company experience in conjunction with market-related data for each of the plans.

In preparation for the Separation, effective January 1, 2020, certain U.S. pension and other postretirement benefit plans previously sponsored by ParentCo (the "U.S. Shared Plans" – see above) were separated into standalone plans for both Arconic (the "New Direct Plans") and Howmet. Accordingly, on January 1, 2020, Arconic recognized an aggregate liability of $1,920, of which $60 was current, reflecting the combined net funded status of the New Direct Plans, comprised of a benefit obligation of $4,255 and plan assets of $2,335, as well as $1,752 (net of tax impact) in Accumulated other comprehensive loss representing a net actuarial loss.

Additionally, effective on the Separation Date, certain other Shared Plans (the "Additional New Direct Plans," and, collectively with the Direct Plans and New Direct Plans, the "Cumulative Direct Plans") were assumed by Arconic. Accordingly, on the Separation Date, Arconic recognized a noncurrent asset of $65 and a noncurrent liability of $15, reflecting the combined net funded status of the Additional New Direct Plans, as well as $50 (net of tax impact) in Accumulated other comprehensive loss representing a net actuarial loss.

The funded status of Arconic's Cumulative Direct Plans is measured as of December 31 each calendar year. All the information that follows for pension and other postretirement benefit plans is only applicable to the Cumulative Direct Plans, as appropriate.

Obligations and Funded Status

	Pension benefits		Other postretirement benefits	
December 31,	**2022**	**2021**	**2022**	**2021**
Change in benefit obligation				
Benefit obligation at beginning of year	$ 2,817	$ 4,081	$ 440	$ 514
Service cost	16	21	5	6
Interest cost	75	63	10	11
Amendments	23	—	—	(30)
Actuarial gains[1]	(722)	(105)	(96)	(23)
Benefits paid	(135)	(183)	(30)	(38)
Settlements	(78)	(1,051)	—	—
Foreign currency translation impact	(33)	(9)	—	—
Divestitures	(2)	—	—	—
Benefit obligation at end of year[2]	$ 1,961	$ 2,817	$ 329	$ 440
Change in plan assets				
Fair value of plan assets at beginning of year	$ 2,124	$ 2,754	$ —	$ —
Actual return on plan assets	(535)	177	—	—
Employer contributions	31	458	—	—
Benefits paid	(127)	(176)	—	—
Settlements	(78)	(1,069)	—	—
Foreign currency translation impact	(32)	(3)	—	—
Administrative expenses	—	(17)	—	—
Fair value of plan assets at end of year[2]	$ 1,383	$ 2,124	$ —	$ —
Funded status[2]	$ (578)	$ (693)	$ (329)	$ (440)
Amounts recognized on the Consolidated Balance Sheet:				
Noncurrent assets	$ 15	$ 32	$ —	$ —
Current liabilities	(7)	(8)	(27)	(29)
Noncurrent liabilities	(586)	(717)	(302)	(411)
Net amount recognized	$ (578)	$ (693)	$ (329)	$ (440)
Amounts recognized in Accumulated Other Comprehensive Loss (pretax):				
Net actuarial loss	$ 1,165	$ 1,389	$ 62	$ 166
Prior service cost (benefit)	20	—	(77)	(85)
Net amount recognized	$ 1,185	$ 1,389	$ (15)	$ 81
Other changes in plan assets and benefit obligations recognized in Other Comprehensive Income (pretax):				
Net actuarial gain	$ (103)	$ (137)	$ (96)	$ (23)
Amortization of net actuarial loss (includes settlements)	(121)	(678)	(8)	(8)
Prior service cost (benefit)	23	—	—	(30)
Amortization of prior service (cost) benefit	(3)	—	8	6
Total	$ (204)	$ (815)	$ (96)	$ (55)

(1) At December 31, 2022 and 2021, actuarial gains for pension benefits include approximately $(725) and $(130), respectively, and for other postretirement benefits includes approximately $(100) and $(15), respectively, attributable to the change in the discount rate used to determine the benefit obligation (see "Assumptions" below).

(2) At December 31, 2022, the benefit obligation, fair value of plan assets, and funded status for U.S. pension plans were $1,716, $1,167, and $549, respectively. At December 31, 2021, the benefit obligation, fair value of plan assets, and funded status for U.S. pension plans were $2,398, $1,744, and $654, respectively.

Pension Plan Benefit Obligations

December 31,	Pension benefits 2022	Pension benefits 2021
The projected benefit obligation and accumulated benefit obligation for all defined benefit pension plans was as follows:		
Projected benefit obligation	$ 1,961	$ 2,817
Accumulated benefit obligation	1,956	2,807
The aggregate projected benefit obligation and fair value of plan assets for defined benefit pension plans with projected benefit obligations in excess of plan assets was as follows:		
Projected benefit obligation	1,759	2,472
Fair value of plan assets	1,167	1,747
The aggregate accumulated benefit obligation and fair value of plan assets for defined benefit pension plans with accumulated benefit obligations in excess of plan assets was as follows:		
Accumulated benefit obligation	1,755	2,464
Fair value of plan assets	1,167	1,747

Components of Net Periodic Benefit Cost

For the year ended December 31,	Pension benefits[1] 2022	Pension benefits[1] 2021	Pension benefits[1] 2020	Other postretirement benefits 2022	Other postretirement benefits 2021	Other postretirement benefits 2020
Service cost	$ 16	$ 21	$ 21	$ 5	$ 6	$ 5
Interest cost	75	63	108	10	11	13
Expected return on plan assets	(87)	(110)	(170)	—	—	—
Amortization of net actuarial loss	74	94	123	8	8	8
Amortization of prior service benefit	3	—	—	(8)	(6)	(4)
Settlements[2]	47	584	199	—	—	—
Net periodic benefit cost[3]	$ 128	$ 652	$ 281	$ 15	$ 19	$ 22

(1) In 2022 and 2021, net periodic benefit cost for U.S pension plans was $127 and $653, respectively.

(2) In 2022, Settlements were due to the payment of lump-sum benefits. In 2021, Settlements were due to the purchase of a group annuity contract ($549 - see U.S. Pension Plan Annuitizations above) and the payment of lump-sum benefits ($35). In 2020, Settlements were due to two separate purchases of a group annuity contract (see U.S. Pension Plan Annuitizations and U.K. Pension Plan Annuitization above).

(3) Service cost was included within Cost of goods sold, Settlements were included within Restructuring and other charges, and all other components were included in Other expenses (income), net on the accompanying Statement of Consolidated Operations.

Assumptions

Weighted average assumptions used to determine benefit obligations and net periodic benefit cost for pension and other postretirement benefit plans were as follows:

	Benefit obligations December 31, 2022	Benefit obligations December 31, 2021	Net periodic benefit cost For the year ended December 31, 2022	Net periodic benefit cost For the year ended December 31, 2021	Net periodic benefit cost For the year ended December 31, 2020
Discount rate—pension plans	5.49 %	2.76 %	2.76 %	2.27 %	2.86 %
Discount rate—other postretirement benefit plans	5.62	2.90	2.99	2.19	2.49
Rate of compensation increase—pension plans	2.67	2.66	2.66	2.54	3.20
Expected long-term rate of return on plan assets—pension plans	—	—	5.18	4.91	6.09

The discount rate is determined using a Company-specific yield curve model (above-median) developed with the assistance of an external actuary. The cash flows of the projected benefit obligations are discounted using a single equivalent rate derived from yields on high quality corporate bonds, which represent a broad diversification of issuers in various sectors. The yield curve model parallels the projected plan cash flows, which have a weighted average duration of 11 years, and the underlying cash flows of the bonds included in the model exceed the cash flows needed to satisfy the plan obligations multiple times. If a deep market of high quality corporate bonds does not exist in a country, then the yield on government bonds is used.

The expected long-term rate of return on plan assets is generally applied to a five-year market-related value of plan assets (the fair value at the plan measurement date is used for certain non-U.S. plans). The process used by management to develop this assumption is one that relies on forward-looking investment returns by asset class. Management incorporates expected future investment returns on current and planned asset allocations using information from various external investment managers and consultants, as well as management's own judgment. For 2022 and 2021, the expected long-term rate of return used by management was based on the prevailing and planned strategic asset allocations, as well as estimates of future returns by asset class. For 2023, management anticipates that the weighted-average expected long-term rate of return will be in the range of 5.50% to 6.50%.

Weighted-average assumed health care cost trend rates for U.S. other postretirement benefit plans were as follows (non-U.S. plans are not material):

	2022	2021
Health care cost trend rate assumed for next year	6.3 %	4.5 %
Rate to which the cost trend rate gradually declines	4.5 %	4.7 %
Year that the rate reaches the rate at which it is assumed to remain	2029	2027

The assumed health care cost trend rate is used to measure the expected cost of gross eligible charges covered by the Company's other postretirement benefit plans. For 2023, a 6.3% trend rate will be used, reflecting management's best estimate of the change in future health care costs covered by the plans.

Plan Assets

Arconic's pension plan investment policy and weighted average asset allocations at December 31, 2022 and 2021, by asset class, were as follows:

		Plan assets at December 31,	
Asset class	Policy maximum	2022	2021
Equities	40%	22 %	31 %
Fixed income	100%	62	56
Other investments	30%	16	13
Total		100 %	100 %

The principal objectives underlying the investment of the pension plan assets are to ensure that Arconic can properly fund benefit obligations as they become due under a broad range of potential economic and financial scenarios, maximize the long-term investment return with an acceptable level of risk based on such obligations, and broadly diversify investments across and within various asset classes to protect asset values against adverse movements. The use of derivative instruments is permitted where appropriate and necessary for achieving diversification across the balance of the asset portfolio. Investment practices comply with the requirements of applicable country laws and regulations.

The following section describes the valuation methodologies used by the trustees to measure the fair value of pension plan assets. For plan assets measured at net asset value, this refers to the net asset value of the investment on a per share basis (or its equivalent) as a practical expedient. Otherwise, an indication of the level in the fair value hierarchy in which each type of asset is generally classified is provided (see Note U for the definition of fair value and a description of the fair value hierarchy).

Equities—These securities consist of: (i) direct investments in the stock of publicly traded U.S. and non-U.S. companies and are valued based on the closing price reported in an active market on which the individual securities are traded (generally classified in Level 1); (ii) the plans' share of commingled funds that are invested in the stock of publicly traded companies and are valued at net asset value; and (iii) direct investments in long/short equity hedge funds and private equity (limited partnerships and venture capital partnerships) and are valued at net asset value.

Fixed income—These securities consist of: (i) U.S. government debt and are generally valued using quoted prices (included in Level 1); and (ii) publicly traded U.S. and non-U.S. fixed interest obligations (principally corporate bonds and debentures) and are valued through consultation and evaluation with brokers in the institutional market using quoted prices and other observable market data (included in Level 2).

Other investments—These investments include, among others: (i) real estate investment trusts valued based on the closing price reported in an active market on which the investments are traded (included in Level 1); (ii) the plans' share of commingled funds that are invested in real estate partnerships and are valued at net asset value; (iii) direct investments in discretionary and systematic macro hedge funds and private real estate (includes limited partnerships) and are valued at net asset value; and (iv) absolute return strategy funds and are valued at net asset value.

The fair value methods described above may not be indicative of net realizable value or reflective of future fair values. Additionally, while Arconic believes the valuation methods used by the plans' trustees are appropriate and consistent with other market participants, the use of different methodologies or assumptions to determine the fair value of certain financial instruments could result in a different fair value measurement at the reporting date.

The following table presents the fair value of pension plan assets classified under either the appropriate level of the fair value hierarchy or net asset value:

December 31, 2022	Level 1	Level 2	Level 3	Net Asset Value	Total
Equities:					
Equity securities	$ 1	$ —	$ —	$ 120	$ 121
Long/short equity hedge funds	—	—	—	23	23
Private equity	—	—	—	156	156
	$ 1	$ —	$ —	$ 299	$ 300
Fixed Income:					
Intermediate and long duration government/credit	$ 113	$ 277	$ —	$ 369	$ 759
Other	5	—	—	91	96
	$ 118	$ 277	$ —	$ 460	$ 855
Other investments:					
Real estate	$ —	$ —	$ —	$ 93	$ 93
Discretionary and systematic macro hedge funds	—	—	—	100	100
Other	—	—	—	29	29
	$ —	$ —	$ —	$ 222	$ 222
Total*	$ 119	$ 277	$ —	$ 981	$ 1,377

December 31, 2021	Level 1	Level 2	Level 3	Net Asset Value	Total
Equities:					
Equity securities	$ 12	$ —	$ —	$ 392	$ 404
Long/short equity hedge funds	—	—	—	24	24
Private equity	—	—	—	224	224
	$ 12	$ —	$ —	$ 640	$ 652
Fixed Income:					
Intermediate and long duration government/credit	$ 95	$ 412	$ —	$ 612	$ 1,119
Other	23	—	—	50	73
	$ 118	$ 412	$ —	$ 662	$ 1,192
Other investments:					
Real estate	$ —	$ —	$ —	$ 108	$ 108
Discretionary and systematic macro hedge funds	—	—	—	99	99
Other	—	—	—	77	77
	$ —	$ —	$ —	$ 284	$ 284
Total*	$ 130	$ 412	$ —	$ 1,586	$ 2,128

* As of December 31, 2022 and 2021, the total fair value of pension plan assets excludes a net receivable of $3 and net payable of $4, respectively, which represents securities not yet settled plus interest and dividends earned on various investments.

Funding and Cash Flows

It is Arconic's policy to contribute amounts to funded defined benefit pension plans sufficient to meet the minimum requirements set forth in applicable country employee benefit and tax regulations, including ERISA for U.S. plans. From time to time, Arconic may contribute additional amounts as deemed appropriate. In 2022, 2021, and 2020, cash contributions to Arconic's funded defined benefit pension plans were $31, $458, and $271, respectively. The 2021 cash contributions include a total of $250 made by the Company in April 2021 to itstwo funded U.S. defined benefit pension plans to maintain the funding level of the remaining plan obligations not transferred under a group annuity contract (see U.S. Pension Plan Annuitizations above). The minimum required contributions to Arconic's funded defined benefit pension plans in 2023 are estimated to be $38, of which $31 is for U.S. plans. Benefit payments expected to be paid to pension (funded and unfunded) and other postretirement benefit plan participants are as follows:

For the year ended December 31,	Pension benefits	Other postretirement benefits
2023	$ 142	$ 27
2024	142	27
2025	143	27
2026	146	27
2027	145	26
2028 through 2032	719	127
	$ 1,437	$ 261

Defined Contribution Plans

Arconic sponsors savings and investment plans in the United States and certain other countries. Prior to the Separation Date, employees attributable to the Arconic Businesses participated in ParentCo-sponsored plans. In the United States, employees may contribute a portion of their compensation to the plans, and Arconic (ParentCo prior to the Separation Date) matches a specified percentage of these contributions in equivalent form of the investments elected by the employee. Also, Arconic (ParentCo prior to the Separation Date) makes contributions to a retirement savings account based on a percentage of eligible compensation for certain U.S. employees. Arconic's expenses (contributions) related to all defined contribution plans were $45 in 2022, $39 in 2021, and $35 in 2020.

I. Income Taxes

(Benefit) Provision for income taxes. The components of loss before income taxes were as follows:

For the year ended December 31,	2022	2021	2020
Domestic - United States	$ (65)	$ (611)	$ (126)
Foreign	(127)	152	18
	$ (192)	$ (459)	$ (108)

(Benefit) Provision for income taxes consisted of the following:

For the year ended December 31,	2022	2021	2020
Current:			
U.S. federal*	$ 8	$ —	$ —
Foreign	23	36	13
U.S. state and local	3	2	4
	34	38	17
Deferred:			
U.S. federal*	(18)	(86)	(12)
Foreign	(1)	(2)	4
U.S. state and local	(26)	(12)	(8)
	(45)	(100)	(16)
Total	$ (11)	$ (62)	$ 1

* Includes U.S. income taxes related to foreign income. Also, in 2020, the Deferred amount includes a $21 charge related to income generated by the Company prior to the Separation Date that was included in ParentCo's 2020 tax return.

A reconciliation of the U.S. federal statutory rate to Arconic's effective tax rate was as follows (the effective tax rate was a benefit on loss in 2022, a benefit on loss in 2021, and a provision on loss in 2020):

For the year ended December 31,	2022	2021	2020
U.S. federal statutory rate	21.0 %	21.0 %	21.0 %
Taxes on foreign operations - rate differential	14.4	0.1	(4.8)
Other taxes related to foreign operations[1]	(2.4)	(5.0)	(9.4)
U.S. state and local taxes, including federal benefit	0.1	2.6	3.3
Statutory tax rate and law changes	(2.8)	—	(2.1)
Changes in valuation allowances	13.1	(0.9)	(7.3)
Non-taxable income - indemnification liability[2]	0.4	0.4	3.8
Subsidiary recapitalizations and reorganizations	—	(1.1)	(3.9)
Impairment of goodwill	—	(3.0)	—
Non-deductible loss related to sale of Russian operations (S)	(37.9)	—	—
Changes in uncertain tax positions	11.2	(0.1)	—
Stock-based compensation	3.1	0.8	0.9
Non-deductible costs related to the Separation (A)	—	—	(2.2)
Write-off of deferred tax assets due to remote utilization[3]	(13.0)	—	—
Non-deductible officer compensation	(1.3)	(0.2)	—
Other	(0.2)	(1.1)	(0.2)
Effective tax rate	5.7 %	13.5 %	(0.9)%

(1) In 2021, this line item includes the impact of incremental income tax expense of $11 related to foreign operations that generated income subject to the global intangible low-taxed income inclusion under the U.S. Internal Revenue Code.

(2) In 2020, this line item reflects the impact of the absence of income tax expense for non-taxable income generated by the reversal of a liability previously established at the Separation Date related to a potential indemnification to Howmet by Arconic for an outstanding income tax matter in Spain (see Note G).

(3) In 2022, this line item reflects the write-off of a foreign subsidiary's deferred tax assets on the basis of remote likelihood of utilization. The deferred tax assets were previously fully offset by a reserve for an uncertain tax position (see Uncertain tax positions below) and a valuation allowance (see Deferred income taxes below). The Changes in valuation allowances and Changes in uncertain tax positions line items include 1.8% and 11.2%, respectively, related to the foreign subsidiary deferred tax write-off.

Deferred income taxes. The components of deferred tax assets and liabilities based on the underlying attributes without regard to jurisdiction were as follows:

	2022		2021	
December 31,	Deferred tax assets	Deferred tax liabilities	Deferred tax assets	Deferred tax liabilities
Employee benefits	$ 264	$ —	$ 331	$ —
Tax loss carryforwards	181	—	206	—
Deferred income/expense*	45	—	47	—
Interest	42	—	44	—
Operating lease right-of-use assets and liabilities	25	25	30	30
Loss provisions	27	—	24	—
Inventory accounting method change*	—	63	—	97
Depreciation	13	245	13	267
Other	4	13	17	11
	$ 601	$ 346	$ 712	$ 405
Valuation allowance	(70)	—	(90)	—
	$ 531	$ 346	$ 622	$ 405

* In 2021, an accounting method change was filed to revoke the U.S. tax LIFO election. In 2022, an accounting method change was filed related to inventory cost capitalization.

The following table details the expiration periods of the deferred tax assets presented above:

December 31, 2022	Expires within 10 years	Expires within 11-20 years	No expiration[1]	Other[2]	Total
Tax loss carryforwards	$ 42	$ 31	$ 108	$ —	$ 181
Employee benefits	—	—	—	264	264
Other	—	1	42	113	156
Valuation allowance	(42)	—	(7)	(21)	(70)
	$ —	$ 32	$ 143	$ 356	$ 531

(1) Deferred tax assets with no expiration may still have annual limitations on utilization.

(2) Employee benefits will become deductible for tax purposes over an extended period of time as contributions are made to employee benefit plans and payments are made to participants. Other represents deferred tax assets whose expiration is dependent upon the reversal of the underlying temporary difference.

The total deferred tax asset (net of valuation allowance) is supported by projections of future taxable income exclusive of reversing temporary differences (35%) and taxable temporary differences that reverse within the carryforward period (65%).

The following table details the changes in the valuation allowance:

December 31,	2022	2021	2020
Balance at beginning of year	$ 90	$ 91	$ 113
Establishment of new allowances[1]	2	3	—
Net change to existing allowances[2]	11	(3)	(16)
Separation-related adjustments	—	—	22
Acquisitions and divestitures	(7)	—	(31)
Release of allowances[3]	(21)	—	—
Foreign currency translation	(5)	(1)	3
Balance at end of year	$ 70	$ 90	$ 91

(1) This line item reflects valuation allowances initially established as a result of a change in management's judgement regarding the realizability of deferred tax assets.

(2) This line item reflects movements in previously established valuation allowances, which increase or decrease as the related deferred tax assets increase or decrease. Such movements occur as a result of remeasurement due to a tax rate change and changes in the underlying attributes of the deferred tax assets, including expiration of the attribute and reversal of the temporary difference that gave rise to the deferred tax assets.

(3) This line item reflects valuation allowances released as a result of a change in management's judgement regarding the realizability of deferred tax assets. In 2022, Arconic released a U.S. state valuation allowance ($18) after completing a legal entity reorganization, which resulted in management concluding it is now more likely than not the Company will realize the benefits of the tax attributes. Additionally, in 2022, the Company released a foreign valuation allowance in connection with the write-off of the related deferred tax assets ($3 – see footnote 3 to the reconciliation of the U.S. federal statutory rate to Arconic's effective tax rate above.)

Undistributed net earnings. Foreign undistributed net earnings that have not otherwise previously been subject to U.S. tax are generally exempt from U.S. tax if repatriated in the future. Such future distributions, as well as distributions of previously taxed foreign earnings, may be subject to U.S. state and/or foreign withholding taxes in certain jurisdictions. Also, foreign currency gains/losses related to the translation of previously taxed foreign earnings from the functional currency to the U.S. dollar may be subject to U.S. tax if such earnings were to be distributed in the future. At this time, management has no plans to repatriate such earnings in the foreseeable future, unless it is tax efficient to do so. Management continuously evaluates the Company's local and global cash needs for future business operations and anticipated debt facilities, which may influence future repatriation decisions. If such earnings were to be distributed in the future, management does not expect the potential U.S. state and/or foreign withholding taxes to be material to the Company's Consolidated Financial Statements.

The undistributed earnings of the Company's Canadian subsidiary are expected to be repatriated to the U.S. in a future period. The associated withholding tax and other costs are immaterial.

Uncertain tax positions. Arconic and its subsidiaries file income tax returns in various U.S. federal, U.S. state, and foreign jurisdictions. For U.S. federal income tax purposes, Arconic's U.S. operations were included in the income tax filings of ParentCo's U.S. consolidated tax group through March 31, 2020. ParentCo's U.S. federal income tax filings have been examined for all periods through 2020. In 2021, the Company's U.S. consolidated tax group filed a nine-month U.S. federal income tax return (April 1, 2020 through December 31, 2020). The Company's U.S. federal income tax filings for 2020 and 2021 are subject to income tax examination. For U.S. state income tax purposes, Arconic and its subsidiaries remain open to examination for the 2019 tax year and forward. For foreign income tax purposes, Arconic and its subsidiaries remain subject to income tax examinations for the 2014 tax year and forward.

A reconciliation of the beginning and ending amount of unrecognized tax benefits (excluding interest and penalties) was as follows:

December 31,	2022	2021	2020
Balance at beginning of year	$ 22	$ 23	$ 21
Additions for tax positions of prior years	—	1	—
Reductions for tax positions of prior years*	(20)	—	—
Settlements with tax authorities	(1)	—	—
Foreign currency translation	(1)	(2)	2
Balance at end of year	$ —	$ 22	$ 23

* See footnote 3 to the reconciliation of the U.S. federal statutory rate to Arconic's effective tax rate abov.

Unrecognized tax benefits, if recognized, would not impact the annual effective tax rate for 2022, 2021, and 2020. Management does not anticipate that changes in the Company's unrecognized tax benefits will have a material impact on the Statement of Consolidated Operations during 2023.

It is Arconic's policy to recognize interest and penalties related to income taxes as a component of the (Benefit) Provision for income taxes on the accompanying Statement of Consolidated Operations. In 2022, 2021, and 2020, Arconic did not recognize any interest or penalties. As of December 31, 2022 and 2021,no interest and penalties were accrued.

J. Earnings Per Share

Basic earnings per share (EPS) amounts are computed by dividing Net loss attributable to Arconic by the weighted-average number of common shares outstanding. Diluted EPS amounts assume the issuance of common stock for all potentially dilutive share equivalents outstanding. Specific to Arconic, such share equivalents consist of outstanding employee stock awards (excluding out-of-the-money stock options – see below). For periods in which the Company generates net income, the diluted weighted-average number of shares include common share equivalents associated with outstanding employee stock awards. For periods in which the Company generates a net loss, common share equivalents are excluded from the diluted weighted-average number of shares as their effect is anti-dilutive.

The share information used to compute basic and diluted EPS attributable to Arconic common stockholders was as follows (shares in millions):

	2022	2021	2020
Weighted-average shares outstanding – basic	103.6	108.7	109.1
Effect of dilutive share equivalents:			
Stock options	—	—	—
Stock units	—	—	—
Weighted-average shares outstanding – diluted	103.6	108.7	109.1
Anti-dilutive share equivalents:			
Stock units	2.6	3.3	2.6
Stock options*:			
In-the-money	0.1	0.1	—
Out-of-the-money	—	—	—
	2.7	3.4	2.6

* Stock options are in-the-money when the respective exercise price of each such option is less than the average market price of the Company's common stock during the applicable period presented. Conversely, stock options are out-of-the-money when the respective exercise price of each such option is more than the average market price of the Company's common stock during the applicable period presented. Out-of-the-money stock options never result in common share equivalents for purposes of diluted EPS regardless of whether a company generates net income or a net loss. As of December 31, 2022 and

December 31, 2020 there were 0.3 million and 0.5 million out-of-the money stock options outstanding, respectively, with a weighted average exercise price of $29.90 and $33.32, respectively.

K. Preferred and Common Stock

Preferred Stock. Arconic is authorized to issue 10,000,000 shares of preferred stock at a par value of $0.01 per share. At December 31, 2022 and 2021, the Company had no issued preferred stock.

Common Stock. Arconic is authorized to issue 150,000,000 shares of common stock at a par value of $0.01 per share. On the Separation Date, the Company distributed 109,021,376 shares of its common stock to ParentCo's stockholders (see Note A). As of December 31, 2022 and 2021, Arconic had 111,280,206 and 110,239,390, respectively, issued and 99,432,194 and 105,326,885, respectively, outstanding shares of common stock. In 2022, 2021, and from the Separation Date through December 31, 2020, the Company issued 1,040,816, 1,034,164, and 183,850, respectively, shares of common stock under its employee stock-based compensation plan (see below).

On May 4, 2021, Arconic announced that its Board of Directors approved a share repurchase program authorizing the Company to repurchase shares of its outstanding common stock up to an aggregate transactional value of $300 over a two-year period expiring April 28, 2023. In 2022 and 2021, Arconic repurchased 4,863,672 and 4,912,505 shares, respectively, of the Company's common stock for $139 and $161, respectively, resulting in completion of the total authorization under this program in August 2022. In connection with the establishment of a new repurchase program (see below), this repurchase program was terminated. Repurchases under the program were made from time to time, as the Company deemed appropriate, based on a variety of factors such as price, capital position, liquidity, financial performance, alternative uses of capital, and overall market conditions. This program was intended to comply with Rule 10b5-1 and all purchases were made in compliance with Rule 10b-18, including without limitation the timing, price, and volume restrictions thereof.

On November 16, 2022, Arconic announced that its Board of Directors approved a share repurchase program authorizing the Company to repurchase shares of its outstanding common stock up to an aggregate transactional value of $200 over a two-year period expiring November 17, 2024. Repurchases under the program may be made from time to time, as the Company deems appropriate, solely through open market repurchases effected through a broker dealer, based on a variety of factors such as price, capital position, liquidity, financial performance, alternative uses of capital, and overall market conditions. There can be no assurance as to the number of shares the Company will purchase. The share repurchase program may be increased or otherwise modified, renewed, suspended or terminated by the Company at any time, without prior notice. This program is intended to comply with Rule 10b5-1 and all purchases shall be made in compliance with Rule 10b-18, including without limitation, the timing, price, and volume restrictions thereof. In 2022, Arconic repurchased 2,071,835 shares of the Company's common stock for $46 under this program.

The Company issues new shares of common stock to satisfy the exercise of stock options and the conversion of stock units granted under its employee stock-based compensation plan. On May 20, 2021, the Company's shareholders approved an amendment to the plan to increase the shares of common stock authorized for issuance by 3,000,000 to 11,500,000 shares and to eliminate the plan's fungible share accounting, with the result that shares issued pursuant to full value stock awards, on or after the date of amendment, will be counted against the share reserve as one share issued under each such award, rather than as one and one-half shares. Shares returned to the plan, on or after the date of amendment, will be counted as one share, regardless of whether such shares were counted as one and one-half shares upon grant based on the prior fungible share accounting convention. In 2022, 2021, and from the Separation Date through December 31, 2020, there were 147,418, 251,919 and 84,959, respectively, stock options exercised and 1,314,188, 1,125,983 and 157,230, respectively, stock units converted (see table below). Additionally, as of December 31, 2022 and 2021, there were 413,592 and 590,906, respectively, stock options and 3,325,170 and 3,913,337, respectively, stock units outstanding (i.e., unexercised and/or unvested) under this plan (see table below). Accordingly, as of December 31, 2022 and 2021, there were 4,051,334 and 4,749,255, respectively, shares of common stock available for issuance under the plan.

Dividends on common stock are subject to authorization by the Company's Board of Directors. Arconic did not declare any dividends in 2022, 2021, and from Separation Date through December 31, 2020.

Stock-based Compensation

In 2022, 2021, and from the Separation Date through December 31, 2020, eligible Arconic employees participated in the Company's stock-based compensation plan. In all periods prior to the Separation Date, eligible employees attributable to the Arconic Corporation Businesses participated in ParentCo's stock-based compensation plan.

Effective on the Separation Date, all outstanding stock options (vested and non-vested) and non-vested stock units originally granted under ParentCo's stock-based compensation plan related to employees of the Arconic Corporation

Businesses, as well as the ParentCo corporate employees that became Arconic employees at Separation, were replaced with similar stock options and stock units under Arconic's stock-based compensation plan. In order to preserve the intrinsic value of these awards, the referenced employees received replacement stock options and stock units under Arconic's stock-based compensation plan at a ratio of 1.07 and 2.18, respectively, compared to the number of stock options and stock units originally granted under ParentCo's stock-based compensation plan. The ratio for stock options was developed by dividing the March 31, 2020 closing market price ($16.06) of ParentCo's common stock by the April 1, 2020 opening market price ($15.00) of Arconic's common stock (the Company's common stock did not trade on a "when issued" basis prior to April 1, 2020). Additionally, the exercise price of stock options was decreased by a ratio of 0.93 developed by dividing $15.00 by $16.06. The ratio for stock units was developed by dividing the March 31, 2020 closing market price of ParentCo's common stock by the volume weighted average trading price ($7.37) of Arconic's common stock during the first five trading days subsequent to March 31, 2020. This resulted in a beginning balance of outstanding stock options and stock units under Arconic's stock-based compensation plan of 1,173,492 and 3,062,013, respectively, as of the Separation Date. The respective fair values of these stock options and units were adjusted accordingly. Arconic did not recognize any immediate incremental stock-based compensation expense as a result of this adjustment.

The following description of Arconic's stock-based compensation plan is not materially different from the description of ParentCo's stock-based compensation plan prior to the Separation.

Stock awards are generally granted in the first quarter of each calendar year to eligible employees at the closing market price of Arconic's common stock on the date of grant. Stock options typically grade-vest over a three-year service period (1/3 each year) with a ten-year contractual term; stock units typically cliff-vest on the third anniversary of the award grant date. Beginning in 2022, stock units either cliff-vest or grade-vest based on the award type and employee's job level. As a condition of Arconic's stock-based compensation plan design, individuals who are retirement-eligible have a six-month requisite service period in the year of grant. In 2022, 2021, and 2020, certain of the stock unit grants also contain both performance and market conditions (for each year, the "performance stock units") and were granted to a limited number of eligible employees, including the Company's executive officers.

For 2022 and 2021, the final number of performance stock units earned is dependent on Arconic's achievement of certain targets (performance condition) and a total stockholder return ("TSR") (market condition) over a three-year measurement period. Specifically, determination of the initial number of stock units earned is based on the Company's achievement of adjusted EBITDA (50%), return on invested capital (25%), and TSR (25%) targets. Beginning in 2021, TSR was added as a stand-alone metric rather than a post-performance period multiplier (see below) to further align executive compensation with the creation of shareholder value as compared to Arconic's peers. For the 2022 and 2021 performance stock units, cumulative three-year performance goals (including for the TSR) were established for the performance period.

In 2020, the final number of performance stock units earned is dependent on Arconic's achievement of certain targets (performance condition) modified by a TSR multiplier (market condition) over a three-year measurement period. Specifically, determination of the initial number of stock units earned is based on the Company's achievement of an adjusted EBITDA target (25%), a controllable free cash flow target (25%), and a pretax return on net assets target (50%). For the 2020 performance stock units, the Compensation Committee of the Company's Board of Directors established three one-year financial targets to address the lack of visibility and challenge in setting long-term financial goals at the outset of the COVID-19 pandemic, while aligning executive compensation to long-term results. This result is then scaled by the TSR multiplier, which is based on the Company's relative three-year (January 1 of the grant year through December 31 of the third year in the service period) performance against the TSRs of a group of peer companies.

In 2022, 2021, and 2020, Arconic recognized stock-based compensation expense of $15 ($12 after-tax), $22 ($17 after-tax), and $23 ($18 after-tax), respectively, of which a minimum of approximately 85% was related to stock units in each period. The amount recognized in 2022 includes a reversal of $4 of expense previously recognized in 2021 related to the performance condition portion of the 2021 performance stock units (see footnote 2 to table below). No stock-based compensation expense was capitalized as an asset in 2022, 2021, or 2020. For periods prior to the Separation, the stock-based compensation expense recorded by Arconic was comprised of two components: (i) the expense associated with employees attributable to the Arconic Corporation Businesses, and (ii) an allocation of expense related to ParentCo corporate employees (see Cost Allocations in Note A). In the 2020 Pre-Separation Period, this allocation was $5 of Arconic's recognized stock-based compensation expense.

Stock-based compensation expense is based on the grant date fair value of the applicable equity grant. For stock units granted with no market condition, the fair value is equivalent to the closing market price of Arconic's or ParentCo's common stock on the date of grant in the respective periods. For stock units granted with a market condition, the fair value is estimated on the date of grant using a Monte Carlo simulation model, which generated a result of $37.40, $46.17, and $10.02 per unit in 2022, 2021, and 2020 respectively. There were no stock options granted in 2022, 2021, or 2020.

To estimate the fair value of a stock unit with a market condition, the Monte Carlo simulation model uses certain assumptions, including a risk-free interest rate and volatility, to estimate the probability of satisfying market conditions. The risk-free interest rate (1.4% in 2022, 0.3% in 2021, and 0.2% in 2020) was based on a yield curve of interest rates at the time of the grant based on the remaining performance period. Volatility was estimated using implied and historical volatility (60.4% in 2022, 50.1% in 2021, and 35.4% in 2020).

The activity in 2022 for stock options and stock units, including performance stock units granted to the Company's executive officers, was as follows:

	Stock options		Stock units	
	Number of options	Weighted average exercise price	Number of units	Weighted average FMV per unit
Outstanding, January 1, 2022	590,906	$ 25.56	3,913,337	$ 15.14
Granted	—	—	1,227,768	25.73
Exercised	(147,418)	20.61	—	—
Converted[1]	—	—	(1,314,188)	9.67
Expired or forfeited	(29,896)	32.25	(194,074)	22.18
Performance share adjustment[2]	—	—	(307,673)	30.84
Outstanding, December 31, 2022	413,592	26.84	3,325,170	19.35

(1) The number of converted units includes 420,790 shares “withheld” to meet the Company’s statutory tax withholding requirements related to the income earned by the employees as a result of vesting in the units.

(2) In 2022, the Company adjusted the target payout of the performance stock units granted in 2022, 2021, and 2020. Since the respective grant date, the fair value related to the performance condition of the 2022 and 2021 performance stock units has been expensed based on a payout at 100% of target. Additionally, the fair value related to the 2020 performance stock units has been expensed based on a payout at 8.3% of target (previously adjusted in the second half of 2021). However, at December 31, 2022, the estimated future payout for the 2022, 2021, and 2020 performance stock units was determined to be at 15.9%, 27.2%, and 7.9% of target, respectively. Accordingly, the number of non-vested performance stock units outstanding were adjusted to reflect the probable payout percentage.

As of December 31, 2022, the 413,592 outstanding stock options had a weighted average remaining contractual life of 2.3 years and a total intrinsic value of $1. Additionally, as of December 31, 2022 all of the total outstanding stock options were fully vested and exercisable. In 2022, 2021, and from the Separation Date through December 31, 2020, cash received from stock option exercises was $3, $6, $1, respectively, and the total intrinsic value of stock options exercised was $1, $2, and $1, respectively.

At December 31, 2022, there was $18 (pretax) of unrecognized compensation expense related to non-vested grants of stock units. This expense is expected to be recognized over a weighted average period of 1.8 years.

L. Accumulated Other Comprehensive Loss

The following table details the activity of the three components that comprise Accumulated other comprehensive loss for Arconic (such activity for Noncontrolling interest was immaterial for all periods presented):

	2022	2021	2020
Pension and other postretirement benefits (H)			
Balance at beginning of period	$ (1,121)	$ (1,791)	$ (43)
Establishment of additional defined benefit plans	—	—	(1,752)
Separation-related adjustments (A)	—	—	(50)
Other comprehensive income:			
Unrecognized net actuarial loss and prior service cost/benefit	176	190	(259)
Tax (expense) benefit	(41)	(43)	62
Total Other comprehensive income (loss) before reclassifications, net of tax	135	147	(197)
Amortization of net actuarial loss and prior service cost/benefit[1]	124	680	326
Tax expense[2]	(30)	(157)	(75)
Total amount reclassified from Accumulated other comprehensive loss, net of tax[5]	94	523	251
Total Other comprehensive income	229	670	54
Balance at end of period	$ (892)	$ (1,121)	$ (1,791)
Foreign currency translation			
Balance at beginning of period	$ 25	$ 29	$ 338
Separation-related adjustments (A)	—	—	(396)
Other comprehensive (loss) income:			
Foreign currency translation[3]	(76)	(4)	65
Net amount reclassified to earnings from Accumulated other comprehensive income[3],[5]	—	—	22
Total Other comprehensive (loss) income	(76)	(4)	87
Balance at end of period	$ (51)	$ 25	$ 29
Cash flow hedges (U)			
Balance at beginning of period	$ (15)	$ 1	$ —
Separation-related adjustments (A)	—	—	(4)
Other comprehensive income (loss):			
Net change from periodic revaluations	95	(161)	(2)
Tax (expense) benefit	(22)	37	1
Total Other comprehensive income (loss) before reclassifications, net of tax	73	(124)	(1)
Net amount reclassified to earnings:			
Aluminum[4]	(48)	142	8
Energy[4]	(20)	—	—
Alloying materials[4]	1	(2)	—
Sub-total	(67)	140	8
Tax benefit (expense)[2]	15	(32)	(2)
Total amount reclassified from Accumulated other comprehensive income, net of tax[5]	(52)	108	6
Total Other comprehensive income (loss)	21	(16)	5
Balance at end of period	$ 6	$ (15)	$ 1
Accumulated other comprehensive loss	$ (937)	$ (1,111)	$ (1,761)

(1) These amounts were included in the non-service component of net periodic benefit cost for pension and other postretirement benefits (seeNote H). In 2022, 2021, and 2020, this amount includes $47, $584, and $199, respectively, related to the settlement of certain employee retirement benefits (seeNote H).

(2) These amounts were reported in (Benefit) Provision for income taxes on the accompanying Statement of Consolidated Operations.

(3) In all periods presented, there were no tax impacts related to rate changes. In 2020, the net amount reclassified to earnings was reported in Restructuring and other charges on the accompanying Statement of Consolidated Operations related to the sale of certain foreign subsidiaries.

(4) A portion of the amounts related to aluminum were reported in each of Sales and Cost of goods sold on the accompanying Statement of Consolidated Operations (see Note U). The amounts related to energy and alloying materials were reported in Cost of goods sold on the accompanying Statement of Consolidated Operations (see Note U).

(5) A positive amount indicates a corresponding charge to earnings and a negative amount indicates a corresponding benefit to earnings. These amounts were reflected on the accompanying Statement of Consolidated Operations in the line items indicated in footnotes 1 through 4.

M. Inventories

December 31,	2022	2021
Finished goods	$ 343	$ 350
Work-in-process	1,106	1,105
Purchased raw materials	118	109
Operating supplies	55	66
	$ 1,622	$ 1,630

In November 2022, Arconic completed the sale of all of its operations in Russia (seeNote S). As of December 31, 2021, Inventories related to the Company's operations in Russia were $102.

N. Properties, Plants, and Equipment, Net

December 31,	2022	2021
Land and land rights	$ 19	$ 22
Structures:		
Rolled Products*	1,022	1,107
Building and Construction Systems	96	96
Extrusions	152	150
Other	149	152
	1,419	1,505
Machinery and equipment:		
Rolled Products*	4,299	4,816
Building and Construction Systems	204	203
Extrusions	523	523
Other	274	291
	5,300	5,833
	6,738	7,360
Less: accumulated depreciation and amortization	4,613	4,878
	2,125	2,482
Construction work-in-progress	236	169
	$ 2,361	$ 2,651

* In November 2022, Arconic completed the sale of all of its operations in Russia (seeNote S).

In 2022, Arconic recognized an impairment charge of $90 for the Extrusions segment as the result of a business review (See 2022 Actions inNote E).

O. Goodwill and Other Intangible Assets

The following table details the changes in the carrying value of Goodwill:

	Rolled Products	Building and Construction Systems	Extrusions	Total
Balances at December 31, 2020:				
Goodwill	$ 254	$ 99	$ 65	$ 418
Accumulated impairment losses	—	(28)	—	(28)
Goodwill, net	254	71	65	390
Impairment (B)	—	—	(65)	(65)
Translation	(1)	(2)	—	(3)
Balances at December 31, 2021:				
Goodwill	253	97	65	415
Accumulated impairment losses	—	(28)	(65)	(93)
Goodwill, net	253	69	—	322
Divestitures (S)	(14)	—	—	(14)
Translation	(15)	(1)	—	(16)
Balances at December 31, 2022:				
Goodwill	224	96	65	385
Accumulated impairment losses	—	(28)	(65)	(93)
Goodwill, net	$ 224	$ 68	$ —	$ 292

In 2021, Arconic recognized an impairment charge of $65 for the Extrusions reporting unit based on the result of the annual review of goodwill for impairment (see Goodwill in Note B).

Other intangible assets, which are included in Other noncurrent assets on the accompanying Consolidated Balance Sheet, were as follows:

December 31, 2022	Gross carrying amount	Accumulated amortization	Net carrying amount
Computer software*	$ 545	$ (521)	$ 24
Patents and licenses	27	(27)	—
Other	21	(18)	3
Total other intangible assets	$ 593	$ (566)	$ 27

December 31, 2021	Gross carrying amount	Accumulated amortization	Net carrying amount
Computer software	$ 555	$ (523)	$ 32
Patents and licenses	27	(27)	—
Other	21	(15)	6
Total other intangible assets	$ 603	$ (565)	$ 38

* In 2022, Arconic recognized an impairment charge of $2 for the Extrusions segment as the result of a business review (See 2022 Actions inNote E).

Computer software consists primarily of software costs associated with an enterprise business solution within Arconic to drive common systems among all businesses.

Amortization expense related to the intangible assets in the tables above for the years ended December 31, 2022, 2021, and 2020 was $13, $17, and $17, respectively. During the next five years, amortization expense related to these intangible assets is expected to decrease from $9 in 2023 to $2 in 2027.

P. Leases

Arconic leases certain land and buildings, plant equipment, vehicles, and computer equipment, which have been classified as operating leases. Operating lease cost, which includes short-term leases and variable lease payments and approximates cash paid, was $54, $59, and $62 in 2022, 2021, and 2020, respectively.

Right-of-use assets obtained in exchange for operating lease obligations in 2022 and 2021 were $34 and $17, respectively.

Future minimum contractual operating lease obligations were as follows:

December 31,	2022
2023	$ 40
2024	31
2025	24
2026	16
2027	7
Thereafter	19
Total lease payments	$ 137
Less: imputed interest	20
Present value of lease liabilities	$ 117

The weighted-average remaining lease term and weighted-average discount rate for Arconic's operating leases at December 31, 2022 and 2021 was 5.2 and 6.1 years, respectively, and 6.3% and 5.8%, respectively.

Q. Debt

December 31,	2022	2021
6.00% Notes, due 2025	$ 700	$ 700
6.125% Notes, due 2028	900	900
Unamortized discounts and deferred financing costs	(3)	(6)
	$ 1,597	$ 1,594

2022 Activity—On February 16, 2022, the Company amended its five-year credit agreement, dated May 13, 2020, with a syndicate of lenders named therein and Deutsche Bank AG New York Branch as administrative agent (the "ABL Credit Agreement") which provides for a senior secured asset-based revolving credit facility (the "ABL Credit Facility") to be used, generally, for working capital or other general corporate purposes. The ABL Credit Agreement was amended to increase the revolving commitments under the ABL Credit Facility to $1,200 from $800. Additionally, the accordion feature of the ABL Credit Facility was revised to provide for the Company to request a further increase to the revolving commitments in an aggregate principal amount equal to the greater of $350 and the excess of the borrowing base over the ABL Credit Facility commitments. Furthermore, the LIBOR-based floating interest rate was replaced with a term SOFR-based interest rate, plus a credit spread adjustment equal to 0.10%, 0.15%, or 0.25% per annum for SOFR-based borrowings with interest periods of one month, three months, or six months, respectively, under the ABL Credit Facility. Arconic paid $1 in upfront costs associated with these amendments.

In 2022, the Company borrowed $275 and repaid $275 under the ABL Credit Facility. These borrowings were designated as SOFR loans with either an initial one-month or three-month interest period. In 2022, the weighted-average interest rate and weighted-average days outstanding of the borrowings was 4.3% and 90 days, respectively.

2021 Activity—On March 3, 2021, the Company completed a Rule 144A (U.S. Securities Act of 1933, as amended) debt offering for an additional $300 aggregate principal amount of 6.125% Senior Secured Second-Lien Notes due 2028 (the "Additional 2028 Notes"). The Additional 2028 Notes were issued under the indenture governing Arconic's existing 6.125% Senior Secured Second-Lien Notes due 2028 (see 2020 Activity below). Other than with respect to the date of issuance and issue price, the Additional 2028 Notes are treated as a single series with and have the same terms as the referenced existing notes. The Additional 2028 Notes were sold at 106.25% of par (i.e., a premium) and, after reflecting a discount to the initial purchasers of the Additional 2028 Notes, the Company received $315 in net proceeds from the debt offering. Arconic used the net proceeds of this issuance to fund an annuitization of certain U.S. defined benefit pension plan obligations (see Note H). The premium ($19) and costs to complete the financing ($5) were deferred and are being amortized to interest expense over the term of the Additional 2028 Notes. The amortization of the premium is reflected as a reduction to interest expense and the amortization of the costs to complete the financing is reflected as an addition to interest expense. Interest on the Additional 2028 Notes is paid semi-annually in February and August and commenced August 15, 2021.

2020 Activity—In connection with the capital structure to be established at the time of the Separation, Arconic secured $1,200 in third-party indebtedness. On February 7, 2020, Arconic completed a Rule 144A (U.S. Securities Act of 1933, as amended) debt offering for $600 of 6.125% Senior Secured Second-Lien Notes due 2028 (the "2028 Notes"). The Company received $593 in net proceeds from the debt offering reflecting a discount to the initial purchasers of the 2028 Notes. Also, on March 25, 2020, Arconic entered into a credit agreement, which provided a $600 Senior Secured First-Lien Term Loan B Facility (variable rate and seven-year term) (the "Term Loan") and a $1,000 Senior Secured First-Lien Revolving Credit Facility (variable rate and five-year term) (the "Credit Facility"), with a syndicate of lenders and issuers named therein (the "Credit Agreement"). The Company received $575 in net proceeds from the Term Loan reflecting upfront fees and costs to enter into the financing arrangement.

The Company used a portion of the $1,168 in net proceeds from the aggregate indebtedness to make a $728 payment to ParentCo on April 1, 2020 to fund the transfer of certain net assets from ParentCo to Arconic in connection with the completion of the Separation (see Note A). The payment to ParentCo was calculated as the difference between (i) the $1,168 of net proceeds from the aggregate indebtedness and (ii) the difference between a beginning cash balance at the Separation Date of $500, as provided for in the Separation and Distribution Agreement, and the amount of cash held by Arconic Businesses at March 31, 2020 ($60 – the sum of this amount and the aggregate indebtedness in (i) equals the sum of Cash and cash equivalents and Restricted cash on the Company's Combined Balance Sheet as of March 31, 2020).

On April 2, 2020, Arconic borrowed $500, which was subject to an interest rate equal to the sum of the three-month LIBOR plus a 2.0% applicable margin, under the Credit Facility. This borrowing was a proactive measure taken by the Company to bolster its liquidity and preserve financial flexibility in light of uncertainties resulting from the COVID-19 pandemic (see Note A).

On May 13, 2020, Arconic executed a refinancing of its existing Credit Agreement in order to provide improved financial flexibility. Arconic completed a Rule 144A (U.S. Securities Act of 1933, as amended) debt offering for $700 of 6.0% Senior Secured First-Lien Notes due 2025 (the "2025 Notes"). The Company received $691 in net proceeds from the debt offering reflecting a discount to the initial purchasers of the 2025 Notes. Additionally, Arconic entered into the ABL Credit Agreement. The ABL Credit Agreement provided for a senior secured asset-based revolving credit facility in an aggregate principal amount of $800 (see 2022 Activity above for amendments to the ABL Credit Agreement), including a letter of credit sub-facility and the ABL Credit Facility. In addition, the ABL Credit Facility includes an accordion feature allowing the Company to request one or more increases to the revolving commitments in an aggregate principal amount up to $350 (see 2022 Activity above for amendment to the ABL Credit Facility).

Arconic used the net proceeds from the new indebtedness, together with cash on hand, to prepay in full the obligations outstanding under both the Term Loan ($600) and Credit Facility ($500) and to terminate in full the commitments under the Credit Agreement.

Descriptions of the 2028 Notes, 2025 Notes, and ABL Credit Agreement are set forth below.

In connection with the issuance of the 2028 Notes and the execution of the Credit Agreement, the Company paid $42 in discounts to the initial purchasers and/or upfront fees and costs (the "debt issuance costs"), of which $30 was attributable to the Term Loan and the Credit Facility. The debt issuance costs were initially deferred and were being amortized to interest expense over the respective terms of the 2028 Notes, the Term Loan, and the Credit Facility. In connection with the issuance of the 2025 Notes and the execution of the ABL Credit Agreement, the Company paid $15 in discounts to the initial purchasers and/or upfront fees and costs (the "new debt issuance costs"). As a result of applying both debt modification and debt extinguishment accounting, as appropriate based on the lender mix for each debt instrument, to the debt refinancing, the Company was required to write off $16 of the $30 in debt issuance costs and immediately expense $3 of the $15 in new debt issuance costs. This $19

was reported within Interest expense on the Company's Statement of Consolidated Operations. The remaining $14 in debt issuance costs continued to be deferred and the remaining $12 in new debt issuance costs were deferred; both are being amortized to interest expense over the respective terms of the 2025 Notes and the ABL Credit Agreement.

2028 Notes—Interest on the 2028 Notes is paid semi-annually in February and August and commenced August 15, 2020.

Arconic has the option to redeem the 2028 Notes on at least10 days, but not more than 60 days, prior notice to the holders of the 2028 Notes under multiple scenarios, including, in whole or in part, at any time or from time to time after February 14, 2023 at a redemption price specified in the indenture (up to 103.063% of the principal amount plus any accrued and unpaid interest in each case). At any time prior to February 15, 2023, the Company may redeem all or a part of the notes at a redemption price equal to 100% of the principal amount of the notes redeemed plus a "make-whole" redemption price determined as the greater of (1)1.0% of the principal amount of such notes and (2) the excess, if any, of (a) the present value at the date of redemption of (i) 103.063% of the principal amount of such notes plus (ii) all required interest payments due on such notes (excluding accrued but unpaid interest to the date of redemption) through February 15, 2023, computed using a discount rate equal to, generally, the yield to maturity of United States Treasury securities with a constant maturity as of the date of redemption plus 50 basis points, over (b) the principal amount of such notes, as of, and accrued and unpaid interest, if any, to, but excluding, the date of redemption. Also, at any time prior to February 15, 2023, Arconic may, on one or more occasions, redeem up to 40% of the aggregate principal amount of the notes at a redemption price equal to106.125% of the principal amount of the notes to be redeemed, plus accrued and unpaid interest, if any, to, but excluding, the redemption date, with the net cash proceeds of certain equity offerings, if at least 60% of the original aggregate principal amount of the notes remains outstanding immediately after such redemption and the redemption occurs within 120 days of the date of such equity offering. Additionally, the 2028 Notes are subject to repurchase upon the occurrence of a change in control repurchase event (as defined in the indenture) at a repurchase price in cash equal to 101% of the aggregate principal amount of the 2028 Notes repurchased, plus any accrued and unpaid interest on the 2028 Notes repurchased.

The 2028 Notes are senior secured obligations of Arconic and do not entitle the holders to any registration rights pursuant to a registration rights agreement. The Company does not intend to file a registration statement with respect to resales of or an exchange offer for the 2028 Notes. The 2028 Notes are guaranteed on a senior secured basis by Arconic and its subsidiaries that are guarantors (the "subsidiary guarantors" and, together with Arconic, the "guarantors") under the ABL Credit Agreement (see below). Each of the subsidiary guarantors will be released from their 2028 Notes guarantees upon the occurrence of certain events, including the release of such guarantor from its obligations as a guarantor under the ABL Credit Agreement.

The 2028 Notes indenture includes several customary affirmative covenants. Additionally, the 2028 Notes indenture contains several negative covenants, that, subject to certain exceptions, limit the Company's ability to, among other things, (i) make investments, loans, advances, guarantees, and acquisitions, (ii) pay dividends on or make other distributions in respect of capital stock and make other restricted payments and investments (as defined in the 2028 Notes), (iii) sell or transfer certain assets, and (iv) create liens on assets to secure debt.

The 2028 Notes rank equally in right of payment with all of Arconic's existing and future senior indebtedness, including the facility under the ABL Credit Agreement (see below); rank senior in right of payment to any future subordinated obligations of Arconic; and are effectively subordinated to Arconic's existing and future secured indebtedness that is secured on a first priority basis, including the 2025 Notes and the facility under the ABL Credit Agreement, to the extent of the value of property and assets securing such indebtedness.

2025 Notes—Interest on the 2025 Notes is paid semi-annually in May and November and commenced November 15, 2020.

Arconic has the option to redeem the 2025 Notes on at least10 days, but not more than 60 days, prior notice to the holders of the 2025 Notes under multiple scenarios, including, in whole or in part, at any time or from time to time after May 14, 2022 at a redemption price specified in the indenture (up to 103.0% of the principal amount plus any accrued and unpaid interest in each case). At any time prior to May 15, 2022, the Company may redeem all or a part of the notes at a redemption price equal to 100% of the principal amount of the notes redeemed plus a "make-whole" redemption price determined as the greater of (1)1.0% of the principal amount of such notes and (2) the excess, if any, of (a) the present value at the date of redemption of (i) 103.0% of the principal amount of such notes plus (ii) all required interest payments due on such notes (excluding accrued but unpaid interest to the date of redemption) through May 15, 2022, computed using a discount rate equal to, generally, the yield to maturity of United States Treasury securities with a constant maturity as of the date of redemption plus 50 basis points, over (b) the principal amount of such notes, as of, and accrued and unpaid interest, if any, to, but excluding, the date of redemption. Also, at any time prior to May 15, 2022, Arconic may, on one or more occasions, redeem up to 40% of the aggregate principal amount of the notes at a redemption price equal to 106.0% of the principal amount of the notes to be redeemed, plus accrued and unpaid interest, if any, to, but excluding, the redemption date, with the net cash proceeds of certain equity offerings, if at least 60% of the original aggregate principal amount of the notes remains outstanding immediately after such redemption and

the redemption occurs within 120 days of the date of such equity offering. Additionally, the 2025 Notes are subject to repurchase upon the occurrence of a change in control repurchase event (as defined in the indenture) at a repurchase price in cash equal to 101% of the aggregate principal amount of the 2025 Notes repurchased, plus any accrued and unpaid interest on the 2025 Notes repurchased.

The 2025 Notes are senior secured obligations of Arconic and do not entitle the holders to any registration rights pursuant to a registration rights agreement. The Company does not intend to file a registration statement with respect to resales of or an exchange offer for the 2025 Notes. The 2025 Notes are guaranteed on a senior secured basis by Arconic and its subsidiaries that are guarantors (the "subsidiary guarantors" and, together with Arconic, the "guarantors") under the ABL Credit Agreement (see below). Each of the subsidiary guarantors will be released from their 2025 Notes guarantees upon the occurrence of certain events, including the release of such guarantor from its obligations as a guarantor under the ABL Credit Agreement.

The 2025 Notes indenture includes several customary affirmative covenants. Additionally, the 2025 Notes indenture contains several negative covenants, that, subject to certain exceptions, limit the Company's ability to, among other things, (i) pay dividends on or make other distributions in respect of capital stock and make other restricted payments and investments (as defined in the 2025 Notes), (ii) sell or transfer certain assets, (iii) incur indebtedness, and (iv) create liens on assets to secure debt.

The 2025 Notes are secured on a first priority basis by certain defined collateral (generally consisting of the Company's and the Guarantors' equipment, material owned U.S. real property, intellectual property, certain stock, and other tangible and intangible personal property, in each case, subject to certain exceptions) and on a second priority basis by certain other assets (generally consisting of substantially all of the accounts receivable, inventory, deposit accounts, securities accounts, commodities accounts, and cash assets of the Company and the Guarantors, and the proceeds thereof).

ABL Credit Agreement—Availability under the ABL Credit Facility is subject to a monthly borrowing base calculation, which, in general, is determined by applying a predetermined percentage to the amount of eligible accounts receivable and inventory, less customary reserves. As of December 31, 2022, the available balance was $1,189.

The ABL Credit Facility is scheduled to mature on May 13, 2025, unless extended or earlier terminated in accordance with the ABL Credit Agreement. Under the provision of the ABL Credit Agreement, Arconic will pay a quarterly commitment fee ranging from 0.250% to 0.375% (based on Arconic's leverage ratio) per annum on the unused portion of the ABL Credit Facility, which will be determined based on the Company's average daily utilization. The ABL Credit Facility was undrawn as of both December 31, 2022 and 2021.

As referenced above (See 2022 Actions), the ABL Credit Agreement was amended on February 16, 2022 to replace the LIBOR-based floating interest rate with a term SOFR-based interest rate. The ABL Credit facility is now subject to an interest rate for U.S. dollar borrowings equal to an applicable margin plus, at the Company's option, of either (a) base rate ("ABR") determined by reference to the highest of (1) Deutsche Bank AG New York Branch's "prime rate," (2) the greater of the federal funds effective rate and the overnight bank funding rate, plus 0.5%, and (3) the term SOFR-based interest rate for a one month interest period, plus1% per annum or (b) a term SOFR-based interest rate (which will not be less than 0.75% per annum), plus a credit spread adjustment equal to 0.10%, 0.15%, or 0.25% per annum for SOFR-based borrowings with interest periods of one month, three months, or six months, respectively. The applicable margin for the ABL Credit Facility is (a)0.75% to 1.25% per annum for ABR loans and (b) 1.75% per annum to 2.25% per annum for SOFR loans based on the average daily excess availability (as defined under the ABL Credit Agreement). Accordingly, the interest rates for the ABL Credit Facility will fluctuate based on changes in the ABR, SOFR, and/or futures changes in the average daily excess availability.

Prior to February 16, 2022, the ABL Credit Facility was subject to an interest rate for U.S. dollar borrowings equal to an applicable margin plus, at the Company's option, of either (a) base rate ("ABR") determined by reference to the highest of (1) Deutsche Bank AG New York Branch's "prime rate," (2) the greater of the federal funds effective rate and the overnight bank funding rate, plus 0.5%, and (3) the one month adjusted LIBO Rate, plus1% per annum or (b) an adjusted LIBO Rate (not less than 0.75% per annum) ("LIBOR"). The applicable margin for the ABL Credit Facility through June 30, 2021 was (a)1.25% for ABR loans and (b) 2.25% for LIBOR loans. Thereafter, the applicable margin for the ABL Credit Facility was (a) 0.75% to 1.25% per annum for ABR loans and (b) 1.75% per annum to 2.25% per annum for LIBOR loans based on the average daily excess availability (as defined under the ABL Credit Agreement).

All obligations under the ABL Credit Facility are unconditionally guaranteed, jointly and severally, by substantially all of the direct and indirect wholly-owned material subsidiaries of the Company that are organized under the laws of the United States, any state thereof or the District of Columbia, subject to certain exceptions (collectively, the "Guarantors"). The Company and the Guarantors entered into a guarantee under the ABL Credit Agreement concurrently with the effectiveness of the ABL Credit Agreement.

Subject to certain limitations, the ABL Credit Facility is secured on a first priority basis by certain defined collateral (generally consisting of substantially all of the accounts receivable, inventory, deposit accounts, securities accounts, commodities accounts, and cash assets of the Company and the Guarantors, and the proceeds thereof) and on a second-priority basis by certain defined collateral under the 2025 Notes (generally consisting of the Company and the Guarantors' equipment, material owned U.S. real property, intellectual property, certain stock, and other tangible and intangible personal property, in each case, subject to exceptions as defined in the 2025 Notes). The Company and the Guarantors entered into collateral agreements concurrently with the effectiveness of the ABL Credit Agreement.

The ABL Credit Facility contains certain affirmative and negative covenants customary for financings of this type that, among other things, limit the Company's and its subsidiaries' ability to incur additional indebtedness or liens, to dispose of assets, to make certain fundamental changes, to enter into restrictive agreements, to make certain investments, loans, advances, guarantees and acquisitions, to prepay certain indebtedness and to pay dividends, to make other distributions or redemptions/repurchases, in respect of the Company's and its subsidiaries' equity interests, to engage in transactions with affiliates and to amend certain material documents.

In addition, the ABL Credit Facility contains a financial maintenance covenant applicable to any fiscal quarter in which the excess availability is less than the greater of (a) 10% of the lesser of (x) the aggregate amount of the commitments under the ABL Credit Facility and (y) the borrowing base and (b) $50. In such circumstances, until such time as excess availability shall have exceeded such threshold for at least 30 consecutive days, the Company would be required to maintain a fixed charge coverage ratio of not less than 1.00 to 1.00. The ABL Credit Facility also requires the Company and its subsidiaries to maintain substantially all of the Company's cash in accounts that are subject to the control of the agent, which control becomes applicable when (a) an event of default under the facility occurs and is continuing until the first day thereafter on which no event of default shall exist or (b) excess availability is less than the greater of (i) 12.5% of the lesser of (x) the aggregate amount of the commitments under the ABL Credit Facility and (y) the borrowing base or (ii) $62.5 for five consecutive business days until the first day thereafter on which excess availability shall have exceeded such threshold for at least 30 consecutive days.

The ABL Credit Facility contains customary events of default, including with respect to a failure to make payments thereunder, cross-default and cross-acceleration, certain bankruptcy and insolvency events, and customary change of control events.

R. Cash Flow Information

Cash paid for interest and income taxes was as follows:

	2022	2021	2020
Interest, net of amount capitalized	$ 99	$ 87	$ 48
Income taxes, net of amount refunded	29	26	27

For all periods presented, both Cash and cash equivalents and restricted cash at beginning of year and Cash and cash equivalents and restricted cash at end of year includes Restricted cash of less than $0.03.

S. Acquisitions and Divestitures

Divestitures

Russia. On November 15, 2022, Arconic completed the sale of all of its operations in Russia to Promishlennie Investitsii LLC, the majority owner of VSMPO-AVISMA Corporation, for cash proceeds of $230. The transaction closed after the Company received all required approvals, resulting in the receipt of the cash consideration in exchange for all of Arconic's net assets in Russia. These net assets included $203 of cash held in Russia that was not available for distribution to the parent company because of injunctions imposed as a result of litigation initiated in March 2020 by the Federal Antimonopoly Service of the Russian Federation(see Litigation in Note T). The legal form of the transaction was a stock sale of all of the Company's Russian subsidiaries.VSMPO-AVISMA Corporation owns a limited portion of one of these legal entities, which was reported as Noncontrolling interest in the accompanying Consolidated Financial Statements prior to consummation of the sale transaction. The decision to pursue a sale was the result of a strategic review of alternatives completed in May 2022 by management with respect to Arconic's Russian operaons in consideration of the sanctions and other trade restrictions levied against Russia beginning in February 2022 and preliminary injunctions imposed on the Company's operations in Russia by the Federal Antimonopoly Service of the Russian Federation, which included a prohibition on the ability of Arconic's Russian subsidiaries to pay dividends to the parent company. In connection with this transaction, the Company recognized a loss of $306 ($304 after-tax), which was recorded in Restructuring and other charges (see Note E) on the accompanying Statement of Consolidated Operations. The loss includes the write-off of goodwill of $14 (see Note O).

The following table presents selected financial information related to Arconic's former operations in Russia:

	November 14, 2022	December 31, 2021
Cash and cash equivalents	$ 203	$ 79
Receivables from customers	100	120
Inventories	129	102
Properties, plants, and equipment, net	184	200
Accounts payable, trade	55	47

	January 1, 2022 through November 14, 2022	For the year ended December 31, 2021
Third-party sales*	$ 903	$ 968
Segment Adjusted EBITDA	71	87

* In both periods presented, Third-party sales includes aluminum products manufactured at the Company's plant in Russia and sold through Arconic's international selling company located in Hungary.

Prior to the sale transaction, the Company continued to conduct business in Russia to fulfill existing obligations in accordance with applicable laws, regulations, and international rules. Arconic's former operations in Russia were comprised of one principal location in Samara, which manufactured sheet, plate, extrusions, and forgings across all of the Company's end markets. The Samara facility continued to operate at relatively normal levels despite the imposition of sanctions and trade restrictions that limit a Russian entity's ability to export goods, as Samara's operations primarily served customer demand within Russia. The Samara facility had approximately 2,900 employees at the time of divestiture.

Arconic's former local Russian management team continued to operate the Samara facility through November 14, 2022 without undue influence imposed by any third-party, including the Russian government. Additionally, other than the prohibition on dividend payments to the parent company, the Company did not encounter any other significant constraints related to its control over the now former Russian subsidiaries. As a result, Arconic reported the financial results of its former Russian operations through November 14, 2022 in the Company's Consolidated Financial Statements and within Arconic's Rolled Products segment. Additionally, the Company did not record an impairment of long-lived or other assets in any prior financial reporting period given the relatively normal operating levels at Samara and lack of any triggering events in such periods.

Building and Construction Systems. On June 6, 2022, the Company announced that it is evaluating strategic options for the businesses that comprise the Building and Construction Systems segment, including exploring a sale of its architectural systems business (Kawneer® brand). Subsequent to this announcement, Arconic initiated a sale process with respect to its architectural systems business, which has six principal locations in the United States, Canada, and Europe. Products manufactured under the Kawneer brand include windows, doors, and curtain walls. This business generated third-party sales of approximately $970 in 2022 and had approximately 2,900 employees as of December 31, 2022.

On August 2, 2022, the Company announced a pause in the sale process of this business due to current economic conditions, particularly uncertainty in the debt markets. This business will remain classified as held and in-use in Arconic's Consolidated Financial Statements and will continue to be reported within the Building and Construction Systems segment.

Itapissuma. On February 1, 2020, Arconic completed the sale of its aluminum rolling mill (aseptic foil and sheet products) in Itapissuma, Brazil to Companhia Brasileira de Alumínio for a net $46 in cash. In December 2020, the Company paid $4 in cash to Companhia Brasileira de Alumínio to settle certain working capital and other post-closing adjustments. Additionally, in June 2021, the Company paid $2 in cash to Companhia Brasileira de Alumínio to settle the remaining working capital and other post-closing adjustments. Arconic has recognized a cumulative loss of $60 (pretax) on this transaction, composed of the following: a charge of $53 in 2019 for the non-cash impairment of the carrying value (i.e., write-down to fair value) of the rolling mill's net assets, primarily properties, plants, and equipment, as a result of entering into an agreement in August 2019 to sell this rolling mill; a charge of $6 in February 2020 for further necessary adjustments upon completion of the divestiture; and a charge of $1 in March 2021 for a then-proposed final settlement of the remaining post-closing adjustments and other items. Each of these amounts were recorded in Restructuring and other charges (see Note E) on the accompanying Statement of Consolidated Operations in the respective reporting periods. This transaction is no longer subject to post-closing adjustments. Prior to the divestiture, this rolling mill's operating results and assets and liabilities were reported in Arconic's Rolled Products segment. The rolling mill generated third-party sales of $143 in 2019 and, at the time of divestiture, had approximately 500 employees.

Changwon. On March 1, 2020, Arconic completed the sale of its hard alloy extrusions plant in South Korea to SeAH Besteel Corporation for a net $5 in cash, resulting in a gain of $31 (pretax), which was recorded in Restructuring and other charges (see Note E) on the accompanying Statement of Consolidated Operations. The gain is net of a $6 write-off of related goodwill. In May 2020, the Company received an additional $1 in cash as a result of a post-closing adjustment, which was previously contemplated in the aforementioned gain. This transaction is no longer subject to post-closing adjustments. Prior to the divestiture, this plant's operating results and assets and liabilities were reported in Arconic's Extrusions segment. The extrusions plant generated third-party sales of $51 in 2019 and, at the time of divestiture, had approximately 160 employees.

Texarkana. In October 2018, Arconic sold its Texarkana (Texas) rolling mill and cast house, which had a combined net book value of $63, to Ta Chen International, Inc. for $302 in cash plus additional contingent consideration of up to $50, of which $25 was received prior to 2020. The contingent consideration related to the achievement of various milestones associated with operationalizing the rolling mill equipment within 36 months of the transaction closing date. In 2020, Arconic received the remaining contingent consideration of $25, which was recorded as a gain in Restructuring and other charges (see Note E) on the accompanying Statement of Consolidated Operations. As of December 31, 2020, there was no remaining contingent consideration associated with this transaction.

T. Contingencies and Commitments

Unless specifically described to the contrary, all matters within Note T are the full responsibility of Arconic pursuant to the Separation and Distribution Agreement (see Note A). Additionally, the Separation and Distribution Agreement provides for cross-indemnities between the Company and Howmet for claims subject to indemnification.

Contingencies

Environmental Matters. Arconic participates in environmental assessments and cleanups at several locations. These include owned or operating facilities and adjoining properties, previously owned or operating facilities and adjoining properties, and waste sites, including Superfund (Comprehensive Environmental Response, Compensation and Liability Act (CERCLA)) sites.

A liability is recorded for environmental remediation when a cleanup program becomes probable and the costs can be reasonably estimated. As assessments and cleanups proceed, the liability is adjusted based on progress made in determining the extent of remedial actions and related costs. The liability can change substantially due to factors such as, among others, the nature and extent of contamination, changes in remedial requirements, and technological changes.

The Company's remediation reserve balance was $85 and $64 (of which $40 and $15, respectively, was classified as a current liability) at December 31, 2022 and 2021, respectively, and reflects the most probable costs to remediate identified environmental conditions for which costs can be reasonably estimated.

In 2022, the remediation reserve was increased by $33 due to a charge of $13 (recorded in Cost of goods sold) for the Massena location (see below); a charge of $14 (recorded in Cost of goods sold) related to the estimated costs of future operations, maintenance, and monitoring activities at several sites, including $9 for a correction of an accrual (see Note A); a charge of $5 (recorded in Cost of goods sold) for the planned removal of polychlorinated biphenyls- (PCBs) contaminated soil from certain portions of a ditch that is adjacent to the Company's facility in Lafayette, Indiana under a previously issued corrective action order with the Indiana Department of Environmental Management (remediation to be completed over a two-year period beginning in 2023); and a charge of $1 (recorded in Restructuring and other charges – see Note E) to reflect an estimate of Arconic's share of newly-identified costs for additional remediation work, which is subject to review by Italy's Ministry of the Environment, related to a recently completed project at a former site where the Company is one of several responsible parties.

In 2021, the remediation reserve was reduced by $5 due to a reversal of an $11 liability (a credit was recorded in Cost of goods sold) previously established for the Massena location (see below) and a charge of $2 (recorded in Restructuring and other charges – see Note E) for the assumption of a remediation obligation related to a legal settlement associated with a former operating site. Additionally, the change to the remediation reserve includes a charge of $4 (recorded in Cost of goods sold) for other items including incremental estimated expenditures associated with active remediation systems and/or monitoring and inspection programs at several sites.

In 2020, the remediation reserve was reduced by $2 due to the reversal of a $5 liability (a credit was recorded in Restructuring and other charges – see Note E) previously established by ParentCo, as the underlying obligation no longer exists based on an assessment completed by Arconic management; a charge of $1 (recorded in Restructuring and other charges – see Note E) to establish a liability related to the divestiture of a rolling mill in Brazil (see Note S); and a charge of $2 (recorded in

Cost of goods sold) for incremental estimated expenditures associated with active remediation systems and/or monitoring and inspection programs at several sites.

Payments related to remediation expenses applied against the reserve were $11, $84, and $82 in 2022, 2021, and 2020, respectively, which include expenditures currently mandated, as well as those not required by any regulatory authority or third party. The change in the reserve in 2022 and 2021 reflects a decrease of $1 and $3 for other items, respectively, and the change in the reserve in 2020 reflects both an increase of $13 for obligations transferred from ParentCo on the Separation Date (see below) and a decrease of $3 for other items.

The Separation and Distribution Agreement includes provisions for the assignment or allocation of environmental liabilities between Arconic and Howmet, including certain remediation obligations associated with environmental matters. In general, the respective parties are responsible for the environmental matters associated with their operations, and with the properties and other assets assigned to each. Additionally, the Separation and Distribution Agreement lists environmental matters with a shared responsibility between the two companies with an allocation of responsibility and the lead party responsible for management of each matter. For matters assigned to Arconic and Howmet under the Separation and Distribution Agreement, the companies have agreed to indemnify each other in whole or in part for environmental liabilities arising from operations prior to the Separation Date.

The following description provides details regarding the Company's largest reserve (next largest is $6), which relates to one of Arconic's current operating locations.

Massena, NY — Arconic has an ongoing remediation project related to the Grasse River, which is adjacent to the Company's Massena plant site. Many years ago, it was determined that sediments and fish in the river contain varying levels of PCBs. The project, which was selected by the U.S. Environmental Protection Agency (EPA) in a Record of Decision issued in April 2013, was aimed at capping PCB contaminated sediments with concentration in excess of one part per million in the main channel of the river and dredging PCB contaminated sediments in the near-shore areas where total PCBs exceed one part per million. The EPA approved the final design phase of the project in March 2019. Following the EPA's approval, the actual remediation fieldwork commenced. In April 2020, the EPA approved an addendum to the final remedial design to address newly-identified matters, including river navigation issues, which resulted in changing the original remedy for a specific segment of the river to dredging from capping. The Company attained substantial completion of remedial construction activities on the Grasse River in September 2021. As a result, along with an assessment of anticipated remaining future costs, primarily for post-construction monitoring, the reserve was reduced by $11.

In March 2022, an ice jam event occurred in a section of the river where the cap was installed. The ice accumulation caused a blockage in the river that restricted flow, which resulted in high forces being placed on the bottom sediments as the river worked its way through the obstruction. Once the ice cleared and it was safe to enter the river, Arconic investigated and analyzed the cap for any damage. It was determined that portions of the cap were damaged and there was disturbance to the underlying sediments. As a result, over the next several months, the Company performed extensive environmental, geotechnical, and ice modeling investigations to support the preparation of a proposed plan to repair the damaged cap and contain the exposed sediment. These activities were completed in September 2022 and led to a completed design and estimated cost of $22 for the proposed repair remedy, which included consideration of a temporary work stoppage due to the winter season extending the repair work into 2023 and the impact of cost inflation for labor and materials. Arconic's existing reserve included consideration of potential future cap repairs given the magnitude and nature of the previously completed remediation project. As a result, in 2022, the Company increased the reserve balance for this matter by $13 for the incremental amount needed to cover the estimated cost of the proposed plan. On October 3, 2022, Arconic submitted an Analysis of Alternatives report to the EPA setting forth four potential remedies, including the Company's proposed plan. On December 22, 2022, at the EPA's request, Arconic submitted a revised Analysis of Alternatives report to address two additional repair remedies for a total of six potential alternatives. Arconic is now waiting on the EPA's response, but continues to maintain an active dialogue with the EPA, as well as other stakeholders, all in support of the EPA making a final decision, which the Company expects over the next several months. Arconic's proposed remedy is consistent with the Record of Decision issued for the original remediation project (see above). In advance of the EPA's decision, the Company performed a portion of the work associated with its proposed remedy in order to reduce the environmental risk associated with the exposed sediments. This work was completed by the end of November 2022.

As the project progresses, further changes to the reserve may be required due to factors such as, among others, the EPA's selection of a remedy that differs from Arconic's proposed plan, additional changes in other remedial requirements, increased site restoration costs, and incremental ongoing operation and maintenance costs.

At December 31, 2022 and 2021, the reserve balance associated with this matter was $38 and $30, respectively. Timing of expenditures is contingent on the EPA's decision with respect to the repair remedy and the subsequent mobilization of third-party contractors.

Litigation.

All references to ParentCo in the matters described under this section Litigation refer to Arconic Inc. only and do not include its subsidiaries, except as otherwise stated.

Federal Antimonopoly Service Of The Russian Federation Litigation—The Federal Antimonopoly Service of the Russian Federation ("FAS") filed a lawsuit on March 17, 2020 with the Arbitrazh (State Commercial) Court of Samara Region against two of the Company's now former subsidiaries, Arconic Rus Investment Holdings LLC ("LLC ARIH") and AlTi Forge Holding Sarl (the "Arconic Russian Holding Companies"), naming Elliott Associates L.P., Elliott International L.P., and Elliot International Capital Advisors Inc. ("Elliott") as third parties. Also named as interested parties are: Parent Co. and certain of its foreign subsidiaries; and Arconic Netherland B.V., the Company's subsidiary that previously directly and indirectly owned LLC ARIH, Arconic SMZ JSC and JSC AlTi Forge (the "Arconic Russian Subsidiaries"). References in this disclosure to the Arconic Russian Holding Companies and/or the Arconic Russian Subsidiaries are applicable only through November 14, 2022. FAS alleges that Elliott indirectly acquired control over the former Arconic Russian Subsidiaries when, in May 2019, directors who had previously been nominated by Elliott and appointed or elected to Parent Co.'s board of directors pursuant to certain settlement agreements among Parent Co. and Elliott constituted a majority of that board as a result of a reduction in the size of the board. FAS claims alleged non-compliance with Russian Federal Law No. 57-FZ, which governs foreign ownership of certain Russian companies and requires certain governmental approvals for a foreign investor to acquire control over strategically important Russian companies. On April 6, 2020, the Samara Court granted preliminary injunctions against the Arconic Russian Holding Companies prohibiting the taking of certain corporate governance actions, including with respect to: (i) the disposal of shares in the Arconic Russian Subsidiaries; and (ii) the making of certain decisions with respect to the Arconic Russian Subsidiaries, including decisions regarding the payment of dividends, placement of bonds, amendment of bylaws and internal documents, the appointment, change and compensation of the Arconic Russian Subsidiaries' CEO, and the election of the Arconic Russian Subsidiaries' board of directors. On April 29, 2020, the Arconic Russian Holding Companies simultaneously filed an appeal and motion to revoke the previously issued injunctions. Both the appeal and motion to revoke were denied. A hearing on the merits of the claim had been postponed several times, most recently until December 22, 2022. As a consequence of the alleged violation, FAS was seeking removal and exclusion of the Arconic Russian Holding Companies from the affairs of the Arconic Russian Subsidiaries, resulting in the deprivation of the benefits of their ownership interests in the Arconic Russian Subsidiaries, including the rights currently restricted in the preliminary injunctions granted on April 6, 2020. On November 15, 2022, the Company completed the sale of all of its operations in Russia to Promishlennie Investitsii LLC, the majority owner of VSMPO-AVISMA Corporation, (see Russia in Note S). At a hearing on December 22, 2022, the Samara Court dismissed the litigation.

Reynobond PE—On June 13, 2017, the Grenfell Tower in London, U.K. caught fire resulting in fatalities, injuries, and damage. A French subsidiary of Arconic Corporation (of ParentCo at that time), Arconic Architectural Products SAS (AAP SAS), supplied a product, Reynobond PE, to its customer, a cladding system fabricator, which used the product as one component of the overall cladding system on Grenfell Tower. The fabricator supplied its portion of the cladding system to the facade installer, who then completed and installed the system under the direction of the general contractor. Neither ParentCo nor AAP SAS was involved in the design or installation of the system used at the Grenfell Tower, nor did it have a role in any other aspect of the building's refurbishment or original design. Regulatory investigations into the overall Grenfell Tower matter are being conducted, including a criminal investigation by the London Metropolitan Police Service (the "Police"), a Public Inquiry by the British government, and a consumer protection inquiry by a French public authority. The Public Inquiry was announced by the U.K. Prime Minister on June 15, 2017 and subsequently was authorized to examine the circumstances leading up to and surrounding the Grenfell Tower fire in order to make findings of fact and recommendations to the U.K. Government on matters such as the design, construction, and modification of the building, the role of relevant public authorities and contractors, the implications of the fire for the adequacy and enforcement of relevant regulations, arrangements in place for handling emergencies, and the handling of concerns from residents, among other things. Hearings for Phase 1 of the Public Inquiry began on May 21, 2018 and concluded on December 12, 2018. Phase 2 hearings of the Public Inquiry began in early 2020 and concluded in 2022, following which a final report will be written and subsequently published. As Phase 2 of the public inquiry concluded, the testimony supported AAP SAS's position that the choice of materials and the responsibility of ensuring compliance of the cladding system with relevant U.K. building code and regulations was with those individuals or entities who designed and installed the cladding system such as the architects, fabricators, contractors and building owners. The ongoing hearings in the U.K. have revealed serious doubts about whether these third parties had the necessary qualifications or expertise to carry out the refurbishment work at Grenfell Tower, adequately oversaw the process, conducted the required fire safety testing or analysis, or otherwise complied with their obligations under U.K. regulations. AAP SAS is participating as a Core Participant in the Public Inquiry and is also cooperating with the ongoing parallel investigation by the Police. Arconic

Corporation does not sell and ParentCo previously stopped selling the PE product for architectural use on buildings. Given the preliminary nature of these investigations and the uncertainty of potential future litigation, Arconic Corporation cannot reasonably estimate at this time the likelihood of an unfavorable outcome or the possible loss or range of losses in the event of an unfavorable outcome.

United Kingdom Litigation. Multiple claimant groups comprised of survivors and estates of decedents of the Grenfell Tower fire have filed claims in the U.K. arising from that fire, including as follows:

- On June 12, 2020, four claimants represented by Birnberg Peirce Ltd filed suit against AAP SAS.
- On June 12, 2020, two claimants represented by Howe & Co Solicitors filed suit against AAP SAS.
- On June 26, 2020, three claimants represented by Russell-Cooke LLP filed suit against AAP SAS.
- On December 23, 2020, several additional suits were filed by claimant groups comprised of survivors and estates of decedents. These suits were all filed against the same group of 23 defendants: AAP SAS, Arconic Corporation, Howmet Aerospace Inc., the Royal Borough of Kensington and Chelsea, the Royal Borough of Kensington and Chelsea Tenant Management Organisation Ltd, the London Fire Commissioner, the UK Home Office, The Ministry of Housing, Communities and Local Government, Rydon Maintenance Ltd, Celotex Ltd, Saint-Gobain Construction Products UK Limited, Kingspan Insulation Limited, Kingspan Group PLC (suits have since been discontinued), Studio E Architects Ltd (In liquidation), Harley Facades Ltd, Harley Curtain Wall Limited (In liquidation), CEP Architectural Facades Ltd, Exova (U.K.) Ltd, CS Stokes & Associates Ltd, Artelia Projects UK Limited (suits have since been discontinued), Whirlpool UK Appliances Limited, Whirlpool Company Polska Sp.z.o.o. and Whirlpool Corporation. These suits include as follows (represent preliminary best estimates of claimants in each suit):
    - Seven claimants represented by Deighton Pierce Glynn;
    - Seven (previously six) claimants represented by SMQ Legal Services;
    - Four (previously three) claimants represented by Scott Moncrieff;
    - Thirty-one (previously twenty-seven) claimants represented by Saunders Law;
    - Thirty-four (previously thirty-three) claimants represented by Russell Cooke LLP. On March 29, 2022, Russell Cooke issued a further suit against the above-mentioned 21 Defendants on behalf of one Claimant. The suits issued by Russell Cooke were consolidated;
    - Forty-seven (previously forty) claimants represented by Imran Khan & Partners;
    - Fifty-seven (previously sixty-one) claimants represented by Howe & Co.;
    - One hundred nineteen (previously one hundred fourteen) claimants represented by Hodge Jones and Allen Solicitor. On March 29, 2022, Hodge Jones and Allen issued a further suit against the above-mentioned 21 Defendants on behalf of four (previously five) Claimants. The suits issued by Hodge Jones and Allen were consolidated;
    - Twenty-three (previously nineteen) claimants represented by Hickman & Rose;
    - Twelve (previously ten) claimants represented by Duncan Lewis Solicitors;
    - One hundred six (previously one hundred thirteen) claimants represented by Birnberg Peirce;
    - Three hundred forty-six (previously three hundred forty-one) claimants represented by Bindmans LLP. On March 31, 2022, Bindmans issued a further suit against the above-mentioned 21 Defendants on behalf of five Claimants. The suits issued by Bindmans were consolidated;
    - Eighty (previously seventy-six) claimants represented by Bhatt Murphy Ltd; and
    - Twenty-seven (previously twenty-six) claimants represented by Slater & Gordon.

Multiple claimant groups comprised of emergency responders who attended the Grenfell Tower fire have also filed claims against AAP SAS arising from that fire, including as follows:

- On June 11, 2020, 98 (previously 80) firefighters represented by Thompsons Solicitors filed suit against AAP SAS, as well as the Royal Borough of Kensington and Chelsea, the Royal Borough of Kensington and Chelsea Tenant Management Organisation Ltd, Celotex Ltd, Exova (U.K.) Ltd, Rydon Maintenance Ltd, Studio E Architects Ltd, Harley Facades Ltd, CEP Architectural Facades Ltd, CS Stokes & Associates Ltd, and the London Fire Commissioner. Since then, another 10 (previously 7) firefighters have sought to be added to the suit.

- On June 12, 2020, 36 (previously 27) police officers represented by Penningtons Manches Cooper LLP filed suit against AAP SAS, as well as the Royal Borough of Kensington and Chelsea, the Royal Borough of Kensington and Chelsea Tenant Management Organisation Ltd, Celotex Ltd, Exova (U.K.) Ltd, Rydon Maintenance Ltd, Studio E Architects Ltd, Harley Facades Ltd, CEP Architectural Facades Ltd, CS Stokes & Associates Ltd, London Fire Commissioner, and the Commissioner of the Police of the Metropolis. Since then, some claimants have withdrawn and others have sought to be added to the suit (currently 33 claimants).

- On June 12, 2020, two firefighters represented by Pattinson and Brewer filed suit against AAP SAS, as well as the Royal Borough of Kensington and Chelsea, the Royal Borough of Kensington and Chelsea Tenant Management Organisation Ltd, Celotex Ltd, Exova (U.K.) Ltd, Rydon Maintenance Ltd, Studio E Architects Ltd, Harley Facades Ltd, CEP Architectural Facades Ltd, CS Stokes & Associates Ltd, and the London Fire Commissioner. A third firefighter, also represented by Pattinson and Brewer, brought a claim against the same defendants on June 15, 2020. One of the original firefighter claimants has now withdrawn from the suit.

On December 17, 2020, a claim was issued by the Royal Borough of Kensington and Chelsea and the Royal Borough of Kensington and Chelsea Tenant Management Organisation Ltd against: (1) Whirlpool Company Polska Spolka z Organiczona; and (2) AAP SAS. The Claimants seek damages in respect of their own losses and/or a contribution to the extent that they are found to be liable by the London High Court for any losses arising out of the Grenfell Tower fire on June 13, 2017. On March 29,2022, the Royal Borough of Kensington and Chelsea and the Royal Borough of Kensington and Chelsea Tenant Management Organisation Ltd sought permission to join two further Defendants to these proceedings, namely: (i) Whirlpool EMEA S.p.A.; and (ii) Whirlpool UK Appliances Limited.

All of these claims were filed in the High Court in London. On October 2, 2020, the High Court ordered that: (a) the suits of the survivors and estates of decedents that were issued in June 2020 be stayed until a hearing scheduled by the High Court for June 9-10, 2021; and (b) that the suits of emergency responders be stayed until a hearing scheduled by the High Court for July 7-8, 2021. The hearing scheduled for June 9-10, 2021 was subsequently vacated by the Court.

The above-mentioned suits brought by: (1) the survivors and estates of decedents; (2) the emergency responders; and (3) the Royal Borough of Kensington and Chelsea for contributions, were heard together at a procedural hearing on July 7-8, 2021, before Senior Master Fontaine. At the hearing, the Senior Master made several directions for the future management of the Grenfell Tower litigation, including staying all suits against Arconic Corporation and its affiliates until the next directions hearing, which was held on April 26, 2022. On July 28, 2022, the Senior Master stayed the cases for another 12 months until the next case management conference, which will be scheduled on a date after April 27, 2023.

The stay is intended to allow Arconic Corporation, along with several other co-defendants to the above-mentioned litigations, to engage in discussions with the claimants' legal representatives in an attempt to reach a mutually agreeable settlement. The parties have agreed to overarching terms as to the form of Alternative Dispute Resolution that they are willing to participate in and discussions are ongoing.

These discussions have resulted in progress toward a settlement to resolve a substantial majority of the claims brought by the survivors and estates of decedents though it is not certain that such a settlement will be concluded. In preparation for a potential settlement, and based on anticipated contribution to a settlement offer, Arconic recorded a liability of $61 for its share and a related receivable of $53 for costs expected to be covered by insurance proceeds if settlement is concluded. Discussions with certain other claimant groups in the above-mentioned litigations are also ongoing though there is no certainty that such discussions will result in any settlement.

Other than the recorded liability related to the claims brought by the survivors and estates of decedents noted above, and given the preliminary nature of these matters and the uncertainty of litigation, Arconic cannot reasonably estimate at this time the likelihood of an unfavorable outcome or the possible loss or range of losses in the event of an unfavorable outcome in the above-referenced disputes.

Behrens et al. v. Arconic Inc. et al. On June 6, 2019, 247 plaintiffs comprised of survivors and estates of decedents of the Grenfell Tower fire filed a complaint against “Arconic Inc., Alcoa Inc., and Arconic Architectural Products, LLC” (collectively, for purposes of the description of such proceeding, the “ParentCo Defendants”), as well as Saint-Gobain Corporation, d/b/a Celotex and Whirlpool Corporation, in the Court of Common Pleas of Philadelphia County. The complaint alleges claims under

Pennsylvania state law for products liability and wrongful death related to the fire. In particular, the plaintiffs allege that the ParentCo Defendants knowingly supplied a dangerous product (Reynobond PE) for installation on the Grenfell Tower despite knowing that Reynobond PE was unfit for use above a certain height. The ParentCo Defendants removed the case to the United States District Court for the Eastern District of Pennsylvania on June 19, 2019. On August 29, 2019, the ParentCo Defendants moved to dismiss the complaint on the bases, among other things, that: (i) the case should be heard in the United Kingdom, not the United States; (ii) there is no jurisdiction over necessary parties; and (iii) Pennsylvania product liability law does not apply to manufacture and sale of product overseas. On December 23, 2019, the Court issued an order denying the motion to dismiss the complaint on bases (ii) and (iii) suggesting a procedure for limited discovery followed by further briefing on those subjects. On September 16, 2020, the Court issued an order granting Defendants' motion to dismiss on forum non conveniens grounds, subject to certain conditions, determining that the United Kingdom, and not the United States, is the appropriate place for plaintiffs to bring their case. Plaintiffs subsequently filed a motion for reconsideration, which the Court denied on November 23, 2020. Plaintiffs are appealing this judgment; ParentCo Defendants are cross-appealing one of the conditions. The briefing was completed and oral argument was held on Plaintiffs' appeal in the Third Circuit on June 7, 2022. On July 8, 2022, the Third Circuit decided the appeal in the Behrens matter in the defendants favor. The Third Circuit affirmed the district court's dismissal of plaintiffs' case on forum non conveniens grounds. The Court also granted ParentCo Defendants' cross-appeal, invalidating one of the trial court's dismissal conditions that would have left open the possibility for Plaintiffs to return to the United States for a trial on damages if defendants were found liable in English courts and if the English court made certain other legal and factual findings. On July 22, 2022, the Plaintiffs filed a petition seeking a panel rehearing, or *en banc* rehearing, of the Third Circuit's July 8, 2022 decision, which the Third Circuit denied on October 7, 2022. On January 5, 2023, the plaintiffs filed a petition seeking review in the U.S. Supreme Court. The Supreme Court has not yet ruled on that petition. Because Plaintiffs have not exhausted all appellate options and the uncertainty of litigation, Arconic Corporation cannot reasonably estimate at this time the likelihood of an unfavorable outcome or the possible loss or range of losses in the event of an unfavorable outcome.

Howard v. Arconic Inc. et al. A purported class action complaint related to the Grenfell Tower fire was filed on August 11, 2017 in the United States District Court for the Western District of Pennsylvania against ParentCo and Klaus Kleinfeld. A related purported class action complaint was filed in the United States District Court for the Western District of Pennsylvania on September 15, 2017, under the caption *Sullivan v. Arconic Inc. et al*., against ParentCo, three former ParentCo executives, several current and former ParentCo directors, and banks that acted as underwriters for ParentCo's September 18, 2014 preferred stock offering (the "Preferred Offering"). The plaintiff in *Sullivan* had previously filed a purported class action against the same defendants on July 18, 2017 in the Southern District of New York and, on August 25, 2017, voluntarily dismissed that action without prejudice. On February 7, 2018, on motion from certain putative class members, the court consolidated *Howard* and *Sullivan*, closed *Sullivan*, and appointed lead plaintiffs in the consolidated case. On April 9, 2018, the lead plaintiffs in the consolidated purported class action filed a consolidated amended complaint. The consolidated amended complaint alleged that the registration statement for the Preferred Offering contained false and misleading statements and omitted to state material information, including by allegedly failing to disclose material uncertainties and trends resulting from sales of Reynobond PE for unsafe uses and by allegedly expressing a belief that appropriate risk management and compliance programs had been adopted while concealing the risks posed by Reynobond PE sales. The consolidated amended complaint also alleged that between November 4, 2013 and June 23, 2017 ParentCo and Kleinfeld made false and misleading statements and failed to disclose material information about ParentCo's commitment to safety, business and financial prospects, and the risks of the Reynobond PE product, including in ParentCo's Form 10-Ks for the fiscal years ended December 31, 2013, 2014, 2015, and 2016, its Form 10-Qs and quarterly financial press releases from the fourth quarter of 2013 through the first quarter of 2017, its 2013, 2014, 2015, and 2016 Annual Reports, its 2016 Annual Highlights Report, and on its official website. The consolidated amended complaint sought, among other things, unspecified compensatory damages and an award of attorney and expert fees and expenses. On June 8, 2018, all defendants moved to dismiss the consolidated amended complaint for failure to state a claim. On June 21, 2019, the Court granted the defendants' motion to dismiss in full, dismissing the consolidated amended complaint in its entirety without prejudice. On July 23, 2019, the lead plaintiffs filed a second amended complaint. The second amended complaint alleges generally the same claims as the consolidated amended complaint with certain additional allegations, as well as claims that the risk factors set forth in the registration statement for the Preferred Offering were inadequate and that certain additional statements in the sources identified above were misleading. The second amended complaint seeks, among other things, unspecified compensatory damages and an award of attorney and expert fees and expenses. On September 11, 2019, all defendants moved to dismiss the second amended complaint. Plaintiffs' opposition to that motion was filed on November 1, 2019 and all defendants filed a reply brief on November 26, 2019. On June 22, 2020, counsel for Arconic Corporation and the individual defendants filed a letter apprising the Court of a recent decision by the Third Circuit and discussing its relevance to the pending motion to dismiss. Pursuant to an Order by the Court directing the plaintiffs to respond to this letter, the plaintiffs filed a letter response on July 9, 2020. On June 23, 2021, the Court granted in part and denied in part the defendants' motion to dismiss the second amended complaint. The Court dismissed with prejudice plaintiffs' claim against ParentCo, certain officers and directors and the underwriters based on the registration statement for the Preferred Offering, with the exception of one statement and only as to purchases made before October 23, 2015. In addition, plaintiffs'

claim based on ParentCo's statements in other SEC filings, in product brochures, and on websites was dismissed in its entirety as to Kleinfeld and dismissed in part and allowed in part as to ParentCo. The Court also dismissed the control-person liability claims in their entirety. On August 11, 2021, ParentCo filed a motion with the district court for certification of an interlocutory appeal and a stay pending appeal. The motion seeks to appeal the aspect of the court's June 23, 2021 opinion concerning the complaint's pleading of ParentCo's alleged scienter. Plaintiffs filed an opposition to the motion on August 17, 2021, and ParentCo filed a reply brief on August 24, 2021. On August 12, 2021, defendants filed an answer to the second amended complaint. A status conference was held before the Court on January 11, 2022 during which the Court heard argument from both parties on the pending motion for certification of an interlocutory appeal. On July 29, 2022, the Court denied the motion for certification of an interlocutory appeal and ordered the parties to submit a proposed scheduling order, which the parties jointly filed on August 29, 2022. The Court held a status conference on September 14, 2022, and on December 2, 2022, the Court issued an Initial Case Management Order ("CMO") setting forth dates for class certification briefing and discovery. Discovery began and is ongoing. Given the preliminary nature of this matter and the uncertainty of litigation, Arconic Corporation cannot reasonably estimate at this time the likelihood of an unfavorable outcome or the possible loss or range of losses in the event of an unfavorable outcome.

Raul v. Albaugh, et al. On June 22, 2018, a derivative complaint was filed nominally on behalf of ParentCo by a purported ParentCo stockholder against the then members of ParentCo's Board of Directors and Klaus Kleinfeld and Ken Giacobbe, naming ParentCo as a nominal defendant, in the United States District Court for the District of Delaware. The complaint raises similar allegations as the consolidated amended complaint and second amended complaint in *Howard*, as well as allegations that the defendants improperly authorized the sale of Reynobond PE for unsafe uses, and asserts claims under Section 14(a) of the Securities Exchange Act of 1934, as amended, and Delaware state law. On July 13, 2018, the parties filed a stipulation agreeing to stay this case until the final resolution of the *Howard* case, the Grenfell Tower Public Inquiry in London, and the investigation by the Police and on July 23, 2018, the Court approved the stay. Given the preliminary nature of this matter and the uncertainty of litigation, Arconic Corporation cannot reasonably estimate at this time the likelihood of an unfavorable outcome or the possible loss or range of losses in the event of an unfavorable outcome.

General. While Arconic believes that all the above referenced Reynobond PE cases are without merit and intends to challenge them vigorously, there can be no assurances regarding the ultimate resolution of these matters.

Tax. Under the terms of the agreements that govern the 2016 Separation Transaction, Arconic may be entitled to future economic benefits resulting from Alcoa Corporation's utilization of certain value-added tax credits that were generated, in part, by the Company's former operations in Brazil in years prior to 2015. Because Arconic is not party to the regulatory filings with the Brazilian government and, therefore, does not have a basis to conclude on the realizability of these value-added tax credits by Alcoa Corporation, the Company will recognize income when amounts are realized, if any.

Other. In 2018, Arconic entered into a consent order with the Iowa Department of Natural Resources (IDNR) to address overflows of stormwater combined with untreated process water from the Company's Davenport plant which the IDNR alleged were unlawful bypasses prohibited by the facility's wastewater discharge permit. These overflows occurred during periodic storm events which Arconic timely reported to the IDNR. The consent order required the Company to submit a feasibility study by November 1, 2022 evaluating the reasonableness, estimated cost, impact, and overall feasibility of all actions that could be implemented to avoid unlawful bypasses from occurring in the future. After conducting extensive monitoring, analysis, and investigative work, the Company submitted the feasibility study to the IDNR in October 2022. Arconic's recommended approach, as documented in the feasibility study, consists of amending its wastewater discharge permit and constructing certain improvements to stormwater and process water systems expecting to result in estimated future capital expenditures of approximately $25 to $30, assuming Arconic's approach is approved. In 2022, the Company established a reserve of $0.5 to cover future operating expenses associated with this proposed approach. Arconic is now in discussions with the IDNR concerning the feasibility study and the Company's proposed approach, following which Arconic is then required to submit a final report for approval. The consent order requires the Company to implement the approved remedy by November 1, 2028. Approval of a final remedy by the IDNR may result in additional expenditure and/or liability.

General. In addition to the matters described above, various other lawsuits, claims, and proceedings have been or may be instituted or asserted against Arconic, including those pertaining to environmental, product liability, safety and health, employment, tax, and antitrust matters. While the amounts claimed in these other matters may be substantial, the ultimate liability is not readily determinable because of the considerable uncertainties that exist. Accordingly, it is possible that the Company's liquidity or results of operations in a reporting period could be materially affected by one or more of these other matters. However, based on facts currently available, management believes that the disposition of these other matters that are pending or asserted will not have a material adverse effect, individually or in the aggregate, on the results of operations, financial position or cash flows of Arconic.

Commitments

Purchase Obligations. Arconic has entered into purchase commitments for raw materials, energy, and other goods and services, which total $909 in 2023, $106 in 2024, $19 in 2025, $15 in 2026, $13 in 2027, and $24 thereafter as of December 31, 2022. Subsequent to December 31, 2022, the Company entered into additional aluminum supply contracts totaling approximately $765, of which approximately $375, $195, and $195 is expected to be purchased in 2023, 2024, and 2025, respectively. Additionally, through February 20, 2023, Arconic is currently negotiating terms for further aluminum supply contracts totaling approximately $2,580.

Operating Leases. See Note P for future minimum contractual obligations under long-term operating leases.

Guarantees, Letters of Credit, and Surety Bonds.

		Guarantees[1]		Letters of Credit[2]		Surety Bonds[3]	
Issuer	Beneficiary	Amount	Expiration Date	Amount	Expiration Date	Amount	Expiration Date
Arconic	Third-parties	$ 2	2023-2029	$ 11	2023	$ 48	2023-2024
Howmet	Arconic	—	—	27	2023	4	2023
Alcoa Corporation	Arconic	8	Open term	—	—	2	2024

(1) Arconic has outstanding bank guarantees related to, among others, tax matters and customs duties. Furthermore, Alcoa Corporation has outstanding bank guarantees related to the Company. In the event Alcoa Corporation would be required to perform under any of these instruments, Alcoa Corporation would be indemnified by Arconic in accordance with the 2016 Separation and Distribution Agreement.

(2) Arconic has outstanding letters of credit primarily related to insurance, environmental, and lease obligations. Additionally, Howmet has outstanding letters of credit related to the Company. In the event Howmet would be required to perform under any of these instruments, Howmet would be indemnified by Arconic in accordance with the Separation and Distribution Agreement.

(3) Arconic has outstanding surety bonds primarily related to customs duties and environmental obligations. Additionally, Howmet has outstanding surety bonds related to the Company. In the event Howmet would be required to perform under any of these instruments, Howmet would be indemnified by Arconic in accordance with the Separation and Distribution Agreement. Furthermore, Alcoa Corporation has outstanding surety bonds related to the Company. In the event Alcoa Corporation would be required to perform under any of these instruments, Alcoa Corporation would be indemnified by Arconic in accordance with the 2016 Separation and Distribution Agreement.

In August 2012, ParentCo and the Iowa Finance Authority entered into a loan agreement for the proceeds from the issuance of $250 in Midwestern Disaster Area Revenue Bonds Series 2012 due 2042 (the "Bonds"). The Bonds were issued by the Iowa Finance Authority pursuant to the Heartland Disaster Tax Relief Act of 2008 for the purpose of financing all or part of the cost of acquiring, constructing, reconstructing, and renovating certain facilities (the "Project") at Arconic's rolling mill plant in Davenport, IA. In accordance with the Separation and Distribution Agreement, as well as a Second Supplemental Tax and Project Certificate and Agreement, dated March 31, 2020, to the Tax Exemption Certificate and Agreement, dated August 14, 2012, (collectively, the "Tax Agreement"), ParentCo remained the borrower associated with the Bonds and Arconic is the legal owner of the Davenport facility, including the Project. The Company has no financial obligations related to the future debt service of the Bonds but is required to continue to operate, and maintain the location of, the Project in accordance with the Tax Agreement.

U. Financial Instruments

Amounts designated below as kmt are thousand metric tons and MMBtu are million British thermal units.

Fair Value—Fair value is defined as the price that would be received to sell an asset or paid to transfer a liability in an orderly transaction between market participants at the measurement date. The fair value hierarchy distinguishes between (i) market participant assumptions developed based on market data obtained from independent sources (observable inputs) and (ii) an entity's own assumptions about market participant assumptions developed based on the best information available in the circumstances (unobservable inputs). The fair value hierarchy consists of three broad levels, which gives the highest priority to

unadjusted quoted prices in active markets for identical assets or liabilities (Level 1) and the lowest priority to unobservable inputs (Level 3). The three levels of the fair value hierarchy are described below:

- Level 1—Unadjusted quoted prices in active markets that are accessible at the measurement date for identical, unrestricted assets or liabilities.
- Level 2—Inputs other than quoted prices included within Level 1 that are observable for the asset or liability, either directly or indirectly, including quoted prices for similar assets or liabilities in active markets; quoted prices for identical or similar assets or liabilities in markets that are not active; inputs other than quoted prices that are observable for the asset or liability (e.g., interest rates); and inputs that are derived principally from or corroborated by observable market data by correlation or other means.
- Level 3—Inputs that are both significant to the fair value measurement and unobservable.

The respective carrying value and fair value of Arconic's financial instruments were as follows:

	2022		2021	
December 31,	**Carrying value**	**Fair value**	**Carrying value**	**Fair value**
Cash and cash equivalents	$ 261	$ 261	$ 335	$ 335
Hedging instruments and derivatives - assets	21	21	1	1
Hedging instruments and derivatives - liabilities	8	8	23	23
Long-term debt	1,597	1,539	1,594	1,692

The following methods were used to estimate the fair value of financial instruments:

Cash and cash equivalents and Short-term debt. The carrying amounts approximate fair value because of the short maturity of the instruments. The fair value amounts for Cash and cash equivalents were classified in Level 1 of the fair value hierarchy and Short-term debt was classified in Level 2 of the fair value hierarchy.

Hedging instruments and derivatives. Arconic is exposed to certain risks relating to its ongoing business operations, including financial, market, political, and economic risks. Information regarding the Company's exposure to risks of changing commodity prices is described below.

Arconic's commodity and hedging activities are subject to the management, direction, and control of the Strategic Risk Management Committee (SRMC), which consists of at least three members, including the Company's chief executive officer and chief financial officer. The remaining member(s) are other Arconic officers and/or employees as the chief executive officer may designate from time to time. Currently, the only other member of the SRMC is the Company's treasurer. The SRMC meets on a periodic basis to review hedging positions and strategy and reports to the Audit and Finance Committee of Arconic's Board of Directors on the scope of its activities.

The Company's hedging instruments are held for purposes other than trading. They are used primarily to mitigate uncertainty and volatility, and to cover underlying exposures. Specifically, these instruments hedge forward sale commitments for aluminum and forward purchase commitments for aluminum, natural gas, and certain alloying materials. Arconic is not involved in trading activities for energy, weather derivatives, or other nonexchange commodity trading activities.

The fair value of the Company's hedging instruments was based on quoted market prices (e.g., aluminum prices on the 10-year London Metal Exchange forward curve) and were classified in Level 1 of the fair value hierarchy. Most of these instruments are comprised of those that were designated as cash flow hedges while the remainder are marked-to-market as they do not qualify for hedge accounting.

The following table presents the fair value and amount of underlying by type for all hedging instruments:

	December 31, 2022			December 31, 2021		
Assets						
Cash flow hedges						
Aluminum	$ —	17	kmt	$ —	—	
Alloying materials*	(1)	1	kmt	—	—	
Marked-to-market						
Aluminum	6	60	kmt	1	14	kmt
	$ 5			$ 1		
Liabilities						
Cash flow hedges						
Aluminum*	$ (14)	368	kmt	$ 18	251	kmt
Energy	5	6.3	MMBtu	2	3.3	MMBtu
Alloying materials	1	3	kmt	—	—	kmt
Marked-to-market						
Aluminum	12	53	kmt	3	19	kmt
Energy	4	0.6	MMBtu	—	—	
	$ 8			$ 23		

* The hedging instruments are classified as assets or liabilities based on the overall position of all instruments held with respective counterparties.

The following table presents the unrealized and realized gains and losses associated with those hedging instruments designated as cash flow hedges:

	2022	2021	2020
Unrealized			
Other comprehensive loss			
Aluminum	$ 81	$ (160)	$ (7)
Energy	16	(2)	—
Alloying materials	(2)	1	5
	$ 95	$ (161)	$ (2)
Realized*			
Sales			
Aluminum	$ 43	$ (150)	$ 2
Cost of goods sold			
Aluminum	(5)	(8)	10
Energy	(20)	—	—
Alloying materials	1	(2)	—
	$ 67	$ (140)	$ (8)

* In all periods presented, these amounts were reclassified from Accumulated other comprehensive loss (see Note L).

For hedging instruments that do not qualify for hedge accounting, in 2022, the Company recognized an unrealized loss of $4 and a realized gain of $6 in Sales for aluminum, and an unrealized loss of $4 and a realized gain of $3 in Cost of goods sold for energy. Unrealized and realized impacts were not material in 2021 and 2020.

The disclosures with respect to commodity price risk do not consider the underlying commitments or anticipated transactions. If the underlying items were included, the gains or losses on the hedging instruments may be offset. Actual results will be determined by several factors that are not under Arconic's control and could vary significantly from those factors disclosed.

The Company is exposed to credit loss in the event of nonperformance by counterparties on the above instruments, as well as credit or performance risk with respect to its hedged customers' commitments. Arconic does not anticipate nonperformance by any of these parties. Contracts are with creditworthy counterparties and are further supported by cash or irrevocable letters of credit issued by carefully chosen banks. In addition, master netting arrangements are in place with counterparties to facilitate settlement of gains and losses on these contracts.

Separately, Arconic has three natural gas supply contracts that are treated as derivatives for accounting purposes as they failed to qualify for the normal purchase normal sale exception due to net settlement provisions. These derivatives also do not qualify for hedge accounting. The Company does not have a regular practice of entering into contracts that are treated as derivatives for accounting purposes. As of December 31, 2022, Arconic's derivatives classified as assets consisted of $16 (12.1 MMBtu). Additionally, in 2022, the Company recognized an unrealized gain of $16, in Cost of goods sold for these derivatives (see Note A).

Long-term debt. The fair value was based on quoted market prices for public debt and were classified in Level 2 of the fair value hierarchy.

V. Receivables

On January 5, 2022, the Company entered into a one-year arrangement with a financial institution to sell certain customer receivables outright without recourse on a continuous basis. All such sales are at Arconic's discretion. Under this arrangement, the Company will serve in an administrative capacity, including collection of the receivables from the respective customers and remittance of these cash collections to the financial institution. Accordingly, upon the sale of customer receivables to the financial institution, Arconic removes the underlying trade receivables from the Consolidated Balance Sheet and includes the reduction as a positive amount in the Decrease (Increase) in receivables line item within Operating Activities on the Statement of Consolidated Cash Flows. At no time can the outstanding balance due to the financial institution exceed $225 (increased in February from original amount of $100). In 2022, the Company sold customer receivables and remitted cash to the financial institution both in the amount of $1,119. Of the total amount of customer receivables sold in 2022, $77 were included in Receivables from customers on the accompanying Consolidated Balance Sheet as of December 31, 2021. In December 2022, this arrangement was amended to provide for an automatic renewal each year unless terminated in accordance with the provisions of the underlying purchase agreement.

W. Subsequent Events

Management evaluated all activity of Arconic and concluded that no subsequent events have occurred that would require recognition in the Consolidated Financial Statements or disclosure in the Notes to the Consolidated Financial Statements.

Item 9. Changes in and Disagreements With Accountants on Accounting and Financial Disclosure.

None.

Item 9A. Controls and Procedures.

(a) Evaluation of Disclosure Controls and Procedures

Arconic’s Chief Executive Officer and Chief Financial Officer have evaluated the Company’s disclosure controls and procedures, as defined in Rules 13a-15(e) and 15d-15(e) of the Securities Exchange Act of 1934, as of the end of the period covered by this report, and they have concluded that these controls and procedures are effective.

(b) Management’s Annual Report on Internal Control over Financial Reporting

Management's Report on Internal Control over Financial Reporting is included in Part II Item 8 of this Form 10-K.

(c) Attestation Report of the Registered Public Accounting Firm

The effectiveness of Arconic’s internal control over financial reporting as of December 31, 2022 has been audited by PricewaterhouseCoopers LLP, an independent registered public accounting firm, as stated in their report, which is included in Part II Item 8 of this Form 10-K.

(d) Changes in Internal Control over Financial Reporting

There have been no changes in internal control over financial reporting during the fourth quarter of 2022 that have materially affected, or are reasonably likely to materially affect, the Company's internal control over financial reporting.

Item 9B. Other Information.

None.

Item 9C. Disclosure Regarding Foreign Jurisdictions that Prevent Inspections.

None.

PART III

Item 10. Directors, Executive Officers and Corporate Governance.

Information regarding the Company's executive officers is included in Part I. Item 1. "Business—Information about our Executive Officers" and is incorporated herein by reference. The other information required by this item will be included under the caption "Corporate Governance" in the Company's Definitive Proxy Statement for its 2023 Annual Meeting of Stockholders (the "2023 Proxy Statement") and is incorporated herein by reference.

Code of Ethics

The Company's Code of Ethics, which applies to our CEO, CFO, principal accounting officer or controller, or persons performing similar functions, is publicly available on the Company's Internet website at http://www.arconic.com under the section "Investors—Corporate Governance—Governance and Policies." The Company intends to disclose any changes in, or waivers from, this Code of Ethics by posting such information on the same website or by filing a Current Report on Form 8-K, in each case to the extent such disclosure is required by SEC or New York Stock Exchange rules and regulations.

Item 11. Executive Compensation.

The information required by this item will be included under the captions "Corporate Governance," "Compensation Discussion and Analysis," "Report of the Compensation and Benefits Committee" and "Executive Compensation" in the 2023 Proxy Statement and is incorporated herein by reference.

Item 12. Security Ownership of Certain Beneficial Owners and Management and Related Stockholder Matters.

The information required by this item will be included under the caption "Information Regarding Security Holders" in the 2023 Proxy Statement and is incorporated herein by reference. Information regarding the Company's equity compensation plans will be included under the caption "Equity Compensation Plan Information" in the 2023 Proxy Statement and is incorporated herein by reference.

Item 13. Certain Relationships and Related Transactions, and Director Independence.

The information required by this item will be included under the caption "Corporate Governance" in the 2023 Proxy Statement and is incorporated herein by reference.

Item 14. Principal Accounting Fees and Services.

The information required by this item will be included under the caption "Principal Accountant Fees and Services" and "Report of the Audit and Finance Committee" in the 2023 Proxy Statement and is incorporated herein by reference.

PART IV

Item 15. Exhibits, Financial Statements Schedules.

(a) The consolidated financial statements and exhibits listed below are filed as part of this Annual Report on Form 10-K.

(1) The Company's consolidated financial statements, the notes thereto and the report of the Independent Registered Public Accounting Firm (PCAOB ID238) are included in Part II. Item 8. "Financial Statements and Supplementary Data."

(2) Financial statement schedules have been omitted because they are not applicable, not required, or the required information is included in the Consolidated Financial Statements or Notes thereto.

(3) Exhibits.

Reference is made to Item 15(b) below.

(b) *Exhibits*. The Exhibit Index, which immediately precedes the signature page, is incorporated by reference into this Annual Report on Form 10-K.

(c) *Financial Statement Schedules*. Reference is made to Item 15(a)(2) above.

Item 16. Form 10-K Summary.

None.

Exhibit Index

Exhibit Number	Exhibit Description
2.1	Separation and Distribution Agreement, dated as of March 31, 2020, by and between Arconic Inc. and Arconic Rolled Products Corporation (incorporated by reference to Exhibit 2.1 to the registrant's Current Report on Form 8-K filed on April 3, 2020)
2.2	Tax Matters Agreement, dated as of March 31, 2020,by and between Arconic Inc. and Arconic Rolled Products Corporation (incorporated by reference to Exhibit 2.2 to the registrant's Current Report on Form 8-K filed on April 3, 2020)
2.3(a)	Employee Matters Agreement, dated as of March 31, 2020, by and between Arconic Inc. and Arconic Rolled Products Corporation (incorporated by reference to Exhibit 2.3 to the registrant's Current Report on Form 8-K filed on April 3, 2020)
2.3(b)	First Amendment to Employee Matters Agreement, dated as of April 10, 2020, by and between Arconic Inc. and Arconic Rolled Products Corporation (incorporated by reference to Exhibit 2.1 to the registrant's Current Report on Form 8-K filed on April 10, 2020)
2.4	Patent Know-How and Trade Secret License Agreement, dated as of March 31, 2020by and between Arconic Inc. and Arconic Corporation (incorporated by reference to Exhibit 2.4 to the registrant's Current Report on Form 8-K filed on April 3, 2020)
2.5(a)	Patent Know-How and Trade Secret License Agreement, dated as of March 31, 2020by and between Arconic Rolled Products Corporation and Arconic Inc. (incorporated by reference to Exhibit 2.5 to the registrant's Current Report on Form 8-K filed on April 3, 2020)
2.5(b)	Amendment No. 1 to Patent Know-How and Trade Secret License Agreement, dated as of August 25, 2020, by and between Arconic Rolled Products Corporation and Arconic Inc. (incorporated by reference to Exhibit 2.5(b) to the registrant's Annual Report on Form 10-K filed on February 22, 2022)
2.6	Trademark License Agreement, dated as of March 31, 2020,by and between Arconic Rolled Products Corporation and Arconic Inc. (incorporated by reference to Exhibit 2.6 to the registrant's Current Report on Form 8-K filed on April 3, 2020)
2.7	Trademark License Agreement, dated as of March 31, 2020,by and between Arconic Inc. and Arconic Rolled Products Corporation (incorporated by reference to Exhibit 2.7 to the registrant's Current Report on Form 8-K filed on April 3, 2020)
2.8	Master Agreement for Product Supply, dated as of March 31, 2020,by and between Arconic Massena LLC, Arconic Lafayette LLC, Arconic Davenport LLC and Arconic Inc. (incorporated by reference to Exhibit 2.8 to the registrant's Current Report on Form 8-K filed on April 3, 2020)
2.9	Metal Supply & Tolling Agreement, dated as of January 9, 2020,by and between Arconic-Köfém Mill Products Hungary Kft and Arconic-Köfém Kft, dated January 1, 2020 (incorporated by reference to Exhibit 2.9 to the registrant's Annual Report on Form 10-K filed on March 30, 2020)
2.10	Use Agreement by and between Arconic-Köfém Székesfehérvári Könnyűfémmű Korlátolt Felelősségű Társaság and Arconic-Köfém Mill Products Hungary Korlátolt Felelősségű Társaság, dated January 6, 2020 (incorporated by reference to Exhibit 2.10 to the registrant's Annual Report on Form 10-K filed on March 30, 2020)

Exhibit Number	Exhibit Description
2.11	Land Use Right Agreement by and between Arconic-Köfém Mill Products Hungary Korlátolt Felelősségű Társaság and Arconic-Köfém Székesfehérvári Könnyűfémmű Korlátolt Felelősségű Társaság, dated January 6, 2020 (incorporated by reference to Exhibit 2.11 to the registrant's Annual Report on Form 10-K filed on March 30, 2020)
2.12	Service Level Agreement for Central Engineering and Maintenance by and between Arconic-Köfém Kft and Arconic-Köfém Mill Products Hungary Kft, dated January 9, 2020 (incorporated by reference to Exhibit 2.12 to the registrant's Annual Report on Form 10-K filed on March 30, 2020)
2.13	Service Level Agreement for Energy, Steam and Water by and between Arconic-Köfém Kft and Arconic-Köfém Mill Products Hungary Kft, dated January 31, 2020 (incorporated by reference to Exhibit 2.13 to the registrant's Annual Report on Form 10-K filed on March 30, 2020)
2.14	Land Use Right Agreement by and between Arconic-Köfém Székesfehérvári Könnyűfémmű Korlátolt Felelősségű Társaság and Arconic-Köfém Mill Products Hungary Korlátolt Felelősségű Társaság, dated January 6, 2020 (incorporated by reference to Exhibit 2.14 to the registrant's Annual Report on Form 10-K filed on March 30, 2020)
2.15(a)	Second Supplemental Tax and Project Certificate and Agreement, dated as of March 31, 2020, by and among Arconic Inc., Arconic Davenport LLC and Arconic Rolled Products Corporation (incorporated by reference to Exhibit 2.9 to the registrant's Current Report on Form 8-K filed on April 3, 2020)
2.15(b)	Third Supplemental Tax and Project Certificate and Agreement, dated as of December 31, 2022, by and among Howmet Aerospace Inc., Arconic US LLC, Arconic Davenport LLC and Arconic Corporation
2.16	Lease and Property Management Agreement, dated as of March 31, 2020, by and between Arconic Inc. and Arconic Massena LLC (incorporated by reference to Exhibit 2.10 to the registrant's Current Report on Form 8-K filed on April 3, 2020)
3.1	Amended and Restated Certificate of Incorporation of Arconic Corporation (incorporated by reference to Exhibit 3.1(b) to the registrant's Current Report on Form 8-K filed on April 3, 2020)
3.2	Amended and Restated Bylaws of Arconic Corporation (incorporated by reference to Exhibit 3.1 to the registrant's Current Report on Form 8-K filed on December 6, 2022)
4.1	Indenture for 6.000% First Lien Notes Due 2025, dated as of May 13, 2020, among Arconic Corporation and U.S. Bank National Association, as trustee, notes collateral agent, registrar, paying agent and authenticating agent (including Form of Note due 2025) (incorporated by reference to Exhibit 4.1 to the registrant's Current Report on Form 8-K filed on May 13, 2020)
4.2(a)	Indenture for 6.125% Second Lien Notes Due 2028, dated as of February 7, 2020, among Arconic Rolled Products Corporation and U.S. Bank National Association, as trustee, notes collateral agent, registrar, paying agent and authenticating agent (including Form of Note due 2028) (incorporated by reference to Exhibit 10.14 to Amendment No. 2 to the registrant's Registration Statement on Form 10 filed on February 7, 2020)
4.2(b)	Supplemental Indenture, dated as of March 30, 2020, among each subsidiary listed therein, U.S. Bank National Association, as trustee, and U.S. Bank National Association, as second priority collateral agent, authenticating agent, registrar and paying agent (incorporated by reference to Exhibit 4.2 to the registrant's Current Report on Form 8-K filed on April 3, 2020)
4.2(c)	Second Supplemental Indenture, dated as of March 3, 2021, among Arconic Corporation and each subsidiary listed therein, U.S. Bank National Association, as trustee, and U.S. Bank National Association, as second priority collateral agent, authenticating agent, registrar and paying agent (incorporated by reference to Exhibit 4.1 to the registrant's Current Report on Form 8-K filed on March 3, 2021)
4.3	Description of Registrant's Securities Registered Pursuant to Section 12 of the Securities Exchange Act of 1934
10.1(a) +	Arconic Corporation 2020 Stock Incentive Plan (incorporated by reference to Exhibit 10.1 to the registrant's Current Report on Form 8-K filed on April 3, 2020)
10.1(b) +	Amended and Restated Arconic Corporation 2020 Stock Incentive Plan (incorporated by reference to Exhibit 99.1 to the registrant's Registration Statement on Form S-8 filed on May 20, 2021)
10.1(c) +	Arconic Corporation Amended and Restated 2020 Stock Incentive Plan (incorporated by reference to Exhibit 104 to the registrant's Current Report on Form 8-K filed on May 20, 2022)
10.2 +	Form of Indemnification Agreement by and between Arconic Corporation and individual directors or officers (incorporated by reference to Exhibit 10.2 to the registrant's registration statement on Form 10 filed on December 17, 2019)
10.3 *	United Company RUSAL – Trading House Agreement for the Supply of Aluminum Products by and between United Company RUSAL — Trading House and Arconic SMZ, dated December 27, 2016 (incorporated by reference to Exhibit 10.3 to the registrant's registration statement on Form 10 filed on December 17, 2019)*
10.4(a) +	Arconic Corporation Deferred Fee Plan for Directors (incorporated by reference to Exhibit 10.6 to the registrant's Current Report on Form 8-K filed on April 3, 2020)

Exhibit Number	Exhibit Description
10.4(b) +	Arconic Corporation Amended and Restated Deferred Fee Plan for Directors (incorporated by reference to Exhibit 10.2 to the registrant's Current Report on Form 8-K filed on December 21, 2020)
10.5(a) +	Arconic Corporation Annual Cash Incentive Plan (incorporated by reference to Exhibit 10.2 to the registrant's Current Report on Form 8-K filed on April 3, 2020)
10.5(b) +	Amended and Restated Arconic Corporation 2020 Annual Cash Incentive Plan (incorporated by reference to Exhibit 10.1 to the registrant's Current Report on Form 8-K filed on May 21, 2021)
10.5(c) +	Amended and Restated Arconic Corporation 2020 Annual Cash Incentive Plan (incorporated by reference to Exhibit 10.3 to the registrant's Current Report on Form 8-K filed on May 20, 2022)
10.6(a) +	Arconic Corporation Non-Employee Director Compensation Policy (incorporated by reference to Exhibit 10.7 to the registrant's Current Report on Form 8-K filed on April 3, 2020)
10.6(b) +	Arconic Corporation Amended and Restated Non-Employee Director Compensation Policy (incorporated by reference to Exhibit 10.1 to the registrant's Current Report on Form 8-K filed on December 21, 2020)
10.6(c) +	Arconic Corporation Amended Non-Employee Director Compensation Policy, effective January 1, 2022 (incorporated by reference to Exhibit 10.1 to the registrant's Current Report on Form 8-K filed on November 2, 2021)
10.7 +	Employment Letter Agreement between Arconic Inc. and Timothy D. Myers, dated as of January 13, 2020 (incorporated by reference to Exhibit 10.7 to Amendment No. 1 to the registrant's registration statement on Form 10 filed on January 22, 2020)
10.8 +	Employment Letter Agreement between Arconic Inc. and Erick R. Asmussen, dated as of January 29, 2020 (incorporated by reference to Exhibit 10.8 to Amendment No. 2 to the registrant's registration statement on Form 10 filed on February 7, 2020)
10.9 +	Employment Letter Agreement between Arconic Corporation and Daniel G. Fayock, dated as of March 7, 2022 (incorporated by reference to Exhibit 10.3 to the registrant's Quarterly Report on Form 10-Q filed on May 3, 2022)
10.10(a) +	Arconic Corporation Change in Control Severance Plan (incorporated by reference to Exhibit 10.3 to the registrant's Current Report on Form 8-K filed on April 3, 2020)
10.10(b) +	Arconic Corporation Amended and Restated Change in Control Severance Plan (incorporated by reference to Exhibit 10.2 to the registrant's Current Report on Form 8-K filed onMay 20, 2022)
10.11(a) +	Arconic Corporation Executive Severance Plan (incorporated by reference to Exhibit 10.4 to the registrant's Current Report on Form 8-K filed on April 3, 2020)
10.11(b) +	Arconic Corporation Amended and Restated Executive Severance Plan (incorporated by reference to Exhibit 10.1 to the registrant's Current Report on Form 8-K filed on May 20, 2022)
10.12 +	Arconic Corporation Deferred Compensation Plan (incorporated by reference to Exhibit 10.12 to Amendment No. 2 to the registrant's registration statement on Form 10 filed on February 7, 2020)
10.13 +	Arconic Corporation Excess Plan C (incorporated by reference to Exhibit 10.13 to Amendment No. 2 to the registrant's registration statement on Form 10 filed on February 7, 2020)
10.14 +	Arconic Corporation Legal Fee Reimbursement Plan (incorporated by reference to Exhibit 10.5 to the registrant's Current Report on Form 8-K filed on April 3, 2020)
10.15(a) +	Terms and Conditions for Restricted Share Units for Annual Director Awards under the Arconic Corporation 2020 Stock Incentive Plan, effective April 8, 2020 (incorporated by reference to Exhibit 10.4 to the registrant's Current Report on Form 8-K filed on April 10, 2020)
10.15(b) +	Terms and Conditions for Restricted Share Units for Annual Director Awards under the Arconic Corporation 2020 Stock Incentive Plan, effective January 1, 2021 (incorporated by reference to Exhibit 10.3 to the registrant's Current Report on Form 8-K filed on December 21, 2020)
10.15(c) +	Terms and Conditions for Deferred Fee Restricted Share Units for Annual Director Awards under the Arconic Corporation 2020 Stock Incentive Plan, effective April 8, 2020 (incorporated by reference to Exhibit 10.5 to the registrant's Current Report on Form 8-K filed on April 10, 2020)
10.15(d) +	Terms and Conditions for Deferred Fee Restricted Share Units for Annual Director Awards under the Arconic Corporation 2020 Stock Incentive Plan, effective January 1, 2021 (incorporated by reference to Exhibit 10.4 to the registrant's Current Report on Form 8-K filed on December 21, 2020)
10.15(e) +	Form of Restricted Share Unit Award Agreement pursuant to the Arconic Corporation 2020 Stock Incentive Plan, effective April 23, 2020 (incorporated by reference to Exhibit 10.1 to the registrant's Current Report on Form 8-K filed on April 23, 2020)
10.15(f) +	Form of Special Retention Award Agreement pursuant to the Arconic Corporation 2020 Stock Incentive Plan, effective April 23, 2020 (incorporated by reference to Exhibit 10.2 to the registrant's Current Report on Form 8-K filed on April 23, 2020)
10.15(g) +	Form of Restricted Share Unit Award Agreement pursuant to the Arconic Corporation 2020 Stock Incentive Plan, effective February 3, 2021 (incorporated by reference to Exhibit 10.1 to the registrant's Current Report on Form 8-K filed on February 8, 2021)

Exhibit Number	Exhibit Description
10.15(h) +	Form of Special Retention Award Agreement pursuant to the Arconic Corporation 2020 Stock Incentive Plan, effective February 3, 2021 (incorporated by reference to Exhibit 10.2 to the registrant's Current Report on Form 8-K filed on February 8, 2021)
10.15(i) +	Form of Restricted Share Unit Award Agreement pursuant to the Arconic Corporation Amended and Restated 2020 Stock Incentive Plan, effective May 19, 2022 (incorporated by reference to Exhibit 10.5 to the registrant's Current Report on Form 8-K filed on May 20, 2022)
10.15(j) +	Form of Special Retention Award Agreement pursuant to the Arconic Corporation Amended and Restated 2020 Stock Incentive Plan, effective May 19, 2022 (incorporated by reference to Exhibit 10.6 to the registrant's Current Report on Form 8-K filed on May 20, 2022)
10.15(k) +	Form of Restricted Share Unit Award Agreement pursuant to the Arconic Corporation Amended and Restated 2020 Stock Incentive Plan, effective March 1, 2023
10.15(l) +	Form of Special Retention Award Agreement pursuant to the Arconic Corporation Amended and Restated 2020 Stock Incentive Plan, effective March 1, 2023
10.16	Credit Agreement, dated as of March 25, 2020, by and among Arconic Rolled Products Corporation, the lenders and issuing banks party thereto, and JPMorgan Chase Bank, N.A., as administrative agent (incorporated by reference to Exhibit 10.1 to the registrant's Current Report on Form 8-K filed on March 26, 2020)
10.17(a)	Credit Agreement, dated May 13, 2020, among Arconic Corporation, a syndicate of lenders named therein and Deutsche Bank AG New York Branch, as administrative agent and collateral agent (incorporated by reference to Exhibit 10.1 to the registrant's Current Report on Form 8-K filed on May 13, 2020)
10.17(b)	Amendment No. 1 to Credit Agreement, dated as of February 16, 2022, among Arconic Corporation, the guarantors party thereto, a syndicate of lenders named therein and Deutsche Bank AG New York Branch, as administrative agent (incorporated by reference to Exhibit 10.1 to the registrant's Current Report on Form 8-K filed on February 17, 2022)
10.17(c)	Amendment No. 2 to Credit Agreement, dated as of February 16, 2022, among Arconic Corporation, the guarantors party thereto, a syndicate of lenders named therein and Deutsche Bank AG New York Branch, as administrative agent (incorporated by reference to Exhibit 10.2 to the registrant's Current Report on Form 8-K filed on February 17, 2022)
10.18 +	Employment Letter Agreement between Arconic Corporation and Diana B. Perreiah, dated as of August 5, 2022 (incorporated by reference to Exhibit 10.1 to the registrant's Quarterly Report on Form 10-Q filed on November 1, 2022)
10.19 +	Employment Letter Agreement between Arconic Corporation and Robert L. Woodall, dated as of August 5, 2022 (incorporated by reference to Exhibit 10.2 to the registrant's Quarterly Report on Form 10-Q filed on November 1, 2022)
21	Subsidiaries of the Registrant
23	Consent of Independent Registered Public Accounting Firm
24	Power of Attorney of Directors of Arconic Corporation
31	Certifications pursuant to Section 302 of the Sarbanes-Oxley Act of 2002
32	Certification pursuant to Section 906 of the Sarbanes-Oxley Act of 2002
101.INS	Inline XBRL Instance Document
101.SCH	Inline XBRL Taxonomy Extension Schema Document
101.CAL	Inline XBRL Taxonomy Extension Calculation Linkbase Document
101.DEF	Inline XBRL Taxonomy Extension Definition Linkbase Document
101.LAB	Inline XBRL Taxonomy Extension Label Linkbase Document
101.PRE	Inline XBRL Taxonomy Extension Presentation Linkbase Document
104	Cover Page Interactive Data File (formatted as Inline XBRL)

* Portions of the exhibit have been omitted to preserve confidentiality in compliance with Item 601 of Regulation S-K.

\+ Management contract or compensatory plan or arrangement.

SIGNATURES

Pursuant to the requirements of Section 13 or 15(d) of the Securities Exchange Act of 1934, the registrant has duly caused this report to be signed on its behalf by the undersigned, thereunto duly authorized.

ARCONIC CORPORATION

February 21, 2023	By	*/s/ Mary E. Zik*
		Mary E. Zik
		Vice President and Controller (Also signing as Principal Accounting Officer)

Pursuant to the requirements of the Securities Exchange Act of 1934, this report has been signed below by the following persons on behalf of the registrant and in the capacities and on the dates indicated.

Signature	Title	Date
/s/ Timothy D. Myers		February 21, 2023
Timothy D. Myers	Chief Executive Officer and Director (Principal Executive Officer)	
/s/ Erick R. Asmussen		February 21, 2023
Erick R. Asmussen	Executive Vice President and Chief Financial Officer (Principal Financial Officer)	

William F. Austen, Christopher L. Ayers, Margaret S. Billson, Jacques Croisetiere, Elmer L. Doty, Carol S. Eicher, Frederick A. Henderson, Ellis A. Jones, E. Stanley O'Neal, and Jeffrey Stafeil as Directors, on February 21, 2023, by Daniel G. Fayock, their attorney-in-fact.

*By */s/ Daniel G. Fayock*
Daniel G. Fayock